



Test Report issued under the responsibility of:



TEST REPORT
IEC 62368-1
Audio/video, information and communication technology equipment
Part 1: Safety requirements

Report Number..... : 21SBCS12047 00481

Date of issue : 2022-04-06

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Name of Testing Laboratory : DongGuan ShuoXin Electronic Technology Co., Ltd.
preparing the Report

Applicant's name..... : SHENZHEN MEGMEET ELECTRICAL CO LTD

Address : 5TH FL, TOWER A B C501-503 D E UNIS INFORPORT, 13
LANGSHAN RD N SEC HI-TECH INDUSTRIAL PARK NAN
SHAN SHENZHEN, GUANGDONG 518057 CHINA

Test specification:

Standard : IEC 62368-1:2018

Test procedure..... : CB Scheme

Non-standard test method..... : N/A

TRF template used : IECEE OD-2020-F1:2020, Ed.1.3

Test Report Form No...... : IEC62368_1E

Test Report Form(s) Originator.... : UL(US)

Master TRF : Dated 2021-02-04

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This report is not valid as a CB Test Report unless signed by an approved CB Testing Laboratory and appended to a CB Test Certificate issued by an NCB in accordance with IECEE 02.

General disclaimer:

The test results presented in this report relate only to the object tested.

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Test item description	SWITCHING POWER SUPPLY	
Trade Mark(s)	MEGMEET	
Manufacturer	Same as applicant	
Model/Type reference	MR484000SG1XXXXXX, MR484000SG2XXXXXX, MR483000SG3XXXXXX, MPS484000HL (X can be A to Z, a to z, 0-9 or blank, indicating the customer code, which do not affect safety and EMC)	
Ratings	For models MR484000SG1XXXXXX, MR484000SG2XXXXXX, MPS484000HL: Input: 100-130V~, 50/60Hz, 21A Output: 53.5Vdc, 32.5A Input: 200-250V~, 50/60Hz, 21A Output: 53.5Vdc, 75A For models MR483000SG3XXXXXX: Input: 100-130V~, 50/60Hz, 17A Output: 53.5Vdc, 25.2A Input: 200-250V~, 50/60Hz, 17A Output: 53.5Vdc, 56.1A	
Responsible Testing Laboratory (as applicable), testing procedure and testing location(s):		
☒ CB Testing Laboratory:	DongGuan ShuoXin Electronic Technology Co., Ltd.	
Testing location/ address	Zone A, 1F, No. 6, XinGang Road YuanGang Street, XinAn District, ChangAn Town, DongGuan City, GuangDong, China	
Tested by (name, function, signature)	Moon Liu / Project handler	
Approved by (name, function, signature) ..	Bruce Zhang / Project reviewer	
Testing procedure: CTF Stage 1:		
Testing location/ address		
Tested by (name, function, signature)		
Approved by (name, function, signature) ..		
Testing procedure: CTF Stage 2:		
Testing location/ address		
Tested by (name, function, signature)		
Witnessed by (name, function, signature) . :		
Approved by (name, function, signature) .. :		
Testing procedure: CTF Stage 3:		
Testing location/ address		
Tested by (name, function, signature)		
Witnessed by (name, function, signature) . :		
Approved by (name, function, signature) .. :		

☐	Testing procedure: CTF Stage 4:		
Testing location/ address			
Tested by (name, function, signature)			
Witnessed by (name, function, signature) .			
Approved by (name, function, signature) ..			
Supervised by (name, function, signature) :			
List of Attachments (including a total number of pages in each attachment):			
- Enclosures (30 pages) - National Differences (35 pages)			
Summary of testing:			
See below for summary and applicable clauses.			
If not otherwise specified, tests were performed on models MR484000SG1, MR484000SG2 and MR483000SG3 to represent other models.			
Tests performed (name of test and test clause):			Testing location:
5.2	Electrical energy source classifications	Unless otherwise indicated, all tests were conducted at DongGuan ShuoXin Electronic Technology Co., Ltd. Zone A, 1F, No. 6, XinGang Road YuanGang Street, XinAn District, ChangAn Town, DongGuan City, GuangDong, China	
5.4.1.3	Test for Hygroscopic Materials		
5.4.1.4, 9.3, B.1.5, B.2.6	Maximum operating temperatures for materials, components and systems		
5.4.1.8	Determination of working voltage		
5.4.2.2, 5.4.2.4 and 5.4.3, T.2	Clearance and Creepage Distance Measurement		
5.4.5.2	Voltage surge test		
5.4.8	Humidity conditioning		
5.4.9	Electric strength test		
5.5.2.2	Capacitor discharge test		
5.6.6.2	Resistance of protective conductors and terminations		
5.7.4	Touch Voltage and Touch Current Measurement – Unearthed Accessible Parts		
5.7.5	Earthed accessible conductive parts		
9.2	Thermal energy source Classifications		
B.2.5	Input tests		
B.3, B.4	Abnormal operating and fault condition tests		
F.3.10	Durability, legibility and permanence of markings		
Annex G.5.3.3	Transformer overload		
T.2	Steady force test, 10 N		

T.5	Steady force test, 250 N	
T.6	Enclosure impact test	
T.8	Stress relief test	

Summary of compliance with National Differences (List of countries addressed):
 EU group differences and national differences, United States of America, Canada, Singapore
 CENELEC member countries (EU group differences): Austria, Belgium, Bulgaria, Croatia, Cyprus, the Czech Republic, Denmark, Estonia, Finland, Former Yugoslav Republic of Macedonia, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, the Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey and the United Kingdom.

☒ The product fulfils the requirements of EN IEC 62368-1:2020+A11:2020
☒ The product fulfils the requirements of BS EN IEC 62368-1:2020+A11:2020
☒ The product fulfils the requirements of UL 62368-1, Third Edition
☒ The product fulfils the requirements of CSA C22.2 No. 62368-1:19

For National Differences see corresponding attachment.

Statement concerning the uncertainty of the measurement systems used for the tests
 (may be required by the product standard or client)

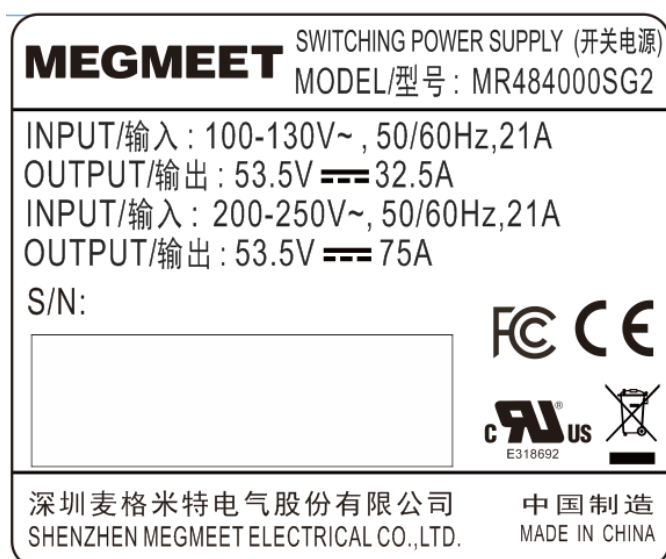
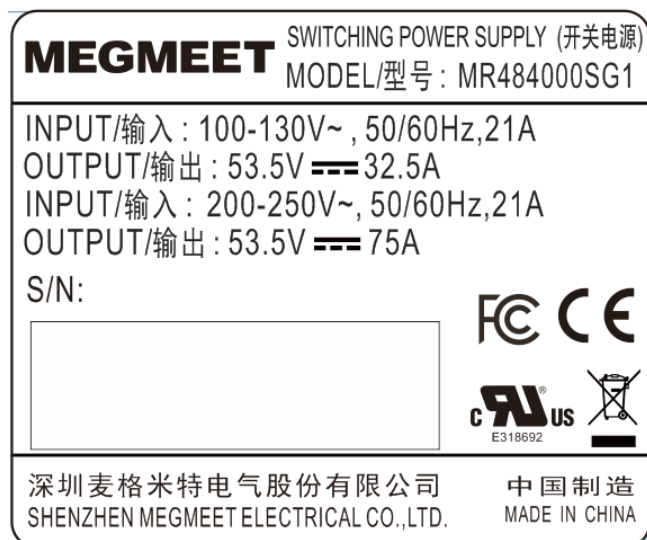
☐ **Internal procedure used for type testing through which traceability of the measuring uncertainty has been established:**
Procedure number, issue date and title:

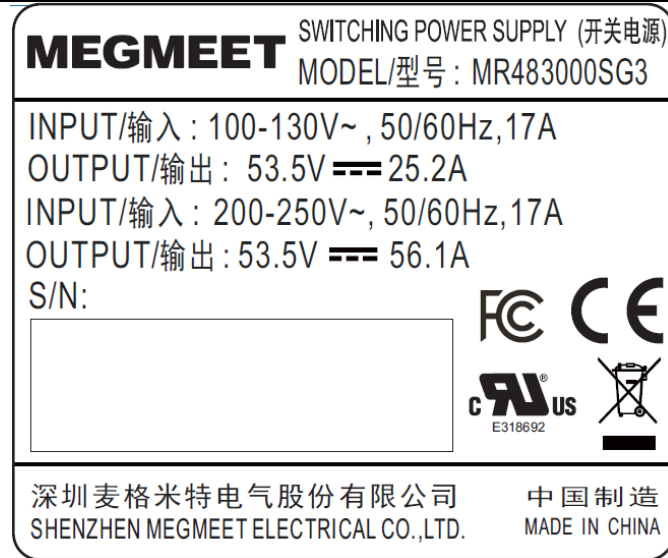
Calculations leading to the reported values are on file with the NCB and testing laboratory that conducted the testing.

☒ **Statement not required by the standard used for type testing**
 (Note: When IEC or ISO standard requires a statement concerning the uncertainty of the measurement systems used for tests, this should be reported above. The informative text in parenthesis should be delete in both cases after selecting the applicable option)

Copy of marking plate:

The artwork below may be only a draft. The use of certification marks on a product must be authorized by the respective NCBs that own these marks.





Test item particulars:	
Product group	☐ end product ☒ built-in component
Classification of use by	☒ Ordinary person ☐ Children likely present ☒ Instructed person ☒ Skilled person
Supply connection	☒ AC mains ☐ DC mains ☐ not mains connected: ☐ ES1 ☐ ES2 ☐ ES3
Supply tolerance	☒ +10%/-10% ☐ +20%/-15% ☐ +__%/-__% ☐ None
Supply connection – type	☐ pluggable equipment type A - ☐ non-detachable supply cord ☐ appliance coupler ☐ direct plug-in ☐ pluggable equipment type B - ☐ non-detachable supply cord ☐ appliance coupler ☐ permanent connection ☐ mating connector ☒ other: be evaluated in end system
Considered current rating of protective device	☒ 25A Location: ☒ building ☐ equipment ☐ N/A
Equipment mobility	☐ movable ☐ hand-held ☐ transportable ☐ direct plug-in ☐ stationary ☒ for building-in ☐ wall/ceiling-mounted ☐ SRME/rack-mounted ☐ other:
Overvoltage category (OVC)	☐ OVC I ☒ OVC II ☐ OVC III ☐ OVC IV ☐ other:
Class of equipment	☒ Class I ☐ Class II ☐ Class III ☐ Not classified ☐
Special installation location	☒ N/A ☐ restricted access area ☐ outdoor location ☐
Pollution degree (PD)	☐ PD 1 ☒ PD 2 ☐ PD 3
Manufacturer's specified T_{ma}	For the maximum operating temperature is 45°C, the equipment can be loaded with maximum rated loading; For the maximum operating temperature is 75°C, see ambient derating table for detail. ☐ Outdoor: minimum °C
IP protection class	☒ IPX0 ☐ IP__
Power systems	☒ TN ☐ TT ☐ IT - V _{L-L} ☐ not AC mains
Altitude during operation (m)	☐ 2000 m or less ☒ 5000 m
Altitude of test laboratory (m)	☒ 2000 m or less ☐ m
Mass of equipment (kg)	Approx. 2.022kg for MR484000SG1XXXXXX, Approx. 2.008kg for MR484000SG2XXXXXX

Possible test case verdicts: - test case does not apply to the test object.....: N/A - test object does meet the requirement.....: P (Pass) - test object does not meet the requirement.....: F (Fail)											
Testing: Date of receipt of test item: 2022-02-17 Date (s) of performance of tests: 2022-02-17 to 2022-03-01											
General remarks: "(See Enclosure #)" refers to additional information appended to the report. "(See appended table)" refers to a table appended to the report. Throughout this report a ☐ comma / ☒ point is used as the decimal separator.											
Manufacturer's Declaration per sub-clause 4.2.5 of IEC62368-1:											
The application for obtaining a CB Test Certificate includes more than one factory location and a declaration from the Manufacturer stating that the sample(s) submitted for evaluation is (are) representative of the products from each factory has been provided:		☐ Yes ☒ Not applicable									
When differences exist; they shall be identified in the General product information section.											
Name and address of factory (ies).....: Zhuzhou Megmeet Electrical Co., Ltd. 1728# Taishan Rd, Tianyuan District, Zhuzhou, Hu'nan, 412007, China.											
General product information and other remarks: 1. The units is SWITCHING POWER SUPPLY intended for building-in except for DC fan side, which is intended to be used for information technology equipment or Audio/Video and similar electronics apparatus. All electronic components are mounted on PWB and housed with in metallic enclosure. 2. Built-in type equipment except for DC fan side, suitable enclosure should be provided by end system. When installing the power supply, all requirements of relevant standard must be fulfilled. 3. Limit Short Circuit tested has not been conducted in PWB trace, need to be considered in end system. 4. Metal enclosure should be permanently connected to earth in end system. 5. The unit is provided with SPD connected to earth, the end system should be provided with reliable earthing in according with Clause 5.6.7 requirement in IEC 62368-1. 6. When installed into the end system, connect earth before connecting L/N conductors, Disconnect earth after disconnecting L/N conductors. 7. Refer to ambient derating table below, following conditions have been evaluated:											
<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 25%;">Model</th> <th style="width: 25%;">Input</th> <th style="width: 25%;">Output (ambient 45°C)</th> <th style="width: 25%;">Output (ambient 75°C)</th> </tr> </thead> <tbody> <tr> <td>MR484000SG1 XXXXXX, MPS484000HL, MR484000SG2</td> <td>100-130Vac, 50/60Hz, 21A</td> <td>53.5Vdc, 32.5A</td> <td>53.5Vdc, 14.9A</td> </tr> </tbody> </table>				Model	Input	Output (ambient 45°C)	Output (ambient 75°C)	MR484000SG1 XXXXXX, MPS484000HL, MR484000SG2	100-130Vac, 50/60Hz, 21A	53.5Vdc, 32.5A	53.5Vdc, 14.9A
Model	Input	Output (ambient 45°C)	Output (ambient 75°C)								
MR484000SG1 XXXXXX, MPS484000HL, MR484000SG2	100-130Vac, 50/60Hz, 21A	53.5Vdc, 32.5A	53.5Vdc, 14.9A								

XXXXXX	200-250Vac, 50/60Hz, 21A	53.5Vdc, 75A	53.5Vdc, 37.38A
MR483000SG3 XXXXXX	100-130Vac, 50/60Hz, 17A	53.5Vdc, 25.2A	--
	200-250Vac, 50/60Hz, 17A	53.5Vdc, 56.1A	--

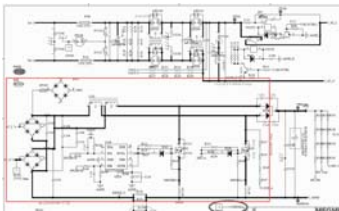
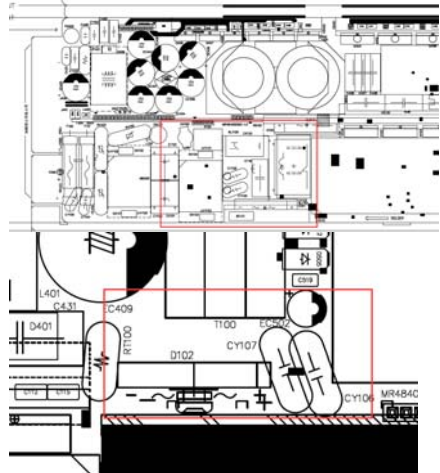
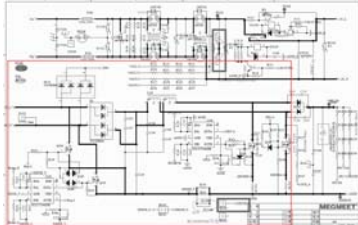
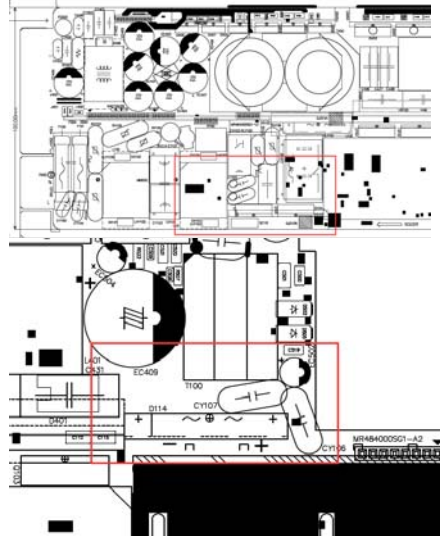
Model Differences:

The models are identical to each other except for model name, Circuit (Rectification method), PCB layout, input current rating and output rating.

MR484000SG1XXXXXX and MPS484000HL are identical to each other except for model name.

MR484000SG2XXXXXX and MR483000SG3XXXXXX are identical to each other except for model name, input current rating and output rating.

MR484000SG1XXXXXX and MR484000SG2XXXXXX are identical to each other except for model name, Circuit (Rectification method) and PCB layout, see below for details:

Model	Circuit	PCB layout
MR484000SG1X XXXXX		
MR484000SG2X XXXXX		

Additional application considerations – (Considerations used to test a component or sub-assembly) –

1. The product was submitted and evaluated for use at the maximum ambient temperature (Tma) permitted by the manufacturer's specification of: For the maximum operating temperature is 45°C, the equipment can be loaded with maximum rated loading; For the maximum operating temperature is 75°C, see ambient derating table for detail.
2. The equipment disconnect device is considered to be: be evaluated in end system
3. Basic insulation is provided between input circuit and PE/metal enclosure. Reinforced insulation is provided between input and output circuit.
4. The openings is considered to be evaluated in end system except for DC fan side.
5. This equipment is intended to operate in an area which has an elevation of maximum 16,405 feet (5,000 meters). Clearance distances were calculated according to Table 16, factor 1.48.

OVERVIEW OF ENERGY SOURCES AND SAFEGUARDS				
Clause	Possible Hazard			
5	Electrically-caused injury			
Class and Energy Source (e.g. ES3: Primary circuit)	Body Part (e.g. Ordinary)	Safeguards		
		B	S	R
ES3: All circuits except for output circuits	Ordinary, Instructed person, Skilled person	See 5.4.2, 5.4.3	PE (Built-in component, consider in end system)	Enclosures are required, other see 5.3.2, 5.4.2, 5.4.3, 5.5.3, 5.5.4
ES3: Store discharge on X-capacitor	Ordinary, Instructed person, Skilled person	N/A	N/A	See 5.5.2
ES2: Store discharge on Y-capacitor	Ordinary, Instructed person, Skilled person	N/A	N/A	See 5.5.2
ES1: Secondary output	Ordinary, Instructed person, Skilled person	N/A	N/A	N/A
6	Electrically-caused fire			
Class and Energy Source (e.g. PS2: 100 Watt circuit)	Material part (e.g. Printed board)	Safeguards		
		B	1 st S	2 nd S
PS3: All circuits	1) All combustible materials/ components; 2) PCB; 3) Internal wiring	1) See 6.3 2) See 6.3 3) See 6.3	1) See, 6.4.5 6.4.6 2) V-1 or better 3) See 6.5	1) N/A 2) N/A 3) N/A
	4) Plastic Enclosure(front side only)	4) See 6.3	4) V-0 or better	4) N/A
7	Injury caused by hazardous substances			
Class and Energy Source (e.g. Ozone)	Body Part (e.g., Skilled)	Safeguards		
		B	S	R
N/A	N/A	N/A	N/A	N/A
8	Mechanically-caused injury			
Class and Energy Source (e.g. MS3: Plastic fan blades)	Body Part (e.g. Ordinary)	Safeguards		
		B	S	R
MS1: Mass of the unit	Ordinary	N/A	N/A	N/A
MS1: Edges and corners	Ordinary	N/A	N/A	N/A

MS2: DC Fan	Ordinary	N/A	N/A	Enclosure and plastic cover
9	Thermal burn			
Class and Energy Source (e.g. TS1: Keyboard caps)	Body Part (e.g., Ordinary)	Safeguards		
		B	S	R
TS3: Internal parts/circuits	Ordinary	N/A	N/A	Suitable enclosure should be provided in end system (except for DC fan side)
TS1: Plastic DC fan side	Ordinary	N/A	N/A	N/A
TS1: Metal screw near DC fan side	Ordinary	N/A	N/A	N/A
10	Radiation			
Class and Energy Source (e.g. RS1: PMP sound output)	Body Part (e.g., Ordinary)	Safeguards		
		B	S	R
RS1: Indicating LED	Ordinary	N/A	N/A	N/A
Supplementary Information: “B” – Basic Safeguard; “S” – Supplementary Safeguard; “R” – Reinforced Safeguard				

ENERGY SOURCE DIAGRAM

Optional. Manufacturers are to provide the energy sources diagram identify declared energy sources and identifying the demarcations are between power sources. Recommend diagram be provided included in power supply and multipart systems.

Insert diagram below. Example diagram designs are; Block diagrams; image(s) with layered data; mechanical drawings

☐ ES ☐ PS ☐ MS ☐ TS ☐ RS

See table OVERVIEW OF ENERGY SOURCES AND SAFEGUARDS for detail

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
4	GENERAL REQUIREMENTS		P
4.1.1	Acceptance of materials, components and subassemblies		P
4.1.2	Use of components	(See appended table 4.1.2)	P
4.1.3	Equipment design and construction		P
4.1.4	Specified ambient temperature for outdoor use (°C) :	For indoor use	N/A
4.1.5	Constructions and components not specifically covered		N/A
4.1.8	Liquids and liquid filled components (LFC)		N/A
4.1.15	Markings and instructions	(See Annex F)	P
4.4.3	Safeguard robustness		P
4.4.3.1	General		P
4.4.3.2	Steady force tests	Building-in equipment, however per applicant's request the appliance enclosure of DC fan side was tested. (See Annex T.2, T.3, T.4, T.5)	P
4.4.3.3	Drop tests		N/A
4.4.3.4	Impact tests	Building-in equipment, however per applicant's request the appliance enclosure of DC fan side was tested. (See Annex T.6)	P
4.4.3.5	Internal accessible safeguard tests		N/A
4.4.3.6	Glass impact tests		N/A
4.4.3.7	Glass fixation tests		N/A
	Glass impact test (1J)		N/A
	Push/pull test (10 N)		N/A
4.4.3.8	Thermoplastic material tests	(See Annex T.8)	P
4.4.3.9	Air comprising a safeguard	Building-in equipment, however per applicant's request the appliance enclosure of DC fan side was tested. (See Annex T)	P
4.4.3.10	Accessibility, glass, safeguard effectiveness		P
4.4.4	Displacement of a safeguard by an insulating liquid		N/A
4.4.5	Safety interlocks		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
4.5	Explosion		P
4.5.1	General		P
4.5.2	No explosion during normal/abnormal operating condition	(See Clause B.2, B.3)	P
	No harm by explosion during single fault conditions	(See Clause B.4)	P
4.6	Fixing of conductors		P
	Fix conductors not to defeat a safeguard		P
	Compliance is checked by test..... :	10 N pull / push test performed	P
4.7	Equipment for direct insertion into mains socket-outlets		N/A
4.7.2	Mains plug part complies with relevant standard .. :		N/A
4.7.3	Torque (Nm)		N/A
4.8	Equipment containing coin/button cell batteries		N/A
4.8.1	General	No batteries	N/A
4.8.2	Instructional safeguard		N/A
4.8.3	Battery compartment door/cover construction		N/A
	Open torque test		N/A
4.8.4.2	Stress relief test		N/A
4.8.4.3	Battery replacement test		N/A
4.8.4.4	Drop test		N/A
4.8.4.5	Impact test		N/A
4.8.4.6	Crush test		N/A
4.8.5	Compliance		N/A
	30N force test with test probe		N/A
	20N force test with test hook		N/A
4.9	Likelihood of fire or shock due to entry of conductive object		N/A
4.10	Component requirements		P
4.10.1	Disconnect Device	Disconnect device should be evaluated in end system	N/A
4.10.2	Switches and relays	No Switches, relays see Clause G.2	P

5	ELECTRICALLY-CAUSED INJURY		P
5.2	Classification and limits of electrical energy sources		P
5.2.2	ES1, ES2 and ES3 limits		P
5.2.2.2	Steady-state voltage and current limits	(See appended table 5.2)	P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.2.2.3	Capacitance limits	(See appended table 5.2)	P
5.2.2.4	Single pulse limits	No single pulse introduced	N/A
5.2.2.5	Limits for repetitive pulses	No repetitive pulses introduced	N/A
5.2.2.6	Ringing signals	No ringing signals.	N/A
5.2.2.7	Audio signals	No audio signals.	N/A
5.3	Protection against electrical energy sources		P
5.3.1	General Requirements for accessible parts to ordinary, instructed and skilled persons	Built-in component, consider in end system except for DC fan side	P
5.3.1 a)	Accessible ES1/ES2 derived from ES2/ES3 circuits		P
5.3.1 b)	Skilled persons not unintentional contact ES3 bare conductors		P
5.3.2.1	Accessibility to electrical energy sources and safeguards	Built-in component, consider in end system except for DC fan side	P
	Accessibility to outdoor equipment bare parts		N/A
5.3.2.2	Contact requirements	Built-in component, consider in end system except for DC fan side	P
	Test with test probe from Annex V	See annex V	—
5.3.2.2 a)	Air gap – electric strength test potential (V)		N/A
5.3.2.2 b)	Air gap – distance (mm)	More than 0.39mm. Built-in component, consider in end system except for DC fan side	P
5.3.2.3	Compliance	Built-in component, consider in end system except for DC fan side	P
5.3.2.4	Terminals for connecting stripped wire		N/A
5.4	Insulation materials and requirements		P
5.4.1.2	Properties of insulating material		P
5.4.1.3	Material is non-hygroscopic		P
5.4.1.4	Maximum operating temperature for insulating materials	(See appended table 5.4.1.4, 9.3, B.1.5, B.2.6)	P
5.4.1.5	Pollution degrees	Pollution degree 2	P
5.4.1.5.2	Test for pollution degree 1 environment and for an insulating compound		N/A
5.4.1.5.3	Thermal cycling test	UL recognized PWB as per clause 5.4.7 required.	N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.4.1.6	Insulation in transformers with varying dimensions		N/A
5.4.1.7	Insulation in circuits generating starting pulses		N/A
5.4.1.8	Determination of working voltage	(See appended table 5.4.1.8)	P
5.4.1.9	Insulating surfaces		N/A
5.4.1.10	Thermoplastic parts on which conductive metallic parts are directly mounted	Phenolic material which is acceptable without test.	P
5.4.1.10.2	Vicat test.....		N/A
5.4.1.10.3	Ball pressure test		N/A
5.4.2	Clearances		P
5.4.2.1	General requirements		P
	Clearances in circuits connected to AC Mains, Alternative method		N/A
5.4.2.2	Procedure 1 for determining clearance		P
	Temporary overvoltage	2000Vpk	—
5.4.2.3	Procedure 2 for determining clearance		P
5.4.2.3.2.2	a.c. mains transient voltage	2500Vpk	—
5.4.2.3.2.3	d.c. mains transient voltage	N/A	—
5.4.2.3.2.4	External circuit transient voltage.....	N/A	—
5.4.2.3.2.5	Transient voltage determined by measurement	N/A	—
5.4.2.4	Determining the adequacy of a clearance using an electric strength test.....	Not such procedure used.	N/A
5.4.2.5	Multiplication factors for clearances and test voltages	Up to 5000m, Factor 1.48	P
5.4.2.6	Clearance measurement	(See appended table 5.4.2)	P
5.4.3	Creepage distances		P
5.4.3.1	General		P
5.4.3.3	Material group	IIIb	—
5.4.3.4	Creepage distances measurement	(See appended table 5.4.3)	P
5.4.4	Solid insulation		P
5.4.4.1	General requirements		P
5.4.4.2	Minimum distance through insulation	(See appended table 5.4.4.2)	P
5.4.4.3	Insulating compound forming solid insulation		P
5.4.4.4	Solid insulation in semiconductor devices	Approved optocoupler used	P
5.4.4.5	Insulating compound forming cemented joints	UL recognized Multi-layer PCB complied to c)	P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.4.4.6	Thin sheet material	Tapes used in and around transformer	P
5.4.4.6.1	General requirements		P
5.4.4.6.2	Separable thin sheet material		P
	Number of layers (pcs).....:	Min. two layers	P
5.4.4.6.3	Non-separable thin sheet material		N/A
	Number of layers (pcs)		N/A
5.4.4.6.4	Standard test procedure for non-separable thin sheet material		N/A
5.4.4.6.5	Mandrel test		N/A
5.4.4.7	Solid insulation in wound components		P
5.4.4.9	Solid insulation at frequencies >30 kHz, E_P , K_R , d , V_{PW} (V)		N/A
	Alternative by electric strength test, tested voltage (V), K_R:	(See appended Table 5.4.4.9)	P
5.4.5	Antenna terminal insulation	Performed between mains and output connector which may be connected to an equipment with antenna terminal	P
5.4.5.1	General		P
5.4.5.2	Voltage surge test	Surge test with 50 discharges at a maximum rate of 12/min from a 1nF capacitor charged to 10kV performed. Measured insulation resistance between mains supply to output terminals after the surge test, see below for details.	P
5.4.5.3	Insulation resistance (M Ω).....:	Between input and output terminal: 500M Ω	P
	Electric strength test		N/A
5.4.6	Insulation of internal wire as part of supplementary safeguard	No such wire.	N/A
5.4.7	Tests for semiconductor components and for cemented joints	UL recognized Multi-layer PCB	P
5.4.8	Humidity conditioning		P
	Relative humidity (%), temperature (°C), duration (h)	95%, 40°C, 120 h	—
5.4.9	Electric strength test		P
5.4.9.1	Test procedure for type test of solid insulation.....:	(See appended table 5.4.9)	P
5.4.9.2	Test procedure for routine test		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.4.10	Safeguards against transient voltages from external circuits	No connection to external circuits with transient voltage.	N/A
5.4.10.1	Parts and circuits separated from external circuits		N/A
5.4.10.2	Test methods		N/A
5.4.10.2.1	General		N/A
5.4.10.2.2	Impulse test		N/A
5.4.10.2.3	Steady-state test.....		N/A
5.4.10.3	Verification for insulation breakdown for impulse test.....		N/A
5.4.11	Separation between external circuits and earth	No connection to external circuits with transient voltage.	N/A
5.4.11.1	Exceptions to separation between external circuits and earth		N/A
5.4.11.2	Requirements		N/A
	SPDs bridge separation between external circuit and earth		N/A
	Rated operating voltage $U_{op}(V)$		—
	Nominal voltage $U_{peak}(V)$		—
	Max increase due to variation ΔU_{sp}		—
	Max increase due to ageing ΔU_{sa}		—
5.4.11.3	Test method and compliance		N/A
5.4.12	Insulating liquid		N/A
5.4.12.1	General requirements		N/A
5.4.12.2	Electric strength of an insulating liquid		N/A
5.4.12.3	Compatibility of an insulating liquid		N/A
5.4.12.4	Container for insulating liquid		N/A
5.5	Components as safeguards		P
5.5.1	General		P
5.5.2	Capacitors and RC units		P
5.5.2.1	General requirement		P
5.5.2.2	Safeguards against capacitor discharge after disconnection of a connector	(See appended table 5.5.2.2)	P
5.5.3	Transformers	Complied with clause annex G.5.3.	P
5.5.4	Optocouplers	(Clause G.12)	P
5.5.5	Relays	Relay not used as a safeguard	N/A
5.5.6	Resistors		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.5.7	SPDs	The earth connection shall comply with 5.6.7; and The varistor shall comply with Clause G.8.	P
5.5.8	Insulation between the mains and an external circuit consisting of a coaxial cable		N/A
5.5.9	Safeguards for socket-outlets in outdoor equipment		N/A
	RCD rated residual operating current (mA).....		—
5.6	Protective conductor		P
5.6.2	Requirement for protective conductors	Should be considered in end system	N/A
5.6.2.1	General requirements	No switches, current limiting devices or over current protective devices use in Protective conductors. Should be considered in end system	N/A
5.6.2.2	Colour of insulation		N/A
5.6.3	Requirement for protective earthing conductors		N/A
	Protective earthing conductor size (mm ²)		—
	Protective earthing conductor serving as a reinforced safeguard		N/A
	Protective earthing conductor serving as a double safeguard		N/A
5.6.4	Requirements for protective bonding conductors		P
5.6.4.1	Protective bonding conductors		P
	Protective bonding conductor size (mm ²).	Comply with sub-clause 5.6.6.2 and built-in component, also considered in end system	—
5.6.4.2	Protective current rating (A).....	25A	P
5.6.5	Terminals for protective conductors		P
5.6.5.1	Terminal size for connecting protective earthing conductors (mm).....		N/A
	Terminal size for connecting protective bonding conductors (mm)	Screws diameter Min. 5.3 mm	P
5.6.5.2	Corrosion		P
5.6.6	Resistance of the protective bonding system		P
5.6.6.1	Requirements		P
5.6.6.2	Test Method	(See appended table 5.6.6)	P
5.6.6.3	Resistance (Ω) or voltage drop.....	(See appended table 5.6.6)	P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.6.7	Reliable connection of a protective earthing conductor	Built-in component, consider in end system	N/A
5.6.8	Functional earthing		N/A
	Conductor size (mm ²)..... :		N/A
	Class II with functional earthing marking :		N/A
	Appliance inlet cl & cr (mm)..... :		N/A
5.7	Prospective touch voltage, touch current and protective conductor current		P
5.7.2	Measuring devices and networks		P
5.7.2.1	Measurement of touch current	(See appended table 5.2)	P
5.7.2.2	Measurement of voltage	(See appended table 5.2)	P
5.7.3	Equipment set-up, supply connections and earth connections		P
5.7.4	Unearthed accessible parts :	(See appended table 5.7.4)	P
5.7.5	Earthed accessible conductive parts :	(See appended table 5.7.5)	P
5.7.6	Requirements when touch current exceeds ES2 limits	Touch current not exceeds ES2 limits	N/A
	Protective conductor current (mA)..... :		N/A
	Instructional Safeguard..... :		N/A
5.7.7	Prospective touch voltage and touch current associated with external circuits		N/A
5.7.7.1	Touch current from coaxial cables		N/A
5.7.7.2	Prospective touch voltage and touch current associated with paired conductor cables		N/A
5.7.8	Summation of touch currents from external circuits		N/A
	a) Equipment connected to earthed external circuits, current (mA) :		N/A
	b) Equipment connected to unearthed external circuits, current (mA)..... :		N/A
5.8	Backfeed safeguard in battery backed up supplies		N/A
	Mains terminal ES..... :		N/A
	Air gap (mm)..... :		N/A
6	ELECTRICALLY- CAUSED FIRE		P
6.2	Classification of PS and PIS		P
6.2.2	Power source circuit classifications..... :	All circuits are considered PS3.	P
6.2.3	Classification of potential ignition sources		P
6.2.3.1	Arcing PIS :	(See appended table 6.2.3.1)	P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
6.2.3.2	Resistive PIS	(See appended table 6.2.3.2)	P
6.3	Safeguards against fire under normal operating and abnormal operating conditions		P
6.3.1	No ignition and attainable temperature value less than 90 % defined by ISO 871 or less than 300 °C for unknown materials	(See appended table 5.4.1.4, 9.3, B.1.5, B.2.6)	P
	Combustible materials outside fire enclosure	Label	P
6.4	Safeguards against fire under single fault conditions		P
6.4.1	Safeguard method	Method of "control of fire spread" is used.	P
6.4.2	Reduction of the likelihood of ignition under single fault conditions in PS1 circuits		N/A
6.4.3	Reduction of the likelihood of ignition under single fault conditions in PS2 and PS3 circuits	Building-in equipment, it should be evaluated in end product.	N/A
6.4.3.1	Supplementary safeguards		N/A
6.4.3.2	Single Fault Conditions		N/A
	Special conditions for temperature limited by fuse		N/A
6.4.4	Control of fire spread in PS1 circuits		N/A
6.4.5	Control of fire spread in PS2 circuits		N/A
6.4.5.2	Supplementary safeguards		N/A
6.4.6	Control of fire spread in PS3 circuits	- Printed board: rated min. V-1 - All other components: at least V-2 except for parts mounted on min. V-1 material or small parts of combustible material (with mass less than 4g) or components complying with relevant IEC standard. - Isolating transformer: complying with G.5.3. - Wire insulation and tubing: complying with Clause 6.5 - Built-in component, suitable enclosure should be considered in end system except for DC fan side.	P
6.4.7	Separation of combustible materials from a PIS		N/A
6.4.7.2	Separation by distance		N/A
6.4.7.3	Separation by a fire barrier		N/A
6.4.8	Fire enclosures and fire barriers	DC fan side considered, Other considered in end system	P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
6.4.8.2	Fire enclosure and fire barrier material properties		P
6.4.8.2.1	Requirements for a fire barrier		N/A
6.4.8.2.2	Requirements for a fire enclosure		P
6.4.8.3	Constructional requirements for a fire enclosure and a fire barrier		P
6.4.8.3.1	Fire enclosure and fire barrier openings	Min. 15mm distance to PS3 circuit.	P
6.4.8.3.2	Fire barrier dimensions		N/A
6.4.8.3.3	Top openings and properties		N/A
	Openings dimensions (mm)..... :		N/A
6.4.8.3.4	Bottom openings and properties		N/A
	Openings dimensions (mm)..... :		N/A
	Flammability tests for the bottom of a fire enclosure		N/A
	Instructional Safeguard..... :		N/A
6.4.8.3.5	Side openings and properties		P
	Openings dimensions (mm)..... :	Min. 15mm distance to PS3 circuit.	P
6.4.8.3.6	Integrity of a fire enclosure, condition met: a), b) or c)..... :		N/A
6.4.8.4	Separation of a PIS from a fire enclosure and a fire barrier distance (mm) or flammability rating :		N/A
6.4.9	Flammability of insulating liquid..... :		N/A
6.5	Internal and external wiring		P
6.5.1	General requirements	VW-1 wires used was considered meet related IEC standard requirements.	P
6.5.2	Requirements for interconnection to building wiring :	See table 4.1.2 for detail	P
6.5.3	Internal wiring size (mm ²) for socket-outlets..... :		N/A
6.6	Safeguards against fire due to the connection to additional equipment		N/A

7	INJURY CAUSED BY HAZARDOUS SUBSTANCES		N/A
7.2	Reduction of exposure to hazardous substances		N/A
7.3	Ozone exposure		N/A
7.4	Use of personal safeguards or personal protective equipment (PPE)		N/A
	Personal safeguards and instructions :		—
7.5	Use of instructional safeguards and instructions		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Instructional safeguard (ISO 7010)		—
7.6	Batteries and their protection circuits		N/A

8	MECHANICALLY-CAUSED INJURY		P
8.2	Mechanical energy source classifications		P
8.3	Safeguards against mechanical energy sources		P
8.4	Safeguards against parts with sharp edges and corners		P
8.4.1	Safeguards		N/A
	Instructional Safeguard		N/A
8.4.2	Sharp edges or corners	Sharp edges and corners classified as MS1. No safeguard is required to be interposed between MS1 and an ordinary person.	P
8.5	Safeguards against moving parts		P
8.5.1	Fingers, jewellery, clothing, hair, etc., contact with MS2 or MS3 parts	The test probe cannot touch the DC fan. The plastic cover used as safeguard. MS2: Plastic fan blade $\frac{N}{44000} + \frac{K}{7200} \leq 1$ m=0.020kg, r=20mm, N=19800rpm (for maximum speed) K = 6 x 10 ⁻⁷ (m r ² N ²)=1882 N/44000+K/7200=0.45+0.26 0.71(for Maximum speed)	P
	MS2 or MS3 part required to be accessible for the function of the equipment		N/A
	Moving MS3 parts only accessible to skilled person		N/A
8.5.2	Instructional safeguard	The moving part is obvious when expose, so the contact is unlikely.	P
8.5.4	Special categories of equipment containing moving parts		N/A
8.5.4.1	General		N/A
8.5.4.2	Equipment containing work cells with MS3 parts		N/A
8.5.4.2.1	Protection of persons in the work cell		N/A
8.5.4.2.2	Access protection override		N/A
8.5.4.2.2.1	Override system		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
8.5.4.2.2.2	Visual indicator		N/A
8.5.4.2.3	Emergency stop system		N/A
	Maximum stopping distance from the point of activation (m).....:		N/A
	Space between end point and nearest fixed mechanical part (mm)		N/A
8.5.4.2.4	Endurance requirements		N/A
	Mechanical system subjected to 100 000 cycles of operation		N/A
	- Mechanical function check and visual inspection		N/A
	- Cable assembly		N/A
8.5.4.3	Equipment having electromechanical device for destruction of media		N/A
8.5.4.3.1	Equipment safeguards		N/A
8.5.4.3.2	Instructional safeguards against moving parts		N/A
8.5.4.3.3	Disconnection from the supply		N/A
8.5.4.3.4	Cut type and test force (N).....:		N/A
8.5.4.3.5	Compliance		N/A
8.5.5	High pressure lamps	No high pressure lamps.	N/A
	Explosion test.....:		N/A
8.5.5.3	Glass particles dimensions (mm)		N/A
8.6	Stability of equipment		N/A
8.6.1	General	MS1	P
	Instructional safeguard	No stability requirements.	P
8.6.2	Static stability		N/A
8.6.2.2	Static stability test		N/A
8.6.2.3	Downward force test		N/A
8.6.3	Relocation stability		N/A
	Wheels diameter (mm)		—
	Tilt test		N/A
8.6.4	Glass slide test		N/A
8.6.5	Horizontal force test		N/A
8.7	Equipment mounted to wall, ceiling or other structure		N/A
8.7.1	Mount means type		N/A
8.7.2	Test methods		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Test 1, additional downwards force (N).....:		N/A
	Test 2, number of attachment points and test force (N).....:		N/A
	Test 3 Nominal diameter (mm) and applied torque (Nm).....:		N/A
8.8	Handles strength		N/A
8.8.1	General		N/A
8.8.2	Handle strength test		N/A
	Number of handles.....:		—
	Force applied (N)		—
8.9	Wheels or casters attachment requirements		N/A
8.9.2	Pull test	No wheels or casters.	N/A
8.10	Carts, stands and similar carriers		N/A
8.10.1	General	No carts or stands or other carriers.	N/A
8.10.2	Marking and instructions.....:		N/A
8.10.3	Cart, stand or carrier loading test		N/A
	Loading force applied (N)		N/A
8.10.4	Cart, stand or carrier impact test		N/A
8.10.5	Mechanical stability		N/A
	Force applied (N)		—
8.10.6	Thermoplastic temperature stability		N/A
8.11	Mounting means for slide-rail mounted equipment (SRME)		N/A
8.11.1	General	No slide-rail	N/A
8.11.2	Requirements for slide rails		N/A
	Instructional Safeguard.....:		N/A
8.11.3	Mechanical strength test		N/A
8.11.3.1	Downward force test, force (N) applied.....:		N/A
8.11.3.2	Lateral push force test		N/A
8.11.3.3	Integrity of slide rail end stops		N/A
8.11.4	Compliance		N/A
8.12	Telescoping or rod antennas		N/A
	Button/ball diameter (mm)	No rod antennas.	—

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
9	THERMAL BURN INJURY		P
9.2	Thermal energy source classifications		P
9.3	Touch temperature limits		P
9.3.1	Touch temperatures of accessible parts	See Energy source identification and classification table.	P
9.3.2	Test method and compliance		P
9.4	Safeguards against thermal energy sources		P
9.5	Requirements for safeguards		P
9.5.1	Equipment safeguard	Considered for DC fan side, Enclosure used as safeguard.	P
9.5.2	Instructional safeguard		N/A
9.6	Requirements for wireless power transmitters		N/A
9.6.1	General		N/A
9.6.2	Specification of the foreign objects		N/A
9.6.3	Test method and compliance		N/A

10	RADIATION		P
10.2	Radiation energy source classification		P
10.2.1	General classification	RS1	P
	Lasers	N/A	—
	Lamps and lamp systems	RS1: LED only used for indicator	—
	Image projectors	N/A	—
	X-Ray	N/A	—
	Personal music player	N/A	—
10.3	Safeguards against laser radiation		N/A
	The standard(s) equipment containing laser(s) comply		N/A
10.4	Safeguards against optical radiation from lamps and lamp systems (including LED types)		N/A
10.4.1	General requirements	The LED only used for indicating, which is considered as low power & inherently exempt group according to IEC 62471	N/A
	Instructional safeguard provided for accessible radiation level needs to exceed		N/A
	Risk group marking and location		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Information for safe operation and installation		N/A
10.4.2	Requirements for enclosures		N/A
	UV radiation exposure		N/A
10.4.3	Instructional safeguard		N/A
10.5	Safeguards against X-radiation		N/A
10.5.1	Requirements	No X-radiation.	N/A
	Instructional safeguard for skilled persons		—
10.5.3	Maximum radiation (pA/kg).....		—
10.6	Safeguards against acoustic energy sources		N/A
10.6.1	General	No acoustics energy.	N/A
10.6.2	Classification		N/A
	Acoustic output $L_{Aeq,T}$, dB(A).....		N/A
	Unweighted RMS output voltage (mV).....		N/A
	Digital output signal (dBFS)		N/A
10.6.3	Requirements for dose-based systems		N/A
10.6.3.1	General requirements		N/A
10.6.3.2	Dose-based warning and automatic decrease		N/A
10.6.3.3	Exposure-based warning and requirements		N/A
	30 s integrated exposure level (MEL30)		N/A
	Warning for MEL ≥ 100 dB(A)		N/A
10.6.4	Measurement methods		N/A
10.6.5	Protection of persons		N/A
	Instructional safeguards		N/A
10.6.6	Requirements for listening devices (headphones, earphones, etc.)		N/A
10.6.6.1	Corded listening devices with analogue input		N/A
	Listening device input voltage (mV)		N/A
10.6.6.2	Corded listening devices with digital input		N/A
	Max. acoustic output $L_{Aeq,T}$, dB(A)		N/A
10.6.6.3	Cordless listening devices		N/A
	Max. acoustic output $L_{Aeq,T}$, dB(A)		N/A

B	NORMAL OPERATING CONDITION TESTS, ABNORMAL OPERATING CONDITION TESTS AND SINGLE FAULT CONDITION TESTS	P
B.1	General	P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
B.1.5	Temperature measurement conditions	(See appended table 5.4.1.4, 9.3, B.1.5, B.2.6)	P
B.2	Normal operating conditions		P
B.2.1	General requirements	(See Test Item Particulars and appended test tables)	P
	Audio Amplifiers and equipment with audio amplifiers	Not such equipment.	N/A
B.2.3	Supply voltage and tolerances	100-130V~ or 200-250V~, Rated voltage +10%/ -10%	P
B.2.5	Input test	(See appended table B.2.5)	P
B.3	Simulated abnormal operating conditions		P
B.3.1	General		P
B.3.2	Covering of ventilation openings	(See appended table B.3, B.4)	P
	Instructional safeguard		N/A
B.3.3	DC mains polarity test	AC mains	N/A
B.3.4	Setting of voltage selector	No voltage selector	N/A
B.3.5	Maximum load at output terminals	(See appended table B.3, B.4)	P
B.3.6	Reverse battery polarity	No battery within the EUT	N/A
B.3.7	Audio amplifier abnormal operating conditions	Not such equipment.	N/A
B.3.8	Safeguards functional during and after abnormal operating conditions.....	All safeguards remained effective.	P
B.4	Simulated single fault conditions		P
B.4.1	General		P
B.4.2	Temperature controlling device		N/A
B.4.3	Blocked motor test	(See appended table B.3, B.4)	P
B.4.4	Functional insulation	(See appended table B.3, B.4)	P
B.4.4.1	Short circuit of clearances for functional insulation	(See appended table B.3, B.4)	P
B.4.4.2	Short circuit of creepage distances for functional insulation	(See appended table B.3, B.4)	P
B.4.4.3	Short circuit of functional insulation on coated printed boards		N/A
B.4.5	Short-circuit and interruption of electrodes in tubes and semiconductors	(See appended table B.3, B.4)	P
B.4.6	Short circuit or disconnection of passive components	(See appended table B.3, B.4)	P
B.4.7	Continuous operation of components		N/A
B.4.8	Compliance during and after single fault conditions	(See appended table B.3, B.4)	P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
B.4.9	Battery charging and discharging under single fault conditions	No batteries.	N/A
C	UV RADIATION		N/A
C.1	Protection of materials in equipment from UV radiation		N/A
C.1.2	Requirements	No UV radiation within the EUT.	N/A
C.1.3	Test method		N/A
C.2	UV light conditioning test		N/A
C.2.1	Test apparatus		N/A
C.2.2	Mounting of test samples		N/A
C.2.3	Carbon-arc light-exposure test		N/A
C.2.4	Xenon-arc light-exposure test		N/A
D	TEST GENERATORS		P
D.1	Impulse test generators		N/A
D.2	Antenna interface test generator		P
D.3	Electronic pulse generator		N/A
E	TEST CONDITIONS FOR EQUIPMENT CONTAINING AUDIO AMPLIFIERS		N/A
E.1	Electrical energy source classification for audio signals		N/A
	Maximum non-clipped output power (W)		—
	Rated load impedance (Ω)		—
	Open-circuit output voltage (V)		—
	Instructional safeguard		—
E.2	Audio amplifier normal operating conditions		N/A
	Audio signal source type		—
	Audio output power (W)		—
	Audio output voltage (V)		—
	Rated load impedance (Ω)		—
	Requirements for temperature measurement		N/A
E.3	Audio amplifier abnormal operating conditions		N/A
F	EQUIPMENT MARKINGS, INSTRUCTIONS, AND INSTRUCTIONAL SAFEGUARDS		P
F.1	General		P
	Language	Reviewed only English markings/instructions. May be provided in other languages upon request from the manufacturer.	—
F.2	Letter symbols and graphical symbols		P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
F.2.1	Letter symbols according to IEC60027-1		P
F.2.2	Graphic symbols according to IEC, ISO or manufacturer specific		P
F.3	Equipment markings		P
F.3.1	Equipment marking locations		P
F.3.2	Equipment identification markings		P
F.3.2.1	Manufacturer identification	See copy of marking plate	P
F.3.2.2	Model identification	See page 2 for details	P
F.3.3	Equipment rating markings		P
F.3.3.1	Equipment with direct connection to mains		P
F.3.3.2	Equipment without direct connection to mains		N/A
F.3.3.3	Nature of the supply voltage	See copy of marking plate	P
F.3.3.4	Rated voltage.....	See copy of marking plate	P
F.3.3.5	Rated frequency	See copy of marking plate	P
F.3.3.6	Rated current or rated power.....	See copy of marking plate	P
F.3.3.7	Equipment with multiple supply connections	No multiple supply connection	N/A
F.3.4	Voltage setting device	No such devices on the equipment.	N/A
F.3.5	Terminals and operating devices	See below.	P
F.3.5.1	Mains appliance outlet and socket-outlet markings	No such devices on the equipment.	NA
F.3.5.2	Switch position identification marking.....		N/A
F.3.5.3	Replacement fuse identification and rating markings	Fuses are not intended to be replaceable.	P
	Instructional safeguards for neutral fuse		N/A
F.3.5.4	Replacement battery identification marking.....	No batteries.	N/A
F.3.5.5	Neutral conductor terminal		N/A
F.3.5.6	Terminal marking location		N/A
F.3.6	Equipment markings related to equipment classification		P
F.3.6.1	Class I equipment	Class I	P
F.3.6.1.1	Protective earthing conductor terminal		N/A
F.3.6.1.2	Protective bonding conductor terminals		N/A
F.3.6.2	Equipment class marking.....		N/A
F.3.6.3	Functional earthing terminal marking		N/A
F.3.7	Equipment IP rating marking	IPX0	N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
F.3.8	External power supply output marking	EUT is for building-in.	N/A
F.3.9	Durability, legibility and permanence of marking		P
F.3.10	Test for permanence of markings	The test is conducted by rubbing the marking by hand without appreciable force for 15s with a piece of cloth soaked with water and on a different sample for 15s with a piece of cloth soaked with the petroleum spirit. After each test, the marking remains legible, no curling and not be removable by hand.	P
F.4	Instructions		P
	a) Information prior to installation and initial use		P
	b) Equipment for use in locations where children not likely to be present		P
	c) Instructions for installation and interconnection		P
	d) Equipment intended for use only in restricted access area		N/A
	e) Equipment intended to be fastened in place		N/A
	f) Instructions for audio equipment terminals		N/A
	g) Protective earthing used as a safeguard	EUT is for building-in.	P
	h) Protective conductor current exceeding ES2 limits		N/A
	i) Graphic symbols used on equipment		P
	j) Permanently connected equipment not provided with all-pole mains switch		N/A
	k) Replaceable components or modules providing safeguard function		N/A
	l) Equipment containing insulating liquid		N/A
	m) Installation instructions for outdoor equipment		N/A
F.5	Instructional safeguards		P
G	COMPONENTS		P
G.1	Switches		N/A
G.1.1	General	No switches.	N/A
G.1.2	Ratings, endurance, spacing, maximum load		N/A
G.1.3	Test method and compliance		N/A
G.2	Relays		P
G.2.1	Requirements	Approved relay is used.	P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
G.2.2	Overload test	Approved relay is used.	P
G.2.3	Relay controlling connectors supplying power to other equipment		N/A
G.2.4	Test method and compliance		N/A
G.3	Protective devices		P
G.3.1	Thermal cut-offs	No thermal cut-offs.	N/A
	Thermal cut-outs separately approved according to IEC 60730 with conditions indicated in a) & b)		N/A
	Thermal cut-outs tested as part of the equipment as indicated in c)		N/A
G.3.1.2	Test method and compliance		N/A
G.3.2	Thermal links	No thermal-links.	N/A
G.3.2.1	a) Thermal links tested separately according to IEC 60691 with specifics		N/A
	b) Thermal links tested as part of the equipment		N/A
G.3.2.2	Test method and compliance		N/A
G.3.3	PTC thermistors	No PTCs	N/A
G.3.4	Overcurrent protection devices	Approved fuse is used. (See appended table 4.1.2)	P
G.3.5	Safeguards components not mentioned in G.3.1 to G.3.4		N/A
G.3.5.1	Non-resettable devices suitably rated and marking provided		N/A
G.3.5.2	Single faults conditions		N/A
G.4	Connectors		N/A
G.4.1	Spacings	Built-in component, considered in end system	N/A
G.4.2	Mains connector configuration.....		N/A
G.4.3	Plug is shaped that insertion into mains socket-outlets or appliance coupler is unlikely		N/A
G.5	Wound components		P
G.5.1	Wire insulation in wound components	Approved TIW used for transformers TD700.	P
G.5.1.2	Protection against mechanical stress	Physical separation is provided by insulation tube for T100 and T400.	P
G.5.2	Endurance test		N/A
G.5.2.1	General test requirements		N/A
G.5.2.2	Heat run test		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Test time (days per cycle)..... :		—
	Test temperature (°C)..... :		—
G.5.2.3	Wound components supplied from the mains		N/A
G.5.2.4	No insulation breakdown		N/A
G.5.3	Transformers		P
G.5.3.1	Compliance method..... :	The transformer meets the requirements given in G.5.3.2 and G.5.3.3.	P
	Position..... :	See table 4.1.2	P
	Method of protection..... :	Over current protection by circuit design.	P
G.5.3.2	Insulation		P
	Protection from displacement of windings..... :	Windings on a single spool with a partition wall, where either the spool and partition wall are pressed or moulded in one piece. Insulation tape is provided between windings extending beyond the end turns of each layer.	—
G.5.3.3	Transformer overload tests	(See appended table B.3, B.4)	P
G.5.3.3.1	Test conditions		P
G.5.3.3.2	Winding temperatures		P
G.5.3.3.3	Winding temperatures - alternative test method		N/A
G.5.3.4	Transformers using FIW	No FIW used	N/A
G.5.3.4.1	General		N/A
	FIW wire nominal diameter..... :		—
G.5.3.4.2	Transformers with basic insulation only		N/A
G.5.3.4.3	Transformers with double insulation or reinforced insulation..... :		N/A
G.5.3.4.4	Transformers with FIW wound on metal or ferrite core		N/A
G.5.3.4.5	Thermal cycling test and compliance		N/A
G.5.3.4.6	Partial discharge test		N/A
G.5.3.4.7	Routine test		N/A
G.5.4	Motors		P
G.5.4.1	General requirements	Approved DC fan used	P
G.5.4.2	Motor overload test conditions		N/A
G.5.4.3	Running overload test		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
G.5.4.4.2	Locked-rotor overload test		N/A
	Test duration (days)		—
G.5.4.5	Running overload test for DC motors		N/A
G.5.4.5.2	Tested in the unit		N/A
G.5.4.5.3	Alternative method		N/A
G.5.4.6	Locked-rotor overload test for DC motors	See appended table B.3, B.4	P
G.5.4.6.2	Tested in the unit	See appended table B.3, B.4	P
	Maximum Temperature	See appended table B.3, B.4	P
G.5.4.6.3	Alternative method		N/A
G.5.4.7	Motors with capacitors		N/A
G.5.4.8	Three-phase motors		N/A
G.5.4.9	Series motors		N/A
	Operating voltage		—
G.6	Wire Insulation		P
G.6.1	General	Approved wire used. See Appended table 4.1.2.	P
G.6.2	Enamelled winding wire insulation	Solvent-based enamel is not considered to provide supplementary insulation or reinforced insulation.	N/A
G.7	Mains supply cords		N/A
G.7.1	General requirements	No such cords provided.	N/A
	Type	See above	—
G.7.2	Cross sectional area (mm ² or AWG)	See above	
G.7.3	Cord anchorages and strain relief for non-detachable power supply cords		N/A
G.7.3.2	Cord strain relief		N/A
G.7.3.2.1	Requirements		N/A
	Strain relief test force (N).....		N/A
G.7.3.2.2	Strain relief mechanism failure		N/A
G.7.3.2.3	Cord sheath or jacket position, distance (mm)		N/A
G.7.3.2.4	Strain relief and cord anchorage material		N/A
G.7.4	Cord Entry		N/A
G.7.5	Non-detachable cord bend protection	Not hand-held equipment	N/A
G.7.5.1	Requirements		N/A
G.7.5.2	Test method and compliance		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Overall diameter or minor overall dimension, D (mm) :		—
	Radius of curvature after test (mm) :		—
G.7.6	Supply wiring space		N/A
G.7.6.1	General requirements		N/A
G.7.6.2	Stranded wire		N/A
G.7.6.2.1	Requirements		N/A
G.7.6.2.2	Test with 8 mm strand		N/A
G.8	Varistors		P
G.8.1	General requirements	Approved varistor used	P
G.8.2	Safeguards against fire	Metal enclosure used	P
G.8.2.1	General		P
G.8.2.2	Varistor overload test		N/A
G.8.2.3	Temporary overvoltage test		N/A
G.9	Integrated circuit (IC) current limiters		N/A
G.9.1	Requirements		N/A
	IC limiter output current (max. 5A)..... :		—
	Manufacturers' defined drift :		—
G.9.2	Test Program		N/A
G.9.3	Compliance		N/A
G.10	Resistors		N/A
G.10.1	General		N/A
G.10.2	Conditioning		N/A
G.10.3	Resistor test		N/A
G.10.4	Voltage surge test		N/A
G.10.5	Impulse test		N/A
G.10.6	Overload test		N/A
G.11	Capacitors and RC units		P
G.11.1	General requirements	Approved X and Y capacitors are used. (See appended table 4.1.2)	P
G.11.2	Conditioning of capacitors and RC units		P
G.11.3	Rules for selecting capacitors		P
G.12	Optocouplers		P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Optocouplers comply with IEC 60747-5-5 with specifics	Approved optocoupler is used. (See appended table 4.1.2)	P
	Type test voltage $V_{ini,a}$:	Considered.	—
	Routine test voltage, $V_{ini,b}$:	Considered.	—
G.13	Printed boards		P
G.13.1	General requirements	Approved Printed board used	P
G.13.2	Uncoated printed boards		P
G.13.3	Coated printed boards		N/A
G.13.4	Insulation between conductors on the same inner surface	Approved Multi-layer board used, complied with 5.4.4.5 and 5.4.7	P
G.13.5	Insulation between conductors on different surfaces		P
	Distance through insulation :	Min. 0.4mm or complied with table G.14	P
	Number of insulation layers (pcs) :	4 layers	—
G.13.6	Tests on coated printed boards		N/A
G.13.6.1	Sample preparation and preliminary inspection		N/A
G.13.6.2	Test method and compliance		N/A
G.14	Coating on components terminals		N/A
G.14.1	Requirements :		N/A
G.15	Pressurized liquid filled components		N/A
G.15.1	Requirements		N/A
G.15.2	Test methods and compliance		N/A
G.15.2.1	Hydrostatic pressure test		N/A
G.15.2.2	Creep resistance test		N/A
G.15.2.3	Tubing and fittings compatibility test		N/A
G.15.2.4	Vibration test		N/A
G.15.2.5	Thermal cycling test		N/A
G.15.2.6	Force test		N/A
G.15.3	Compliance		N/A
G.16	IC including capacitor discharge function (ICX)		N/A
G.16.1	Condition for fault tested is not required		N/A
	ICX with associated circuitry tested in equipment		N/A
	ICX tested separately		N/A
G.16.2	Tests		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Smallest capacitance and smallest resistance specified by ICX manufacturer for impulse test..... :		—
	Mains voltage that impulses to be superimposed on :		—
	Largest capacitance and smallest resistance for ICX tested by itself for 10000 cycles test..... :		—
G.16.3	Capacitor discharge test..... :		N/A
H	CRITERIA FOR TELEPHONE RINGING SIGNALS		N/A
H.1	General		N/A
H.2	Method A		N/A
H.3	Method B		N/A
H.3.1	Ringing signal		N/A
H.3.1.1	Frequency (Hz) :		—
H.3.1.2	Voltage (V) :		—
H.3.1.3	Cadence; time (s) and voltage (V) :		—
H.3.1.4	Single fault current (mA): :		—
H.3.2	Tripping device and monitoring voltage		N/A
H.3.2.1	Conditions for use of a tripping device or a monitoring voltage		N/A
H.3.2.2	Tripping device		N/A
H.3.2.3	Monitoring voltage (V)..... :		N/A
J	INSULATED WINDING WIRES FOR USE WITHOUT INTERLEAVED INSULATION		P
J.1	General		P
	Winding wire insulation :	Approved triple insulated winding wiring used as reinforced safeguard in the isolating transformer TD700	—
	Solid round winding wire, diameter (mm) :	See transformer spec for detail	P
	Solid square and rectangular (flatwise bending) winding wire, cross-sectional area (mm ²)..... :		N/A
J.2/J.3	Tests and Manufacturing	Approved triple insulated winding wiring used	—
K	SAFETY INTERLOCKS		N/A
K.1	General requirements		N/A
	Instructional safeguard :		N/A
K.2	Components of safety interlock safeguard mechanism		N/A
K.3	Inadvertent change of operating mode		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
K.4	Interlock safeguard override		N/A
K.5	Fail-safe		N/A
K.5.1	Under single fault condition		N/A
K.6	Mechanically operated safety interlocks		N/A
K.6.1	Endurance requirement		N/A
K.6.2	Test method and compliance..... :		N/A
K.7	Interlock circuit isolation		N/A
K.7.1	Separation distance for contact gaps & interlock circuit elements		N/A
	In circuit connected to mains, separation distance for contact gaps (mm) :		N/A
	In circuit isolated from mains, separation distance for contact gaps (mm) :		N/A
	Electric strength test before and after the test of K.7.2..... :		N/A
K.7.2	Overload test, Current (A)..... :		N/A
K.7.3	Endurance test		N/A
K.7.4	Electric strength test		N/A
L	DISCONNECT DEVICES		N/A
L.1	General requirements	Considered in end system	N/A
L.2	Permanently connected equipment		N/A
L.3	Parts that remain energized		N/A
L.4	Single-phase equipment		N/A
L.5	Three-phase equipment		N/A
L.6	Switches as disconnect devices		N/A
L.7	Plugs as disconnect devices		N/A
L.8	Multiple power sources		N/A
	Instructional safeguard :		N/A
M	EQUIPMENT CONTAINING BATTERIES AND THEIR PROTECTION CIRCUITS		N/A
M.1	General requirements		N/A
M.2	Safety of batteries and their cells		N/A
M.2.1	Batteries and their cells comply with relevant IEC standards :		N/A
M.3	Protection circuits for batteries provided within the equipment		N/A
M.3.1	Requirements		N/A
M.3.2	Test method		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Overcharging of a rechargeable battery		N/A
	Excessive discharging		N/A
	Unintentional charging of a non-rechargeable battery		N/A
	Reverse charging of a rechargeable battery		N/A
M.3.3	Compliance		N/A
M.4	Additional safeguards for equipment containing a portable secondary lithium battery		N/A
M.4.1	General		N/A
M.4.2	Charging safeguards		N/A
M.4.2.1	Requirements		N/A
M.4.2.2	Compliance :		N/A
M.4.3	Fire enclosure :		N/A
M.4.4	Drop test of equipment containing a secondary lithium battery		N/A
M.4.4.2	Preparation and procedure for the drop test		N/A
M.4.4.3	Drop, Voltage on reference and dropped batteries (V); voltage difference during 24 h period (%): :		N/A
M.4.4.4	Check of the charge/discharge function		N/A
M.4.4.5	Charge / discharge cycle test		N/A
M.4.4.6	Compliance		N/A
M.5	Risk of burn due to short-circuit during carrying		N/A
M.5.1	Requirement		N/A
M.5.2	Test method and compliance		N/A
M.6	Safeguards against short-circuits		N/A
M.6.1	External and internal faults		N/A
M.6.2	Compliance		N/A
M.7	Risk of explosion from lead acid and NiCd batteries		N/A
M.7.1	Ventilation preventing explosive gas concentration		N/A
	Calculated hydrogen generation rate :		N/A
M.7.2	Test method and compliance		N/A
	Minimum air flow rate, Q (m ³ /h) :		N/A
M.7.3	Ventilation tests		N/A
M.7.3.1	General		N/A
M.7.3.2	Ventilation test – alternative 1		N/A
	Hydrogen gas concentration (%) :		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
M.7.3.3	Ventilation test – alternative 2		N/A
	Obtained hydrogen generation rate		N/A
M.7.3.4	Ventilation test – alternative 3		N/A
	Hydrogen gas concentration (%)		N/A
M.7.4	Marking		N/A
M.8	Protection against internal ignition from external spark sources of batteries with aqueous electrolyte		N/A
M.8.1	General		N/A
M.8.2	Test method		N/A
M.8.2.1	General		N/A
M.8.2.2	Estimation of hypothetical volume V_Z (m ³ /s).....		—
M.8.2.3	Correction factors		—
M.8.2.4	Calculation of distance d (mm)		—
M.9	Preventing electrolyte spillage		N/A
M.9.1	Protection from electrolyte spillage		N/A
M.9.2	Tray for preventing electrolyte spillage		N/A
M.10	Instructions to prevent reasonably foreseeable misuse		N/A
	Instructional safeguard		N/A
N	ELECTROCHEMICAL POTENTIALS		P
	Material(s) used	Aluminium zinc alloy	—
O	MEASUREMENT OF CREEPAGE DISTANCES AND CLEARANCES		P
	Value of X (mm).....	Measurement is in accordance with applicable figures.	—
P	SAFEGUARDS AGAINST CONDUCTIVE OBJECTS		P
P.1	General	DC fan side was considered, other sides shall be considered in end system.	P
P.2	Safeguards against entry or consequences of entry of a foreign object		P
P.2.1	General		P
P.2.2	Safeguards against entry of a foreign object		N/A
	Location and Dimensions (mm)		—
P.2.3	Safeguards against the consequences of entry of a foreign object		P
P.2.3.1	Safeguard requirements		P

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The ES3 and PS3 keep-out volume in Figure P.3 not applicable to transportable equipment	For DC fan side: within the projected volume as depicted in Figure P.3, there are no bare conductive parts of ES3 or PS3 circuits.	P
	Transportable equipment with metalized plastic parts		N/A
P.2.3.2	Consequence of entry test.....		N/A
P.3	Safeguards against spillage of internal liquids		N/A
P.3.1	General	No internal liquids.	N/A
P.3.2	Determination of spillage consequences		N/A
P.3.3	Spillage safeguards		N/A
P.3.4	Compliance		N/A
P.4	Metallized coatings and adhesives securing parts		N/A
P.4.1	General		N/A
P.4.2	Tests		N/A
	Conditioning, T _C (°C).....		—
	Duration (weeks).....		—
Q	CIRCUITS INTENDED FOR INTERCONNECTION WITH BUILDING WIRING		N/A
Q.1	Limited power sources		N/A
Q.1.1	Requirements		N/A
	a) Inherently limited output		N/A
	b) Impedance limited output		N/A
	c) Regulating network limited output		N/A
	d) Overcurrent protective device limited output		N/A
	e) IC current limiter complying with G.9		N/A
Q.1.2	Test method and compliance.....		N/A
	Current rating of overcurrent protective device (A)		N/A
Q.2	Test for external circuits – paired conductor cable		N/A
	Maximum output current (A)		N/A
	Current limiting method.....		—
R	LIMITED SHORT CIRCUIT TEST		N/A
R.1	General	Consider in end system	N/A
R.2	Test setup		N/A
	Overcurrent protective device for test.....		—

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
R.3	Test method		N/A
	Cord/cable used for test..... :		—
R.4	Compliance		N/A
S	TESTS FOR RESISTANCE TO HEAT AND FIRE		N/A
S.1	Flammability test for fire enclosures and fire barrier materials of equipment where the steady state power does not exceed 4 000 W		N/A
	Samples, material :		—
	Wall thickness (mm) :		—
	Conditioning (°C) :		—
	Test flame according to IEC 60695-11-5 with conditions as set out		N/A
	- Material not consumed completely		N/A
	- Material extinguishes within 30s		N/A
	- No burning of layer or wrapping tissue		N/A
S.2	Flammability test for fire enclosure and fire barrier integrity		N/A
	Samples, material :		—
	Wall thickness (mm) :		—
	Conditioning (°C) :		—
S.3	Flammability test for the bottom of a fire enclosure		N/A
S.3.1	Mounting of samples		N/A
S.3.2	Test method and compliance		N/A
	Mounting of samples :		—
	Wall thickness (mm) :		—
S.4	Flammability classification of materials		N/A
S.5	Flammability test for fire enclosure materials of equipment with a steady state power exceeding 4 000 W		N/A
	Samples, material :		—
	Wall thickness (mm) :		—
	Conditioning (°C) :		—
T	MECHANICAL STRENGTH TESTS		P
T.1	General		P
T.2	Steady force test, 10 N :	(See appended table T.2, T.3, T.4, T.5)	P
T.3	Steady force test, 30 N :		N/A
T.4	Steady force test, 100 N :		N/A

IEC 62368-1			
Clause	Requirement + Test		Verdict
T.5	Steady force test, 250 N	(See appended table T.2, T.3, T.4, T.5), also consider in end system	P
T.6	Enclosure impact test	(See appended table T.6, T.9), also consider in end system	P
	Fall test		P
	Swing test		N/A
T.7	Drop test	Built-in component	N/A
T.8	Stress relief test	(See appended table T.8)	P
T.9	Glass Impact Test	Not applicable.	N/A
T.10	Glass fragmentation test		N/A
	Number of particles counted	No glass.	N/A
T.11	Test for telescoping or rod antennas		N/A
	Torque value (Nm)		N/A
U	MECHANICAL STRENGTH OF CATHODE RAY TUBES (CRT) AND PROTECTION AGAINST THE EFFECTS OF IMPLOSION		N/A
U.1	General		N/A
	Instructional safeguard :	No CRTs.	N/A
U.2	Test method and compliance for non-intrinsically protected CRTs		N/A
U.3	Protective screen		N/A
V	DETERMINATION OF ACCESSIBLE PARTS		P
V.1	Accessible parts of equipment		P
V.1.1	General	DC fan side was considered, no access with test probes to any hazardous parts. Other sides shall be considered in end system.	P
V.1.2	Surfaces and openings tested with jointed test probes		P
V.1.3	Openings tested with straight unjointed test probes		P
V.1.4	Plugs, jacks, connectors tested with blunt probe		N/A
V.1.5	Slot openings tested with wedge probe		N/A
V.1.6	Terminals tested with rigid test wire		N/A
V.2	Accessible part criterion		P
X	ALTERNATIVE METHOD FOR DETERMINING CLEARANCES FOR INSULATION IN CIRCUITS CONNECTED TO AN AC MAINS NOT EXCEEDING 420 V PEAK (300 V RMS)		N/A
	Clearance		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict
Y	CONSTRUCTION REQUIREMENTS FOR OUTDOOR ENCLOSURES		N/A
Y.1	General		N/A
Y.2	Resistance to UV radiation		N/A
Y.3	Resistance to corrosion		N/A
Y.3	Resistance to corrosion		N/A
Y.3.1	Metallic parts of outdoor enclosures are resistant to effects of water-borne contaminants by..... :		N/A
Y.3.2	Test apparatus		N/A
Y.3.3	Water – saturated sulphur dioxide atmosphere		N/A
Y.3.4	Test procedure :		N/A
Y.3.5	Compliance		N/A
Y.4	Gaskets		N/A
Y.4.1	General		N/A
Y.4.2	Gasket tests		N/A
Y.4.3	Tensile strength and elongation tests		N/A
	Alternative test methods :		N/A
Y.4.4	Compression test		N/A
Y.4.5	Oil resistance		N/A
Y.4.6	Securing means		N/A
Y.5	Protection of equipment within an outdoor enclosure		N/A
Y.5.1	General		N/A
Y.5.2	Protection from moisture		N/A
	Relevant tests of IEC 60529 or Y.5.3 :		N/A
Y.5.3	Water spray test		N/A
Y.5.4	Protection from plants and vermin		N/A
Y.5.5	Protection from excessive dust		N/A
Y.5.5.1	General		N/A
Y.5.5.2	IP5X equipment		N/A
Y.5.5.3	IP6X equipment		N/A
Y.6	Mechanical strength of enclosures		N/A
Y.6.1	General		N/A
Y.6.2	Impact test :		N/A

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

5.2		TABLE: Classification of electrical energy sources					P
Supply Voltage	Location (e.g. circuit designation)	Test conditions	Parameters				ES Class
			U (V)	I (mA)	Type ¹⁾	Additional Info ²⁾	
Model: MR484000SG1							
275Vac/60Hz	Sec. output + to -	Normal	53.58Vd _c	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Abnormal Overload	53.58Vd _c	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – Q404 pin 2-3 SC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – Q408 pin 2-3 SC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – D501 SC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – D502 SC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – D505 SC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U500 pin 1-2 SC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U500 pin 3-4 SC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U500 pin 1 OC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U500 pin 3 OC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U700 pin 1-2 SC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U700 pin 3-4 SC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U700 pin 1 OC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U700 pin 3 OC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U702 pin 1-2 SC	0Vdc	--	SS	--	ES1

IEC 62368-1							
Clause	Requirement + Test			Result - Remark			Verdict
275Vac/60Hz	Sec. output + to -	Single fault – U702 pin 3-4 SC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U702 pin 1 OC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U702 pin 3 OC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U703 pin 1-2 SC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U703 pin 3-4 SC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U703 pin 1 OC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – U703 pin 3 OC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – R150 SC	53.58Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – RC100 SC	53.58Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – RC500 SC	0Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output + to -	Single fault – earth OC	53.58Vdc	--	SS	--	ES1
275Vac/60Hz	Sec. output +/- to earth	Normal	--	0.424mApeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Abnormal Overload	--	0.424mApeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – Q404 pin 2-3 SC	--	0.044mApeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – Q408 pin 2-3 SC	--	0.044mApeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – D501 SC	--	0.044mApeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – D502 SC	--	0.044mApeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – D505 SC	--	0.044mApeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U500 pin 1-2 SC	--	0.424mApeak	SS	60Hz	ES1

IEC 62368-1							
Clause	Requirement + Test			Result - Remark			Verdict
275Vac/60Hz	Sec. output +/- to earth	Single fault – U500 pin 3-4 SC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U500 pin 1 OC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U500 pin 3 OC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U700 pin 1-2 SC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U700 pin 3-4 SC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U700 pin 1 OC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U700 pin 3 OC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U702 pin 1-2 SC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U702 pin 3-4 SC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U702 pin 1 OC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U702 pin 3 OC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U703 pin 1-2 SC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U703 pin 3-4 SC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U703 pin 1 OC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – U703 pin 3 OC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – R150 SC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – RC100 SC	--	0.424m Apeak	SS	60Hz	ES1
275Vac/60Hz	Sec. output +/- to earth	Single fault – RC500 SC	--	0.424m Apeak	SS	60Hz	ES1

IEC 62368-1							
Clause	Requirement + Test			Result - Remark			Verdict
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Normal	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Abnormal Overload	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – Q404 pin 2-3 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – Q408 pin 2-3 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – D501 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – D502 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – D505 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U500 pin 1-2 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U500 pin 3-4 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U500 pin 1 OC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U500 pin 3 OC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U700 pin 1-2 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U700 pin 3-4 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U700 pin 1 OC	--	0.026m A peak	SS	60Hz	ES1

IEC 62368-1							
Clause	Requirement + Test			Result - Remark			Verdict

275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U700 pin 3 OC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U702 pin 1-2 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U702 pin 3-4 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U702 pin 1 OC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U702 pin 3 OC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U703 pin 1-2 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U703 pin 3-4 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U703 pin 1 OC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – U703 pin 3 OC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – R150 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – RC100 SC	--	0.026m A peak	SS	60Hz	ES1
275Vac/60Hz	Plastic enclosure to earth (with a metal foil)	Single fault – RC500 SC	--	0.026m A peak	SS	60Hz	ES1

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

275Vac/60Hz	L to N (CX100, CX103, CX101, CX102)	Normal	389	--	CP	CX100 Max. 2.2 μ F $\pm$ 10%, CX101, CX103 Each Max. 2.0 μ F $\pm$ 10%, CX102 Max. 0.68 μ F $\pm$ 10%	ES3
		Abnormal	--	--	--	--	
		Single fault – SC/OC	--	--	--	--	
275Vac/60Hz	L /N to earth (CY100, CY104, CY106, CY107, CY102, CY105, CY101, CY103)	Normal	389	--	CP	CY100, CY104: 2.2nF $\pm$ 10%, CY106, CY107, CY102, CY105, : 4.7nF $\pm$ 10%, CY101, CY103:10nF $\pm$ 10%	ES2
		Abnormal	--	--	--	--	
		Single fault – SC/OC	--	--	--	--	

Supplementary information:

- 1) Type: Steady state (SS), Capacitance (CP), Single pulse (SP), Repetitive pulses (RP), etc.
- 2) Additional Info: Frequency, Pulse duration, Pulse off time, Capacitance value, etc.
- 3) Metal enclosure connected to earth.

5.4.1.8	TABLE: Working voltage measurement				P
Location		RMS voltage (V)	Peak voltage (V)	Frequency (KHz)	Comments
Model MR484000SG1, Input voltage: 250Vac, 60Hz					
T400 pin 5,10-1		178	384	Max.78.2	--
T400 pin 5,10-2,3		161	356		--
T400 pin 5,10-4		200	420		--
T400 pin 6,9-1		163	332		--
T400 pin 6,9-2,3		175	400		--
T400 pin 6,9-4		238	472		--
T400 pin 7,8-1		164	376		--
T400 pin 7,8-2,3		201	444		--

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

T400 pin 7,8-4	280	516		Max Vrms and Vpk
T100 pin 7-1	172	376	Max.95.8	--
T100 pin 7-2	172	376		--
T100 pin 7-3	174	388		--
T100 pin 7-4	304	560		--
T100 pin 7-6	314	528		--
T100 pin 8-1	172	436		--
T100 pin 8-2	172	372		--
T100 pin 8-3	173	376		--
T100 pin 8-4	307	560		--
T100 pin 8-6	313	488		--
T100 pin 9-1	174	456		--
T100 pin 9-2	172	396		--
T100 pin 9-3	172	372		--
T100 pin 9-4	319	588		Max. Vpk
T100 pin 9-6	313	456		--
T100 pin 10-1	108	256		--
T100 pin 10-2	108	224		--
T100 pin 10-3	108	228		--
T100 pin 10-4	173	572		--
T100 pin 10-6	107	224		--
T100 pin 11-1	109	292		--
T100 pin 11-2	108	240		--
T100 pin 11-3	107	220		--
T100 pin 11-4	178	584		--
T100 pin 11-6	107	216		--
TD700 pin 2-5	296	468	Max.85.6	--
TD700 pin 4-5	301	468		--
TD700 pin 2-6	301	472		--
TD700 pin 4-6	312	492		--
TD700 pin 2-7	296	472		--
TD700 pin 4-7	310	476		--
TD700 pin 2-8	296	476		--

IEC 62368-1				
Clause	Requirement + Test		Result - Remark	Verdict
TD700 pin 4-8	296	488	0.06	--
U700 pin 1-3	220	372		--
U700 pin 1-4	220	372		--
U700 pin 2-3	220	372		--
U700 pin 2-4	220	372		--
U702 pin 1-3	221	368		--
U702 pin 1-4	221	368		--
U702 pin 2-3	221	368		--
U702 pin 2-4	221	368		--
U703 pin 1-3	221	368		--
U703 pin 1-4	220	364		--
U703 pin 2-3	221	368		--
U703 pin 2-4	220	364		--
CY500	172	372		--
CY106	108	256		--
CY107	108	180		--
CY100	250	356		--
CY104	0	0	--	--
CY102	250	356	0.06	--
CY105	0	0	--	--
Primary trace under RV103 to secondary trace under L400	221	364	--	--
Primary F101 to secondary C417	239	370	--	--
CY107 primary PIN to T100 secondary PIN	221	372	--	--
R75 to primary heatsink HS102	68	124	--	--
Input voltage: 100Vac, 60Hz				
T400 pin 7,8-4	268	456	Max 78.2	--
T100 pin 9-4	340	576	Max 95.8	Max Vrms
TD700 pin 4-6	315	492	Max 85.6	Max Vrms and Vpk
Supplementary information:				

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

5.4.1.10.2	TABLE: Vicat softening temperature of thermoplastics			N/A
Method.....:		ISO 306 / B50		—
Object/ Part No./Material	Manufacturer/trademark	Thickness (mm)	T softening (°C)	
--	--	--	--	
--	--	--	--	
Supplementary information:				

5.4.1.10.3	TABLE: Ball pressure test of thermoplastics			N/A
Allowed impression diameter (mm)		≤2mm		—
Object/Part No./Material	Manufacturer/trademark	Thickness (mm)	Test temperature (°C)	Impression diameter (mm)
Supplementary information:				
Note: Phenolic material, which acceptable without test.				

5.4.2, 5.4.3	TABLE: Minimum Clearances/Creepage distance							P
Clearance (cl) and creepage distance (cr) at/of/between:	U _p (V)	U _{rms} (V)	Freq ¹⁾ (Hz)	Required cl (mm)	cl (mm)	E.S. ²⁾ (V)	Required cr (mm)	cr (mm)
Line to Neutral before fuse F100 (FI)	420	250	60	2.3 (1.5*1.48)	2.9	--	2.5	2.9
Different polarities of fuse F100 (FI)	420	250	60	2.3 (1.5*1.48)	2.8	--	2.5	2.8
Line and Neutral to PE (BI)	420	250	60	2.3 (1.5*1.48)	3.1	--	2.5	3.1
Two terminals of (CY104,CY100) (BI)	420	250	60	2.3 (1.5*1.48)	3.6	--	2.5	3.6
Two terminals of (CY101,CY103) (BI)	420	250	60	2.3 (1.5*1.48)	3.0	--	2.5	3.0
Primary trace under RV103 to secondary trace under L400(RI)	420	250	60	4.5 (3*1.48)	5.1	--	5.0	5.1
Primary trace under RT100 to CN100(BI)	420	250	60	2.3 (1.5*1.48)	3.0	--	2.5	3.0

IEC 62368-1								
Clause	Requirement + Test			Result - Remark				Verdict
Primary trace under CX101 to CN100(BI)	420	250	60	2.3 (1.5*1.48)	3.0	--	2.5	3.0
T400 core to primary heatsink HS102 (BI)	516	280	78.2K	2.3 (1.5*1.48)	3.2	--	2.8	3.2
T400 core to secondary Heatsink HS400(FI)	516	280	78.2K	2.3 (1.5*1.48)	3.3	--	2.8	3.3
T400 core winding to secondary component EC406(FI)	516	280	78.2K	2.3 (1.5*1.48)	3.5	--	2.8	3.5
T400 core to primary component C426 (BI)	516	280	78.2K	2.3 (1.5*1.48)	3.5	--	2.8	3.5
T400 primary winding to earth(FG501)(BI)	516	280	78.2K	2.3 (1.5*1.48)	2.9	--	2.8	2.9
T400 primary winding to earth(FG508)(BI)	516	280	78.2K	2.3 (1.5*1.48)	3.5	--	2.8	3.5
T400 core to metal enclosure (BI)	516	280	78.2K	2.3 (1.5*1.48)	3.0	--	2.8	3.0
T100 core to primary component EC409 (BI)	588	340	95.8K	2.3 (1.5*1.48)	4.0	--	3.4	4.0
T100 secondary winding to primary component EC409 (RI)	588	340	95.8K	4.5 (3*1.48)	4.7	--	6.8	7.0
T100 core to CY500 secondary PIN(FI)	588	340	95.8K	2.3 (1.5*1.48)	3.0	--	3.4	4.0
T100 core to secondary component EC502 (FI)	588	340	95.8K	2.3 (1.5*1.48)	6.0	--	3.4	6.0
TD700 primary windings to secondary windings (RI)	492	315	85.6K	4.5 (3*1.48)	11.0	--	6.4	11.0
TD700 primary winding/pins to core (BI)	492	315	85.6K	2.3 (1.5*1.48)	7.0	--	3.2	7.0
TD700 secondary winding/pins to core (SI)	492	315	85.6K	2.3 (1.5*1.48)	7.0	--	3.2	7.0
TD700 core to primary X capacitor (BI)	492	315	85.6K	1.9 (1.27*1.48)	2.8	--	3.2	10.0

IEC 62368-1								
Clause	Requirement + Test			Result - Remark				Verdict
Two terminals of CY106 & CY107 (BI)	420	250	60	2.3 (1.5*1.48)	5.1	--	2.5	5.1
CY107 primary PIN to T100 secondary PIN(RI)	420	250	60	4.5 (3*1.48)	5.5	--	5.0	5.5
CY107 primary PIN to DC FAN lead wire (FI)	420	250	60	2.3 (1.5*1.48)	4.0	--	2.5	4.0
CY106 to 7 (BI)	420	250	60	2.3 (1.5*1.48)	5.1	--	2.5	5.1
Secondary component D700 to primary trace of U703(RI)	420	250	60	4.5 (3*1.48)	5.5	--	5.0	5.5
Secondary component U601 to primary Component RV100(RI)	420	250	60	4.5 (3*1.48)	7.0	--	5.0	7.0
Secondary component C707 to primary Component C701(RI)	420	250	60	4.5 (3*1.48)	6.5	--	5.0	6.5
PCB under Primary F101 to secondary C417(RI)	420	250	60	4.5 (3*1.48)	5.2	--	5.0	5.2
PCB under T100 (RI)	588	340	95.8K	4.5 (3*1.48)	7.8	--	6.8	7.8
T100 primary windings to secondary windings (RI)	588	340	95.8K	4.5 (3*1.48)	7.0	--	6.8	7.0
T100 primary winding/pins to core (BI)	588	340	95.8K	2.3 (1.5*1.48)	4.3	--	3.4	5.0
T100 secondary winding/pins to core (SI)	588	340	95.8K	2.3 (1.5*1.48)	4.3	--	3.4	5.0
PCB under T400 (RI)	516	280	78.2K	4.5 (3*1.48)	5.7	--	5.6	5.7
T400 primary winding/pins to core (BI)	516	280	78.2K	2.3 (1.5*1.48)	6.2	--	2.8	6.2
T400 secondary winding/pins to core (SI)	516	280	78.2K	2.3 (1.5*1.48)	6.2	--	2.8	5.9

IEC 62368-1								
Clause	Requirement + Test			Result - Remark				Verdict

T400 primary winding/pins to Secondary component C454 (BI)	516	280	78.2K	2.3 (1.5*1.48)	3.4	--	2.8	3.4
T400 primary secondary to primary C426 (RI)	516	280	78.2K	4.5 (3*1.48)	6.0	--	5.6	6.0
U706 to primary heatsink HS102(RI)	420	250	60	4.5 (3*1.48)	5.7	--	5.0	5.7
Primary PIN CN700 to FG508 (BI)	420	250	60	2.3 (1.5*1.48)	4.8	--	2.8	4.8
Internal PCB input to earth (same inner surface) (BI)	420	250	60	0.4*	1.0	--	0.4*	1.0
Internal PCB input to output (same inner surface) (RI)	420	250	60	0.4*	0.5	--	0.4*	0.5

Supplementary information:

1) Both frequencies lower than 30 kHz and higher than 30 kHz are present. Clearance distance limit from Table 14 which is higher than Table 10/11 based on the temporary overvoltage (2000Vpeak). Creepage distance limit from Table 17 which is higher than Table 18.

2) Complete Electric Strength voltage (E.S. (V) when 5.4.2.4 applied)

- Provide Material Group: IIIb
- FI=Functional insulation; BI=Basic insulation; RI=Reinforced insulation; SI=Supplementary insulation
- The T100/ T400 core of transformer is considered as middle part, magnet wire used in primary windings and secondary windings.
- The TD700 core of transformer is considered as middle part, and Triple insulated wire used in primary windings and secondary windings.
- The equipment is operated up to 16,405 feet (5,000 meters) above sea level as declared by manufacturer. Clearances have been evaluated according to table 16 with a multiplication factor of 1.48 throughout this report.
- All sources of PCB material were tested according to 5.4.4.5 c).

5.4.4.2	TABLE: Minimum distance through insulation				P
Distance through insulation (DTI) at/of	Peak voltage (V)	Insulation	Required DTI (mm)	Measured DTI (mm)	
Bobbin of transformer T100, TD700	588	Reinforced	Min. 0.4	See Table 4.1.2	
Opto-coupler(U700, U702, U703) (Reinforced insulation) (for all sources)	420	Reinforced	Min. 0.4	See Table 4.1.2	
Internal PCB input to output (same inner surface) (for all sources)	588	Reinforced	Min. 0.4	See Table 4.1.2	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

Insulation tape for T100, T400 (Reinforced insulation) (for all sources)	588	Reinforced	Min. two layers	Min. two layers
Heat shrinkable tube (used on DC Fan wire, R100, DX100, RV100, RV103)	588	Supplementary	Min. 0.4	See Table 4.1.2
Mylar sheet(primary circuit to metal enclosure)	588	Basic	--	See Table 4.1.2
Insulation tape used on EC901	588	Reinforced	Min. two layers	Min. two layers
Thermal sheet for on T400	588	Basic	--	See Table 4.1.2
Tubing of T100	588	Basic	--	See Table 4.1.2
Front plastic enclosure Inside to outside	420	Reinforced	Min. 0.4	See Table 4.1.2
Mylar sheet used between secondary circuit and fuse F100,F101)	420	Reinforced	Min. 0.4	See Table 4.1.2
Supplementary information:				
Note 1: Electric strength tests are also conducted after sub-clause 5.4.8 for all sources, details in table of 5.4.9.				
All sources of each material have been considered, see table 4.1.2 for details.				

5.4.4.9	TABLE: Solid insulation at frequencies >30 kHz					P
Insulation material	E_P	Frequency (kHz)	K_R	Thickness d (mm)	Insulation	V_{PW} (Vpk)
Bobbin of transformer T100, TD700	--	95.8	0.71	min. 0.45	Reinforced	588
Insulation tape of transformer T100, T400	--	95.8	0.46	min. two layers	Reinforced	588
Heat shrinkable tube (used on DC Fan wire, R100, DX100, RV100, RV103)	--	95.8	0.35	min. 0.40	Supplementary	588
Mylar sheet(primary circuit to metal enclosure)	--	95.8	0.35	min. 0.25	Reinforced	588
Insulation tape used on EC901	--	95.8	0.46	min. two layers	Reinforced	588
Thermal sheet for on T400	--	95.8	0.46	min. 0.23	Basic	588
Tubing of T100	--	95.8	0.35	--	Basic	588
Front plastic enclosure Inside to outside	--	0.06	0.35	min. 1.5	Reinforced	420
Mylar sheet used between secondary	--	0.06	0.35	min. 0.4	Reinforced	420

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

circuit and fuse F100,F101)						
Supplementary information:						
The mains frequency test voltage shall be as follows: for Basic insulation: $1,2 \times V_{PW} / K_R$; for reinforced insulation: $1,2 \times 2 \times V_{PW} / K_R$ Bobbin: 1988Vpk or dc Insulation tapes 3068Vpk or dc of transformer: Heat shrinkable(used on DC Fan, R100, DX100, RV100, RV103) :2016Vpk or dc Insulation sheets(primary circuit to metal enclosure): 2016Vpk or dc Insulation tapes used on EC901: 3068Vpk or dc Thermal sheet for on T400: 1534Vpk or dc Tubing of T100:1534Vpk or dc Front plastic enclosure Inside to outside: 2880Vpk or dc Mylar sheet used between secondary circuit and fuse F100,F101): 2880Vpk or dc Above test voltage of bobbin and insulation tapes is lower than table 5.4.9, the electric strength tests according to table 5.4.9. All sources of each material have been considered to conduct test, see table 4.1.2 for details.						

5.4.9	TABLE: Electric strength tests			P
Test voltage applied between:	Voltage shape (Surge, Impulse, AC, DC, etc.)	Test voltage (V)	Breakdown Yes/No	
Functional:				
Line to Neutral (with Fuse disconnect)	DC	2500	No	
Mains input(L/N) to Earth pin /metal enclosure(removed DX100)	DC	2500	No	
Basic/supplementary:				
Mylar sheet(primary circuit to metal enclosure)	DC	2500	No	
Thermal sheet for on T400	DC	2500	No	
TD700: primary to core	DC	2500	No	
TD700: secondary to core	DC	2500	No	
T100: primary windings to core	DC	2500	No	
T100: secondary windings to core	DC	2500	No	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

T400: primary windings to core	DC	2500	No
T400: secondary windings to core	DC	2500	No
Tubing of T100	DC	2500	No
Reinforced:			
Mains input(L/N) to secondary terminal (removed metal enclosure, DX100)	DC	4000	No
TD700: primary to secondary	DC	4000	No
T100: primary to secondary	DC	4000	No
T400: primary to secondary	DC	4000	No
One layer Insulation tape of all transformer	DC	4000	No
Mains input (L/N) to front plastic enclosure with metal foil	DC	4000	No
Front plastic enclosure Inside to outside	DC	4000	No
Mylar sheet used between secondary circuit and fuse F100,F101)	DC	4000	No
Insulation tape used on EC901	DC	4000	No
Supplementary information:			
All testing including after Humidity required of clause 5.4.8.			
All sources of each material have been considered to conduct test, see table 4.1.2 for details.			
The T100/ T400 core of transformer is considered as middle part.			
The TD700 core of transformer is considered as middle part.			

5.5.2.2	TABLE: Stored discharge on capacitors				P
Location	Supply voltage (V)	Operating and fault condition ¹⁾	Switch position	Measured voltage (Vpk)	ES Class
Phase to Neutral	275Vac,60Hz	N	--	36Vdc	ES1
Phase to Neutral	275Vac,60Hz	S (R103 open)	--	52Vdc	ES1
Phase to Neutral	275Vac,60Hz	S (R113 open)	--	56Vdc	ES1
Phase to Neutral	275Vac,60Hz	S (R118 open)	--	56Vdc	ES1
L/N to earth	275Vac,60Hz	N	--	60Vdc	ES1
Supplementary information:					
X-capacitors installed for testing: see table 4.1.2					
[X] bleeding resistor rating: see table 4.1.2					
[] ICX:					
1) Normal operating condition (e.g., normal operation, or open fuse), SC= short circuit, OC= open circuit					

5.6.6	TABLE: Resistance of protective conductors and terminations	P
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IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

Location	Test current (A)	Duration (min)	Voltage drop (V)	Resistance (Ω)
AC input earthed to furthest end of metal enclosure	50	4	0.040	0.003
Supplementary information:				
Note: Less than 0.1ohm				

5.7.4	TABLE: Unearthed accessible parts					P
Location	Operating and fault conditions	Supply Voltage (V)	Parameters			ES class
			Voltage (V_{rms} or V_{pk})	Current (A_{rms} or A_{pk})	Freq. (Hz)	
#	--	--	--	--	--	--
Supplementary information:						
Abbreviation: SC= short circuit; OC= open circuit						
#: See table 5.2 for detail						

5.7.5	TABLE: Earthed accessible conductive part			P
Supply voltage (V)	275Vac			—
Phase(s)	[X] Single Phase; [] Three Phase: [] Delta [] Wye			
Power Distribution System	[X] TN [] TT [] IT			
Location	Fault Condition No in IEC 60990 clause 6.2.2	Touch current (mA)	Comment	
Metal enclosure	1	Normal: 4.20mApeak. Reversed: 4.20mApeak.	Pass	
Supplementary Information:				
Fault condition 1: Single-phase EQUIPMENT not reliably earthed shall be tested with loss of protective earth connection (switch e) in combination with normal and reverse polarity (switch p)				
Fault condition 2. Single-phase EQUIPMENT shall be tested with neutral open (switch n), with earth intact and in normal polarity. and again in reverse polarity (switch p).				

5.8	TABLE: Backfeed safeguard in battery backed up supplies					N/A
Location	Supply voltage (V)	Operating and fault condition	Time (s)	Open-circuit voltage (V)	Touch current (A)	ES Class
--	--	--	--	--	--	--
Supplementary information:						
Abbreviation: SC= short circuit, OC= open circuit						

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

6.2.2	TABLE: Power source circuit classifications					P
Location	Operating and fault condition	Voltage (V)	Current (A)	Max. Power ¹⁾ (W)	Time (S)	PS class
All circuits	All conditions	--	--	--	--	PS3 (declared)
Supplementary information:						
Abbreviation: SC= short circuit; OC= open circuit						
1) Measured after 3 s for PS1 and measured after 5 s for PS2 and PS3.						

6.2.3.1	TABLE: Determination of Arcing PIS				P
Location	Open circuit voltage after 3s(Vpk)	Measured r.m.s current(A)	Calculated value	Arcing PIS? Yes / No	
All circuits/components	--	--	--	Yes (declare)	
Supplementary information:					

6.2.3.2	TABLE: Determination of resistive PIS			P
Location		Operating and fault condition	Dissipate power (W)	Arcing PIS? Yes / No
All internal circuits/components		All conditions	--	Yes (declare)
Supplementary information:				
Abbreviation: SC= short circuit; OC= open circuit				

8.5.5	TABLE: High pressure lamp				N/A
Lamp manufacturer	Lamp type	Explosion method	Longest axis of glass particle (mm)	Particle found beyond 1 m Yes / No	
--	--	--	--	--	
Supplementary information:					

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

9.6	TABLE: Temperature measurements for wireless power transmitters							N/A
Supply voltage (V)..... :								—
Max. transmit power of transmitter (W)..... :								—
Foreign objects	w/o receiver and direct contact		with receiver and direct contact		with receiver and at distance of 2 mm		with receiver and at distance of 5 mm	
	Object (°C)	Ambient (°C)	Object (°C)	Ambient (°C)	Object (°C)	Ambient (°C)	Object (°C)	Ambient (°C)
--	--	--	--	--	--	--	--	--
Supplementary information:								

5.4.1.4, 9.3, B.1.5, B.2.6	TABLE: Temperature measurements							P
Supply voltage (V)..... :			See below	See below	See below	See below		—
Ambient temperature during test T_{amb} (°C) :			See below	See below	See below	See below		—
Maximum measured temperature T of part/at:			T (°C)					Allowed T_{max} (°C)
Model: MR484000SG1			90Vac/60 Hz (load with 53.5Vdc, 32.5A)	90Vac/60 Hz (load with 53.5Vdc, 14.9A)	143Vac/60 Hz (load with 53.5Vdc, 32.5A)	143Vac/60 Hz (load with 53.5Vdc, 14.9A)		--
CX100			60.1	90.1	71.1	87.0		110
RV100			57.7	88.4	68.1	87.1		105
PCB under LHY100			59.7	90.6	72.4	88.0		130
CY101			56.2	86.8	65.4	85.0		110
CX101			55.7	87.2	65.1	86.0		110
PCB under LHY103			61.4	92.6	74.8	89.1		130
CY102			57.3	88.8	67.7	88.1		125
CX102			53.0	84.1	60.5	84.3		110
CX103			51.7	82.7	58.4	83.2		110
RLY101 body			53.2	81.5	61.3	80.1		85
PCB under Q105			67.0	96.9	80.3	93.1		130
PCB under G100			68.9	98.1	82.5	94.5		130
PCB under Q102			52.0	83.0	59.7	83.2		130

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict
L102 coil	57.1	87.5	66.8	86.2	120
L102 core	56.8	87.0	65.8	85.7	120
EC901	47.8	77.4	49.1	77.5	105
T100 coil	49.2	78.0	48.1	78.0	90
T100 core	48.0	77.4	47.7	77.5	90
CY107	45.9	75.6	45.8	75.6	125
EC409	49.9	77.5	48.9	78.0	105
L401 coil	71.5	81.1	59.5	82.7	120
L401 core	63.2	79.5	55.0	80.9	120
PCB under Q408	55.2	79.4	52.4	80.8	130
PCB under Q404	59.4	82.6	59.3	85.7	130
T400 coil	74.7	92.2	75.3	95.7	130
T400 core	61.2	95.5	75.1	98.7	130
EC406	59.6	82.8	59.5	85.5	105
TD700 coil	57.5	85.7	67.3	86.0	90
U702	57.0	84.6	65.8	85.1	110
L400	63.4	86.5	68.4	88.2	130
Mylar sheet near T400 top	52.0	82.6	58.2	82.3	105
Mylar sheet near T400 bottom	51.3	82.0	56.8	82.1	105
Internal wire	48.4	77.5	48.7	77.8	80
FAN	47.1	76.8	46.8	76.8	Ref.
Plastic enclosure inside near EC901	46.3	76.2	48.0	76.1	80
Ambient	45.0	75.0	45.0	75.0	--
For accessible parts at 25°C test:	--	--	--	--	--
Metal enclosure on T400 top	29.9	30.2	33.7	30.6	60
Plastic enclosure outside	26.5	27.9	26.5	26.3	77
Ambient	25.0	25.0	25.0	25.0	--
Model: MR484000SG1	180Vac/60Hz (load with 53.5Vdc, 75A)	180Vac/60Hz (load with 53.5Vdc, 37.38A)	275Vac/60Hz (load with 53.5Vdc, 75A)	275Vac/60Hz (load with 53.5Vdc, 37.38A)	--
CX100	102.6	94.3	83.5	88.7	110
RV100	87.5	91.5	76.0	87.6	105
PCB under LHY100	96.3	93.7	81.1	88.4	130
CY101	80.9	89.0	69.7	85.3	110

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict

CX101	79.6	89.3	69.7	85.8	110
PCB under LHY103	103.7	96.0	87.4	89.9	130
CY102	81.1	91.0	70.5	87.4	125
CX102	71.7	85.9	63.5	83.7	110
CX103	69.4	84.6	60.6	82.6	110
RLY101 body	73.5	81.9	63.1	80.4	85
PCB under Q105	99.0	99.8	83.5	92.7	130
PCB under G100	106.5	102.5	88.4	95.6	130
PCB under Q102	75.9	85.9	62.9	83.7	130
L102 coil	92.2	91.3	73.0	86.2	120
L102 core	91.5	90.8	72.1	85.8	120
EC901	52.2	78.3	50.9	77.7	105
T100 coil	49.3	79.0	49.7	78.7	90
T100 core	48.0	78.2	48.6	77.9	90
CY107	45.4	75.9	46.0	75.6	125
EC409	51.2	78.6	50.8	78.3	105
L401 coil	84.4	89.7	85.4	91.6	120
L401 core	70.2	84.7	71.0	85.9	120
PCB under Q408	58.8	82.1	59.5	82.4	130
PCB under Q404	76.0	88.3	76.9	88.5	130
T400 coil	120.8	105.8	120.1	105.1	130
T400 core	96.7	103.1	95.0	101.4	130
EC406	75.4	88.1	75.5	87.9	105
TD700 coil	76.5	86.1	70.5	85.1	90
U702	76.4	85.2	70.1	84.5	110
L400	103.5	96.4	100.0	95.1	130
Mylar sheet near T400 top	72.8	84.9	61.3	82.0	105
Mylar sheet near T400 bottom	70.2	84.0	59.6	81.7	105
Internal wire	50.7	78.3	50.4	77.8	80
FAN	46.9	77.2	47.3	77.0	Ref.
Plastic enclosure inside near EC901	48.3	76.1	49.4	76.3	80
Ambient	45.0	75.0	45.0	75.0	--
For accessible parts at 25°C test:	--	--	--	--	--
Metal enclosure on T400 top	42.6	31.8	36.3	30.3	60

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict

Plastic enclosure	26.7	26.9	26.7	26.4	77
Ambient	25.0	25.0	25.0	25.0	--
Model: MR484000SG2	180Vac/60Hz (load with 53.5Vdc, 75A)	--	--	--	--
CX100	97.7	--	--	--	110
RV100	86.7	--	--	--	105
PCB under LHY100	95.0	--	--	--	130
CY101	78.6	--	--	--	110
CX101	77.2	--	--	--	110
PCB under LHY103	103.3	--	--	--	130
CY102	79.3	--	--	--	125
CX102	70.1	--	--	--	110
CX103	69.0	--	--	--	110
RLY101 body	72.4	--	--	--	85
PCB under Q105	98.7	--	--	--	130
PCB under D1	106.9	--	--	--	130
PCB under Q102	75.4	--	--	--	130
L102 coil	90.3	--	--	--	120
L102 core	89.5	--	--	--	120
EC901	52.0	--	--	--	105
T100 coil	48.2	--	--	--	90
T100 core	47.8	--	--	--	90
CY107	43.9	--	--	--	125
EC409	50.6	--	--	--	105
L401 coil	83.3	--	--	--	120
L401 core	69.1	--	--	--	120
PCB under Q408	57.9	--	--	--	130
PCB under Q404	74.8	--	--	--	130
T400 coil	116.1	--	--	--	130
T400 core	95.0	--	--	--	130
EC406	72.5	--	--	--	105
TD700 coil	76.1	--	--	--	90
U702	75.5	--	--	--	110

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

L400			103.5	--	--	--	130
Mylar sheet near T400 top			71.8	--	--	--	105
Mylar sheet near T400 bottom			70.4	--	--	--	105
Internal wire			50.9	--	--	--	80
FAN			46.8	--	--	--	Ref.
Plastic enclosure inside near EC901			47.3	--	--	--	80
Ambient			45.0	--	--	--	--
For accessible parts at 25°C test:			--	--	--	--	--
Metal enclosure on T400 top			40.3	--	--	--	60
Plastic enclosure			26.4	--	--	--	77
Ambient			25.0	--	--	--	--
Temperature T of winding:	t ₁ (°C)	R ₁ (Ω)	t ₂ (°C)	R ₂ (Ω)	T (°C)	Allowed T _{max} (°C)	Insulation class
--	--	--	--	--	--	--	--
Supplementary information:							
Note 1: Tma should be considered as directed by appliable requirement.							
Note 2: Tma is not included in assessment of Touch Temperatures (Clause 9).							
Note 3: The product was submitted and evaluated for use at the maximum ambient temperature (Tma) permitted by the manufacturer's specification of: For the maximum operating temperature is 45°C, the equipment can be loaded with maximum rated loading; For the maximum operating temperature is 75°C, see ambient derating table for detail.							
Note 4: During the tests with DC fan manufacture: NINGBO SHENGJIU THERMAL TECHNOLOGY CO LTD, model: SE120428BM.							
All values for T(°C) are re-calculated from Tamb respectively.							
Winding components (providing safety isolation):							
- Class A Tmax=90°C							
- Class F Tmax=140°C-10°C= 130°C							

B.2.5		TABLE: Input test						P
U (V)	Hz	I (A)	I rated (A)	P (W)	P rated (W)	Fuse No	I fuse (A)	Condition/status
Model: MR484000SG1								
90	50	21.04	--	1890	--	F100, F101	21.04	Output loading 53.5Vdc, 32.5A
90	60	21.06	--	1891	--	F100, F101	21.06	
100	50	18.94	21	1881	--	F100, F101	18.94	
100	60	18.96	21	1881	--	F100, F101	18.96	

IEC 62368-1								
Clause		Requirement + Test				Result - Remark		Verdict
130	50	14.33	21	1844	--	F100, F101	14.33	
130	60	14.35	21	1844	--	F100, F101	14.35	
143	50	12.92	--	1836	--	F100, F101	12.92	
143	60	12.94	--	1836	--	F100, F101	12.94	
180	50	23.95	--	4280	--	F100, F101	23.95	Output loading 53.5Vdc, 75A
180	60	23.97	--	4280	--	F100, F101	23.97	
200	50	21.43	21	4256	--	F100, F101	21.43	
200	60	21.43	21	4251	--	F100, F101	21.43	
250	50	17.05	21	4211	--	F100, F101	17.05	
250	60	17.05	21	4251	--	F100, F101	17.05	
275	50	18.26	--	4217	--	F100, F101	18.26	
275	60	18.27	--	4225	--	F100, F101	18.27	
Model: MR484000SG2								
90	50	20.96	--	1886	--	F100, F101	20.96	Output loading 53.5Vdc, 32.5A
90	60	20.99	--	1888	--	F100, F101	20.99	
100	50	18.65	21	1863	--	F100, F101	18.65	
100	60	18.67	21	1863	--	F100, F101	18.67	
130	50	14.23	21	1829	--	F100, F101	14.23	
130	60	14.25	21	1830	--	F100, F101	14.25	
143	50	12.90	--	1821	--	F100, F101	12.90	
143	60	12.92	--	1821	--	F100, F101	12.92	
180	50	23.87	--	4274	--	F100, F101	23.87	Output loading

IEC 62368-1								
Clause		Requirement + Test				Result - Remark		Verdict
180	60	23.89	--	4275	--	F100, F101	23.89	53.5Vdc, 75A
200	50	21.35	21	4244	--	F100, F101	21.35	
200	60	21.36	21	4245	--	F100, F101	21.36	
250	50	17.02	21	4206	--	F100, F101	17.02	
250	60	17.04	21	4206	--	F100, F101	17.04	
275	50	17.73	--	4202	--	F100, F101	17.73	
275	60	17.80	--	4202	--	F100, F101	17.80	
Model: MR483000SG3								
90	50	16.13	--	1449	--	F100, F101	16.13	Output loading 53.5Vdc, 25.2A
90	60	16.17	--	1449	--	F100, F101	16.17	
100	50	14.48	17	1437	--	F100, F101	14.48	
100	60	14.50	17	1437	--	F100, F101	14.50	
130	50	10.96	17	1416	--	F100, F101	10.96	
130	60	10.98	17	1416	--	F100, F101	10.98	
143	50	9.952	--	1411	--	F100, F101	9.952	
143	60	9.959	--	1411	--	F100, F101	9.959	
180	50	17.57	--	3156	--	F100, F101	17.57	Output loading 53.5Vdc, 56.1A
180	60	17.60	--	3156	--	F100, F101	17.60	
200	50	15.81	17	3144	--	F100, F101	15.81	
200	60	15.84	17	3145	--	F100, F101	15.84	
250	50	12.55	17	3123	--	F100, F101	12.55	
250	60	12.59	17	3124	--	F100, F101	12.59	

IEC 62368-1			
Clause	Requirement + Test		Verdict

275	50	12.69	--	3125	--	F100, F101	12.69	
275	60	12.88	--	3126	--	F100, F101	12.88	

Supplementary information:

The measured input current did not exceed the marked input rating by more than 10 percent when the apparatus was operated to produce the maximum normal input current.

B.3, B.4		TABLE: Abnormal operating and fault condition tests					P
Ambient temperature T _{amb} (°C)..... :					See below		—
Power source for EUT: Manufacturer, model/type, outputrating .. :					--		—
Component No.	Condition	Supply voltage (V)	Test time	Fuse no.	Fuse current (A)	Observation	
Model: MR484000SG1							
RV101	SC	275Vac	1S	F100, F101	18.27→ 0	# F100, F101 open, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc	
G100 pin 3-4	SC	275Vac	1S	F100, F101	18.27→ 0	# F100, F101 open, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc	
C100	SC	275Vac	1S	F100, F101	18.27→ 0	# F100, F101 open, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc	
Q102 pin 1-2	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q102 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc	
Q102 pin 2-3	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc	
Q102 pin 1-3	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q102 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc	
RC100	SC	275Vac	30mins	F100, F101	18.27→ 18.27	Normal operation, no hazard Touch Current (Sec. output to earth): 0.424mA peak	

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

						Sec. output + to -: 53.58Vdc Normal operation, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 53.58Vdc
R150	SC	275Vac	30mins	F100, F101	18.27→ 18.27	
Q406 pin 1-2	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q402, Q403, Q406, Q408 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
Q406 pin 2-3	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q402, Q403 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
Q406 pin 1-3	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q402, Q403, Q406, Q408 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
T400 pin 1-2,3	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q402, Q403, Q406, Q408 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
T400 pin 2,3-4	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q402, Q403, Q406, Q408 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
T400 pin 1-4	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q402, Q403, Q406, Q408 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
T400 pin 5,10- 6,9	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q402, Q403, Q406, Q408 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
T400 pin 7,8- 6,9	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q402, Q403, Q406, Q408 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
Q404 pin 2-3	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q402, Q403, Q406, Q408

IEC 62368-1						
Clause	Requirement + Test			Result - Remark		Verdict
						damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
Q408 pin 2-3	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q402, Q403, Q406, Q408 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
Q500 pin 1-2	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
Q500 pin 2-3	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
Q500 pin 1-3	SC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
RC500	SC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
T100 pin 1-3	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
T100 pin 2-3	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
T100 pin 4-6	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
T100 pin 7-9	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
T100 pin 8-9	SC	275Vac	1S	F100, F101	18.27→ 0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

T100 pin 10-11	SC	275Vac	1S	F100, F101	18.27→0	Sec. output + to -: 0Vdc #F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
D501	SC	275Vac	1S	F100, F101	18.27→0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
D502	SC	275Vac	1S	F100, F101	18.27→0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
D505	SC	275Vac	1S	F100, F101	18.27→0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
U501 pin 2-5	SC	275Vac	10mins	F100, F101	18.27→0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U501 pin 2-6	SC	275Vac	10mins	F100, F101	18.27→0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U500 pin 1-2	SC	275Vac	10mins	F100, F101	18.27→0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U500 pin 3-4	SC	275Vac	10mins	F100, F101	18.27→0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U500 pin 1	OC	275Vac	10mins	F100, F101	18.27→0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U500 pin 3	OC	275Vac	10mins	F100, F101	18.27→0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
TD700 pin 2-4	SC	275Vac	10mins	F100, F101	18.27→0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

						to earth): 0.424mA peak Sec. output + to -: 0Vdc
TD700 pin 5-6	SC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
TD700 pin 7-8	SC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U700 pin 1-2	SC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U700 pin 3-4	SC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U700 pin 1	OC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U700 pin 3	OC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U702 pin 1-2	SC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U702 pin 3-4	SC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U702 pin 1	OC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U702 pin 3	OC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U703 pin 1-2	SC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard

IEC 62368-1						
Clause	Requirement + Test				Result - Remark	Verdict
						Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U703 pin 3-4	SC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U703 pin 1	OC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U703 pin 3	OC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
Sec. Output	SC	275Vac	10mins	F100, F101	18.27→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
Sec. Output	OL	275Vac	4h 56mins	F100, F101	18.27→ 19.50→ 20.34→ 0.030	Output overload to 83.5A(Constant Temperatures were obtained) and shut down at 83.7A. Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 53.58Vdc Temp.(°C): 1.PCB under D1:73.3 2.T100 coil:30.8 3.T100 core:29.7 4.PCB under Q404:62.5 5.T400 coil:113.9 6.T400 core:82.2 7.TD700 coil:53.7 8.Metal enclosure on T400 top:38.6 9.Plastic enclosure:28.6 10.Ambient:24.8
DC FAN	Locked	275Vac	7h	F100, F101	18.27→ 0.030	After normal operation for 2 minutes and shut down, Recoverable, no hazard. Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 53.58Vdc Temp.(°C): 1.PCB under D1:38.7 2.T100 coil:39.3 3.T100 core:38.7 4.PCB under Q404:35.8

IEC 62368-1						
Clause	Requirement + Test			Result - Remark		Verdict
						5.T400 coil:34.8 6.T400 core:31.9 7.TD700 coil:34 8.Metal enclosure on T400 top:29 9.Plastic enclosure:26.1 10.Ambient:24.6
Ventilation openings	Covered	275Vac	1h 52mins	F100, F101	18.27→ 0.030	After normal operation for 6 minutes and shut down, Recoverable, no hazard. Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 53.58Vdc Temp.(°C): 1.PCB under D1:118.3 2.T100 coil:59.6 3.T100 core:57.6 4.PCB under Q404:118.4 5.T400 coil:93.1 6.T400 core:68 7.TD700 coil:66.1 8.Metal enclosure on T400 top:42.9 9.Plastic enclosure:41.5 10.Ambient:24.6
T400 pin 6,9-5,10/7,8 after Q404/Q408 pin 3	OL	275Vac	4h 38mins	F100, F101	18.27→ 19.58→ 20.36→ 0.030	(Output normal load: 53.5Vdc, 75A) T400 pin 6,9-5,10/7,8 after Q404/Q408 pin 3 overload to 8.8A(Constant Temperatures were obtained) and shut down at 9.0A. Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 53.58Vdc Temp.(°C): 1.PCB under D1:73.8 2.T100 coil:31.0 3.T100 core:30.2 4.PCB under Q404:62.7 5.T400 coil:114.3 6.T400 core:82.5 7.TD700 coil:53.9 8.Metal enclosure on T400 top:38.8 9.Plastic enclosure:29.2 10.Ambient:24.5
T100 pin 11-10 after D501	OL	275Vac	3h 38mins	F100, F101	18.27→ 18.38→ 0.030	(Output normal load: 53.5Vdc, 75A) T100 pin 11-10 after D501 overload to 1.6A(Constant Temperatures were obtained) and shut down at 1.7A.

IEC 62368-1						
Clause	Requirement + Test			Result - Remark		Verdict
						Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to-: 53.58Vdc Temp.(°C): 1.PCB under D1:70.2 2.T100 coil:31.6 3.T100 core:30.1 4.PCB under Q404:59.2 5.T400 coil:102.0 6.T400 core:76.1 7.TD700 coil:52.7 8.Metal enclosure on T400 top:37.2 9.Plastic enclosure:28.0 10.Ambient:24.6
T100 pin 9-7 after D502	OL	275Vac	3h 44mins	F100, F101	18.27→ 18.45→ 0.030	(Output normal load: 53.5Vdc, 75A) T100 pin 9-7 after D502 overload to 2.3A(Constant Temperatures were obtained) and shut down at 2.4A. Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to-: 53.58Vdc Temp.(°C): 1.PCB under D1:71.9 2.T100 coil:33 3.T100 core:30.6 4.PCB under Q404:60.4 5.T400 coil:103.8 6.T400 core:76.9 7.TD700 coil:53.3 8.Metal enclosure on T400 top:37.4 9.Plastic enclosure:28.3 10.Ambient:24.6
Sec. Output	OL	143Vac	3h 36mins	F100, F101	12.94→ 19.62→ 20.88→ 0.030	Output overload to 55.8A(Constant Temperatures were obtained) and shut down at 56.0A. Touch Current (Sec. output to earth): 0.208mA peak Sec. output + to-: 53.58Vdc Temp.(°C): 1.PCB under D1:80.1 2.T100 coil:29.0 3.T100 core:27.9 4.PCB under Q404:45.2 5.T400 coil:71.4 6.T400 core:61.0 7.TD700 coil:54.3 8.Metal enclosure on T400

IEC 62368-1						
Clause	Requirement + Test			Result - Remark		Verdict
						top:38.2 9.Plastic enclosure:27.0 10.Ambient:24.7
DC FAN	Locked	143Vac	7h	F100, F101	12.94→ 0.030	After normal operation for 2 minutes and shut down, Recoverable, no hazard. Touch Current (Sec. output to earth): 0.208mA peak Sec. output + to -: 53.58Vdc Temp.(°C): 1.PCB under D1:110.5 2.T100 coil:54.3 3.T100 core:53.2 4.PCB under Q404:113.1 5.T400 coil:73.5 6.T400 core:65.8 7.TD700 coil:59.4 8.Metal enclosure on T400 top:39 9.Plastic enclosure:36.2 10.Ambient:24.7
Ventilation openings	Covered	143Vac	1h 52mins	F100, F101	12.94→ 0.030	After normal operation for 6 minutes and shut down, Recoverable, no hazard. Touch Current (Sec. output to earth): 0.208mA peak Sec. output + to -: 53.58Vdc Temp.(°C): 1.PCB under D1:37.7 2.T100 coil:38.8 3.T100 core:37.9 4.PCB under Q404:35.3 5.T400 coil:34.3 6.T400 core:31.5 7.TD700 coil:33.4 8.Metal enclosure on T400 top:28.5 9.Plastic enclosure:25.2 10.Ambient:24.5
Model: MR484000SG2						
T400 pin 5,10-6,9	SC	275Vac	1S	F100, F101	17.80→0	#F100, F101 open, Q402, Q403, Q406, Q408 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
T400 pin 7,8-6,9	SC	275Vac	1S	F100, F101	17.80→0	#F100, F101 open, Q402, Q403, Q406, Q408 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
Q404 pin 2-3	SC	275Vac	1S	F100,	17.80→0	#F100, F101 open, Q402,

IEC 62368-1						
Clause	Requirement + Test			Result - Remark		Verdict
				F101		Q403, Q406, Q408 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
Q408 pin 2-3	SC	275Vac	1S	F100, F101	17.80→0	#F100, F101 open, Q402, Q403, Q406, Q408 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
T100 pin 7-9	SC	275Vac	1S	F100, F101	17.80→0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
T100 pin 8-9	SC	275Vac	1S	F100, F101	17.80→0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
T100 pin 10-11	SC	275Vac	1S	F100, F101	17.80→0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
D501	SC	275Vac	1S	F100, F101	17.80→0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
D502	SC	275Vac	1S	F100, F101	17.80→0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
D505	SC	275Vac	1S	F100, F101	17.80→0	#F100, F101 open, Q500, Q501 damaged, no hazard Touch Current (Sec. output to earth): 0.044mA peak Sec. output + to -: 0Vdc
U500 pin 1-2	SC	275Vac	10mins	F100, F101	17.80→0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U500 pin 1	OC	275Vac	10mins	F100, F101	17.80→0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
TD700 pin 5-6	SC	275Vac	10mins	F100, F101	17.80→0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

						to earth): 0.424mA peak Sec. output + to -: 0Vdc
TD700 pin 7-8	SC	275Vac	10mins	F100, F101	17.80→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U700 pin 1-2	SC	275Vac	10mins	F100, F101	17.80→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U700 pin 1	OC	275Vac	10mins	F100, F101	17.80→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U702 pin 1-2	SC	275Vac	10mins	F100, F101	17.80→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U702 pin 1	OC	275Vac	10mins	F100, F101	17.80→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U703 pin 1-2	SC	275Vac	10mins	F100, F101	17.80→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
U703 pin 1	OC	275Vac	10mins	F100, F101	17.80→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc
Sec. Output	SC	275Vac	10mins	F100, F101	17.80→ 0.030	Unit shutdown, Recoverable, no hazard Touch Current (Sec. output to earth): 0.424mA peak Sec. output + to -: 0Vdc

Supplementary information:

Results Key: S-C – Short Circuit; O-C – Open Circuit; O-L- Overload; NB=No indication of dielectric breakdown; NC=Cheesecloth remained intact; NT=Tissue paper remained intact; IP=Internal protection operated (list component); CD=Components damaged (list damaged components);

1. After each of above test, unit can pass the dielectric strength test specified in table 5.4.9.

2. #: The tests where fuse opened were repeated with each source listed in table 4.1.2 and same result observed.

3. No ignition during and after all tests;

4. Output voltage comply with ES1 during and after all tests.

5. Transformer TD700 can't be overload. RT100 only for suppress current impact, no temperature control.

6. During the tests with DC fan manufacture: NINGBO SHENGJIU THERMAL TECHNOLOGY

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

CO LTD, model: SE120428BM.

M.3	TABLE: Protection circuits for batteries provided within the equipment						N/A
Is it possible to install the battery in a reverse polarity position? :				--		—	
Equipment Specification	Charging						
	Voltage (V)			Current (A)			
	--			--			
Manufacturer/type	Battery specification						
	Non-rechargeable batteries		Rechargeable batteries				
	Discharging current (A)	Unintentional charging current (A)	Charging		Discharging current (A)	Reverse charging current (A)	
			Voltage (V)	Current (A)			
--	--	--	--	--	--	--	
Note: The tests of M.3.2 are applicable only when above appropriate data is not available.							
Specified battery temperature (°C)..... :				--			
Component No.	Fault condition	Charge/ discharge mode	Test time	Temp. (°C)	Current (A)	Voltage (V)	Observation
--	--	--	--	--	--	--	--
Supplementary information:							
Abbreviation: SC= short circuit; OC= open circuit NL= no chemical leakage; NS= no spillage of liquid; NE= no explosion; NF= no emission of flame or expulsion of molten metal.							

M.4.2	TABLE: Charging safeguards for equipment containing a secondary lithium battery					N/A
Maximum specified charging voltage (V)					--	—
Maximum specified charging current (A)					--	—
Highest specified charging temperature (°C)					--	
Lowest specified charging temperature (°C)					--	
Battery manufacturer/type	Operating and fault condition	Measurement			Observation	
		Charging voltage (V)	Charging current (A)	Temp. (°C)		
--	--	--	--	--	--	
Supplementary information:						
Abbreviation: SC= short circuit; OC= open circuit; MSCV= maximum specified charging voltage; MSCC= maximum specified charging current; HSCT= highest specified charging temperature; LSCT= lowest specified charging temperature						

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

Q.1	TABLE: Circuits intended for interconnection with building wiring (LPS)						N/A
Output Circuit	Condition	U _{oc} (V)	Time (s)	I _{sc} (A)		S (VA)	
				Meas.	Limit	Meas.	Limit
--	--	--	--	--	--	--	--
--	--	--	--	--	--	--	--
--	--	--	--	--	--	--	--
Supplementary Information:							
SC: ShortCircuit, OC: OpenCircuit							

T.2, T.3, T.4, T.5	TABLE: Steady force test						P
Location/Part	Material	Thickness (mm)	Probe	Force (N)	Test Duration(s)	Observation	
Plastic enclosure near DC fan side(T.5)	See table 4.1.2	Min. 1.5	--	250	5	Enclosure remained intact, no crack/opening developed. Internal ES3, TS3 were not accessible after test. No insulation breakdown.	
Internal components (T.2)	--	--	--	10	5	No insulation breakdown. No reduction the clearances and creepage distances	
Supplementary information:							
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T.6, T.9	TABLE: Impact test				P
Location/Part	Material	Thickness (mm)	Height (mm)	Observation	
Enclosure front (FAN side) (T.6)	See table 4.1.2	Min. 1.5	1300	Enclosure remained intact, no crack/opening developed. Internal ES3, TS3 were not accessible after test. No insulation breakdown.	
Supplementary information:					
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IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

T.7	TABLE: Drop test				N/A
Location/Part		Material	Thickness (mm)	Height (mm)	Observation
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--		--	--	--	--
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Supplementary information:					

T.8	TABLE: Stress relief test				P
Location/Part	Material	Thickness (mm)	Oven Temperature (°C)	Duration (h)	Observation
Plastic Enclosure	See table 4.1.2	Min. 1.5	87	7	No damage, no hazard.
Supplementary information:					

X	TABLE: Alternative method for determining minimum clearances distances			N/A
Clearance distanced between:	Peak of working voltage (V)	Required cl (mm)	Measured cl (mm)	
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Supplementary information:				

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

4.1.2	TABLE: Critical components information					P
Object / part No.	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾	
PCB(for power board and control board only)	Shenzhen Jove Enterprise Ltd	JVE-M2, JVE-M4, JVE-M4A	V-0, 130°C, min. 0.4mm for each layer, CEMENTED JOINT tested per UL60950-1, Sec 2.10.5.5, Working Voltage: 1550 V	UL 796 UL60950	UL	
(alternative)	SHENZHEN BOMIN XING ELECTRONIC CO LTD	BMX-02	V-0, 130°C, min. 0.4mm for each layer, CEMENTED JOINT tested per UL60950-1 and UL62368-1, Test Voltage: 4904 Vac	UL 796 UL60950 or UL62368	UL	
(alternative)	Interchangeable	Interchangeable	Rated V-0, 130°C. Conducted with cemented joint test with min. working voltage 540 V and provided with reinforced insulation with distance through insulation min. 0.4 mm.	UL 796 IEC/EN/UL 62368-1 IEC/EN/UL 60950-1	UL	
Metal enclosure	Interchangeable	Interchangeable	Aluminium zinc alloy, min. thickness 0.8mm.	IEC/EN 62368-1	Tested in appliance	
PCB(Except for power board and control board only)	Interchangeable	Interchangeable	V-0, min. 130 °C.	UL796	UL	
Fuse (F100, F101)	CONQUER ELECTRONICS CO LTD	APA-A	T30A, 250Vac	IEC/EN 60127-1, IEC/EN 60127-7, UL 248-1, UL 248-14	UL TUV RH	
(alternative)	XC ELECTRONICS (SHENZHEN) CORP LTD	6H	T30A, 250Vac	IEC/EN 60127-1, IEC/EN 60127-7, UL 248-1, UL 248-14	UL TUV RH	

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict
(alternative)	DONGGUAN BETTER ELECTRONICS TECHNOLOGY CO LTD	616	T30A, 250Vac	IEC/EN 60127-1, IEC/EN 60127-7, UL 248-1, UL 248-14	UL TUV RH
(alternative)	ADVANCED SURGETECH MATERIALS LTD	HV630	T30A, 250Vac	IEC/EN 60127-1, IEC/EN 60127-7, UL 248-1, UL 248-14	UL TUV RH
(alternative)	HOLLYLAND CO LTD	65TS(P)	T30A, 250Vac	IEC/EN 60127-1, IEC/EN 60127-7, UL 248-1, UL 248-14	UL TUV RH
Plastic of DC Fan cover	KINGFA SCI & TECH CO LTD	FW-620HT	Rated V-0, min. thickness 1.5mm, 80°C	UL94, UL746B	UL
(alternative)	SABIC JAPAN L L C	EXL9330(X)(f1)(GG)(B1)(IP)	Rated V-0, min. thickness 1.5mm, 110°C	UL94, UL746B	UL
DC FAN	NINGBO SHENGJIU THERMAL TECHNOLOGY CO LTD	SE120428BM	12Vdc, min. 0.5A, min. CFM 23.33	IEC/EN 60950-1 UL 507	UL TUV
(alternative)	PROTECHNIC ELECTRIC CO.,LTD	MGT4012UB-W28	12Vdc, min. 0.5A, min. CFM 25.19	IEC/EN 60950-1 UL 507	UL TUV
(alternative)	NINGBO SHENGJIU THERMAL TECHNOLOGY CO LTD	SE120428BH	12Vdc, min. 0.5A, min. CFM 23.98	IEC/EN 60950-1 UL 507	UL TUV
Varistor (RV100, RV101, RV103) optional for model MR484000SG2 XXXXXX and for model MR484000SG1 XXXXXX	THINKING ELECTRONIC INDUSTRIAL CO LTD	TVR20621	105°C, 395Vac, coating min. V-1, (tested for 6KV/3KA combination pulse)	IEC/EN 61051-1 IEC 61051-2 IEC 61051-2-2 UL 1449	UL VDE
(alternative)	TDK (Zhuhai FTZ) Co., Ltd.	S20K385	Min.105°C, min.385Vac, (tested for 6KV/3KA combination pulse) , Min.V-1 coating	IEC/EN 61051-1 IEC/EN 61051-2 UL 1449	UL VDE

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict
(alternative)	GUANGDONG HUIWAN ELECTRONICS TECHNOLOGY CO LTD	V-621K-20D	Min.125°C, min.385Vac, (tested for 6KV/3KA combination pulse) , Min.V-1 coating	IEC/EN 61051-1 IEC/EN 61051-2 UL 1449	UL VDE
Varistor (RV114, RV105, RV106) optional for model MR484000SG2 XXXXXX	THINKING ELECTRONIC INDUSTRIAL CO LTD	TVR10471, TVR10511, TVR14511	Min.105°C, min.300Vac, coating min. V-1, (tested for min.2.5KV/1.25KA combination pulse)	IEC/EN 61051-1 IEC 61051-2 IEC 61051-2-2 UL 1449	UL VDE
(alternative)	CENTRA SCIENCE CORP	CNR-10D471K, CNR-10D511K, CNR-14D511K	Min.105°C, min.300Vac, coating min. V-1, (tested for min.2.5KV/1.25KA combination pulse)	IEC/EN 61051-1 IEC 61051-2 IEC 61051-2-2 UL 1449	UL VDE
(alternative)	BRIGHTKING (SHENZHEN) CO LTD	471KD10, 511KD10, 511KD14	Min.105°C, min.300Vac, coating min. V-1, (tested for min.2.5KV/1.25KA combination pulse)	IEC/EN 61051-1 IEC 61051-2 IEC 61051-2-2 UL 1449	UL VDE
(alternative)	BESTBRIGHT ELECTRONICS CO LTD	511KD10(*)	Min.105°C, min.300Vac, coating min. V-1, (tested for min.2.5KV/1.25KA combination pulse)	IEC/EN 61051-1 IEC 61051-2 IEC 61051-2-2 UL 1449	UL VDE
(alternative)	Shenzhen ChenJu Electronic Technology Co., Ltd.	CNJU-10D471K, CNJU-10D511K, CNJU-14D511K	Min.105°C, min.300Vac, coating min. V-1, (tested for min.2.5KV/1.25KA combination pulse)	IEC/EN 61051-1 IEC 61051-2 IEC 61051-2-2 UL 1449	UL VDE
(alternative)	GUANGDONG FENGHUA ADVANCED TECHNOLOGY HOLDING CO LTD. XIANHUA NEW SENSITIVE COMPONENTS BRANCH	FNR-10K471, FNR-10K511, FNR-14K511	Min.105°C, min.300Vac, coating min. V-1, (tested for min.2.5KV/1.25KA combination pulse)	IEC/EN 61051-1 IEC 61051-2 IEC 61051-2-2 UL 1449	UL VDE
Gas tube (DX100)	BRIGHTKING (SHENZHEN) CO LTD	2RM600L	Min.275Vac	UL 1449	UL
(alternative)	Thinking Electronic Industrial Co., Ltd.	GD82R600	Min.300Vac	UL 1449	UL

IEC 62368-1			
Clause	Requirement + Test	Result - Remark	Verdict

(alternative)	TDK components Taiwan Co Ltd	EM600XHC	Min.275Vac	UL 1449	UL
X-Capacitor (CX100,CX101, CX103, CX102)	NISTRONICS (JIANGXI) CO LTD	MPR	CX100 Max. 2.2μF±10%, CX101, CX103 Each Max. 2.0μF±10%, CX102 Max. 0.68μF±10%, min. 250Vac, 110 degree C, X2 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	XIAMEN FARATRONIC CO LTD	MKP62	CX100 Max. 2.2μF±10%, CX101, CX103 Each Max. 2.0μF±10%, CX102 Max. 0.68μF±10%, min. 250Vac, 110 degree C, X2 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	SHENZHEN YIMANFENG SCIENCE AND TECHNOLOGY CO LTD	MPX/MKP	CX100 Max. 2.2μF±10%, CX101, CX103 Each Max. 2.0μF±10%, CX102 Max. 0.68μF±10%, min. 250Vac, 110 degree C, X2 type	IEC/EN/UL 60384-14	VDE UL
(alternative)	YANGZHOU NISSEI ELECTRONICS CO LTD	MP1	CX100 Max. 2.2μF±10%, CX101, CX103 Each Max. 2.0μF±10%, CX102 Max. 0.68μF±10%, min. 250Vac, 110 degree C, X2 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	SHENZHEN JINGHAO CAPACITOR CO LTD	CBB62B	CX100 Max. 2.2μF±10%, CX101, CX103 Each Max. 2.0μF±10%, CX102 Max. 0.68μF±10%, min. 250Vac, 110 degree C, X2 type	IEC/EN/UL 60384-14	UL VDE

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict
(alternative)	AIHUA INFINITY CAPACITORS (SUZHOU) CO., LTD	FXT	CX100 Max. 2.2μF±10%, CX101, CX103 Each Max. 2.0μF±10%, CX102 Max. 0.68μF±10%, min. 250Vac, 110 degree C, X2 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	EUROPTRONIC (SUZHOU) CO LTD	MPX2, MPX	CX100 Max. 2.2μF±10%, CX101, CX103 Each Max. 2.0μF±10%, CX102 Max. 0.68μF±10%, min. 250Vac, 110 degree C, X2 type	IEC/EN/UL 60384-14	UL VDE
Bleeder resister (R103,R104,R106,R107,R108, R109,R152,R155,R156)	Interchangeable	Interchangeable	Each: max.200KΩ min.1/4W	IEC/EN 62368-1	Tested in appliance
Bleeder resister (R112, R113, R114, R115, R117, R118, R119, R120)	Interchangeable	Interchangeable	Each: max. 360KΩ min.1/4W	IEC/EN 62368-1	Tested in appliance
Bleeder resister (R320, R312, R322, R315, R334, R335, R336, R337) for model MR484000SG2 XXXXXX	Interchangeable	Interchangeable	Each: max. 75KΩ min.1/4W	IEC/EN 62368-1	Tested in appliance
Bleeder resister (R228, R215)	Interchangeable	Interchangeable	Each: max. 10KΩ min.1/4W	IEC/EN 62368-1	Tested in appliance
Relay (RLY101)	XIAMEN HONGFA ELECTROACOUSTIC CO LTD	HF161F-W	Min. 277Vac, Min.31A, 85°C. Coil: 12Vdc	IEC/EN 61810-1:2015 UL 508	UL VDE
(alternative)	PANASONIC CORPORATION	ALFG2PF12	Min. 277Vac, 31A, min. 85°C Coil: 12Vdc	IEC/EN 61810-1:2015 UL 508	UL VDE
(Alternative)	SANYOU CORPORATION LIMITED	SFK-112DMP	Min. 250Vac, 33A, min. 85°C Coil: 12Vdc	IEC/EN 61810-1:2015 UL 508	UL VDE
(Alternative)	DONGGUAN CHUOD ELECTRONICS CO LTD	CHFN-V-112HA	Min. 277Vac, 32A, min. 85°C Coil: 12Vdc	IEC/EN 61810-1:2015 UL 508	UL TUV

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict
Relay (RLY100) (optional)	XIAMEN HONGFA ELECTROACOUSTIC CO LTD	HF32F/012-HSLQ3, JZC-32F/012-HS	Min. 250Vac, Min.5A, min. 85°C	IEC/EN 61810-1:2015 UL 508	UL VDE
(alternative)	NINGBO TIANBO GANGLIAN ELECTRONICS CO LTD	TRG1D-12VDC-S-ZTBF-1	Min. 250Vac, Min.3A, min. 85°C	IEC/EN 61810-1:2015 UL 508	UL TUV RH
(Alternative)	SANYOU CORPORATION LIMITED	SJE-S-112DM	Min. 250Vac, Min.5A, min. 85°C	IEC/EN 61810-1:2015 UL 508	UL VDE
Inductor (L100, L101)	Shenzhen Megmeet Electrical Co., Ltd.	MR484000SG 1-L100	130°C	IEC/EN 62368-1	Tested in appliance
(Alternative)	Interchangeable	Interchangeable	130°C	IEC/EN 62368-1	Tested in appliance
-Magnet Wire	Interchangeable	Interchangeable	Min.130°C	UL1446	UL
- PWB	Interchangeable	Interchangeable	Min. V-1, Min. 130°C	UL 94, UL746B	UL
Inductor (L102)	Shenzhen Megmeet Electrical Co., Ltd.	MR484000SG 1-L102	130°C	IEC/EN 62368-1	Tested in appliance
(Alternative)	Interchangeable	Interchangeable	130°C	IEC/EN 62368-1	Tested in appliance
-Magnet Wire	Interchangeable	Interchangeable	Min.130°C	UL1446	UL
-Bobbin	Interchangeable	Interchangeable	Phenolic,V-0, min.150°C , min. thickness 0.45mm	UL 94, UL 746B	UL
-Insulation tape	Interchangeable	Interchangeable	Min.130°C	UL 510A	UL
-Tube	Interchangeable	Interchangeable	Min.300V, 200°C,VW-1	UL 224	UL
Inductor (L401)	Shenzhen Megmeet Electrical Co., Ltd.	MR484000SG 1-L401	130°C	IEC/EN 62368-1	Tested in appliance
(Alternative)	Interchangeable	Interchangeable	130°C	IEC/EN 62368-1	Tested in appliance
-Magnet wire	Interchangeable	Interchangeable	Min.130°C	UL 1446	UL
-Insulation tape	Interchangeable	Interchangeable	Min.130°C	UL 510A	UL
-Bobbin	Interchangeable	Interchangeable	Phenolic,V-0, min.150°C , min. thickness 0.45mm	UL 94, UL 746B	UL

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict
Transformer (TD700)	Shenzhen Megmeet Electrical Co., Ltd	MR483000HG -TD700	Class A	IEC/EN 62368-1	Tested in appliance
-Triple Insulated Wire	NEW ENGLAND WIRE TECHNOLOGIES CORP	W35T1.5PXX XTC1A, W32T1.5PX XXTC1A	180°C	IEC/EN 62368-1 UL 2353	VDE UL
(alternative)	RUBADUE WIRE CO INC	T-32-X-XX-P-XXX-1.5	180°C	IEC/EN 62368-1 UL 2353	VDE UL
-Double insulating wire	NEW ENGLAND WIRE TECHNOLOGIES CORP	W35D1.5PXX TC1A	180°C	IEC/EN 62368-1 UL 2353	VDE UL
-Bobbin	SUMITOMO BAKELITE CO LTD	PM-9630, PM-9820, PM-9823	Phenolic, V-0, min. 150°C, min. thinness 0.45mm	UL 94, UL 746B	UL
(alternative)	CHANG CHUN PLASTICS CO LTD	T375J(G5)(G6), T375HF	Phenolic, V-0, min. 150°C, min. thinness 0.75mm	UL 94, UL 746B	UL
Transformer (T100)	Shenzhen Megmeet Electrical Co., Ltd.	IPT483K-T100	Class A	IEC/EN 62368-1	Tested in appliance
-Bobbin material of T100	SUMITOMO BAKELITE CO LTD	PM-9820, PM-9630	Phenolic, V-0, min. 155°C, min. thinness 0.51mm	UL 94, UL 746B	UL
(alternative)	SUMITOMO BAKELITE CO LTD	PM-9823	Phenolic, V-0, min. 150°C, min. thinness 0.7mm	UL 94, UL 746B	UL
(alternative)	Chang Chun Plastics Co Ltd	T375J(G5)(G6), T375HF	Phenolic, V-0, 150°C, min. thinness 0.75mm	UL 94, UL 746B	UL
-Magnet wire of T100	Interchangeable	Interchangeable	Min. 130°C	UL 1446	UL
-Insulation tape	XINYU SHENGDAFENG ELECTRIC MATERIAL CO LTD	SDF-312	130°C	UL 510A	UL
(alternative)	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD	CT* (b)(g)	130°C	UL 510A	UL
- Margin tape of T100	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD	WF* (c)(h)	130°C	UL 510A	UL

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict
(alternative)	DONGGUAN SHIN YAHUA ELECTRONIC MATERIAL CO LTD	WF* (c)(h)	130°C	UL 510A	UL
-Tubing of T100	GREAT HOLDING INDUSTRIAL CO LTD	TFT	300V, 200°C,VW-1	UL 224	UL
(alternative)	CHANGYUAN ELECTRONICS GROUP CO LTD	CB-TT-T	300V, 200°C,VW-1	UL 224	UL
-Varnish of T100	SUZHOU TAIHU ELECTRIC ADVANCED MATERIAL CO LTD	T-4260(a)	Min.130°C	UL 1446	UL
(alternative)	HANG CHEUNG COATINGS (HUIYANG) LTD	8562*	Min. 155°C	UL 1446	UL
(alternative)	ELANTAS PDG, Inc.	V1630FS	Min. 155°C	UL 1446	UL
(alternative)	ZHUHAI CHANGXIAN NEW MATERIALS TECHNOLOGY CO LTD	E962	Min.130°C	UL 1446	UL
(alternative)	JOHN C DOLPH CO	BC-346A	Min. 155°C	UL 1446	UL
Transformer (T400)	Shenzhen Megmeet Electrical Co., Ltd.	MR484000SG 1-T400	Class F	IEC/EN 62368-1	Tested in appliance
Insulation system	Shenzhen Megmeet Electrical Co., Ltd.	G-F	Class F	UL 1446	UL
- Magnet wire	Interchangeable	Interchangeable	MW 79, MW 80, MW 82, MW 83, MW 85, min. 155°C	UL 1446	UL
-Copper foil	Interchangeable	Interchangeable	min. 155°C	IEC/EN 62368-1	Tested in appliance
- Insulation tape	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD	PF* (d)(g)	180°C	UL 510A	UL

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict
(alternative)	XINFENG GHILLIE ELECTRICAL MATERIALS CO LTD	JL-N408	180°C	UL 510A	UL
- EPOXY	DONG GUAN SHI PAI HUA CHUANG MATERIAL	808A/B-H, H907-H##	Min. 130°C	UL 94	UL
(alternative)	DONGGUAN EATTO ELECTRONIC MATERIAL CO LTD	E-500(XX)	Min. 130°C	UL 94	UL
- Tubing	CHANGYUAN ELECTRONICS GROUP CO LTD	CB-TT-T	200°C, VW-1	UL 224	UL
Bridge rectifier (G100, G101) for model MR484000SG1 XXXXXX	Interchangeable	Interchangeable	Min.35A, min. 800V	IEC/EN 62368-1	Tested in appliance
Diode (D1) for model MR484000SG2 XXXXXX	Interchangeable	Interchangeable	Each Min. 30A, min. 600V	IEC/EN 62368-1	Tested in appliance
Electrolytic capacitors (EC901, EC902)	Interchangeable	Interchangeable	Max. 470µF, min. 450V, min.105°C	IEC/EN 62368-1	Tested in appliance
Electrolytic capacitors (EC409)	Interchangeable	Interchangeable	Max.120µF, min. 450V, min.105°C	IEC/EN 62368-1	Tested in appliance
MOSFET (Q102, Q103) for model MR484000SG1 XXXXXX	Interchangeable	Interchangeable	Min. 54A, min. 600V	IEC/EN 62368-1	Tested in appliance
MOSFET (Q102, Q103, Q104, Q105) for model MR484000SG2 XXXXXX	Interchangeable	Interchangeable	Min. 54A, min. 600V	IEC/EN 62368-1	Tested in appliance
MOSFET (Q402, Q403, Q406, Q408)	Interchangeable	Interchangeable	Min. 54A, min. 600V	IEC/EN 62368-1	Tested in appliance
MOSFET (Q500)	Interchangeable	Interchangeable	Min. 6.5A, min. 600V	IEC/EN 62368-1	Tested in appliance

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict
Y- Capacitor (CY500)	Walsin Technology Corp.	AH	Max.2200pF, Min.400Vac; 125°C, Min.Y1 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	Yinan Don's Electronic Component Co., Ltd.	CT81	Max.2200pF, Min.400Vac; 125°C, Min.Y1 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	Xiamen Wanming Electronics Co., Ltd.	HJ	Max.2200pF, Min.400Vac; 125°C, Min.Y1 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	HAOHUA ELECTRONIC CO	CT7	Max.2200pF, Min.400Vac; 125°C, Min.Y1 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	TDK Corporation	CD	Max.2200pF, Min.400Vac; 125°C, Min.Y1 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	GUANGDONG HUIWAN ELECTRONICS TECHNOLOGY CO LTD	AR	Max.2200pF, Min.400Vac; 125°C, Min.Y1 type	IEC/EN/UL 60384-14	UL VDE
Y- Capacitor (CY100, CY104, CY106, CY107, CY102, CY105)	WALSIN TECHNOLOGY CORP	AC Series (&)	CY100, CY104 Each Max.2200pF CY106, CY107, CY102, CY105 Each Max.4700pF, Min.250Vac; 125°C, Min.Y2 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	Yinan Don's Electronic Component Co., Ltd.	CT81	CY100, CY104 Each Max.2200pF CY106, CY107, CY102, CY105 Each Max.4700pF, Min.250Vac; 125°C, Min.Y2 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	TDK CORPORATION	CS	CY100, CY104 Each Max.2200pF CY106, CY107, CY102, CY105 Each Max.4700pF, Min.250Vac; 125°C, Min.Y2 type	IEC/EN/UL 60384-14	UL VDE

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict
(alternative)	GUANGDONG HUIWAN ELECTRONICS TECHNOLOGY CO LTD	AB	CY100, CY104 Each Max.2200pF CY106, CY107, CY102, CY105 Each Max.4700pF, Min.250Vac; 125°C, Min.Y2 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	Xiamen Wanming Electronics Co., Ltd.	HM	CY100, CY104 Each Max.2200pF CY106, CY107, CY102, CY105 Each Max.4700pF, Min.250Vac; 125°C, Min.Y2 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	Kunshan Wansheng Electronics Co., Ltd.	CT7, CC7	CY100, CY104 Each Max.2200pF CY106, CY107, CY102, CY105 Each Max.4700pF, Min.250Vac; 125°C, Min.Y2 type	IEC/EN/UL 60384-14	UL VDE
Y-capacitor (CY101,CY103)	Xiamen Faratronic Co. Ltd.	MKP63 or C43	Max.10000pF, Min.250Vac; 110°C, Min.Y2 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	NISTRONICS (JIANGXI) CO LTD	MPY	Max.10000pF, Min.250Vac; 110°C, Min.Y2 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	TDK (Zhuhai FTZ) Co., Ltd.	B3202	Max.10000pF, Min.250Vac; 110°C, Min.Y2 type	IEC/EN/UL 60384-14	UL VDE
(alternative)	AIHUA INFINITY CAPACITORS (SUZHOU) CO., LTD	FY2	Max.10000pF, Min.250Vac; 110°C, Min.Y2 type	IEC/EN/UL 60384-14	UL VDE
Opto-coupler (U700, U702, U703)	LITE-ON Technology Corporation	LTV-10xx for VDE, LTV-10XX(X=0, 1, 2, 3, 4, 5, 6, 7, 8, 9) for UL	Reinforced insulation, Ext. Cr>=8.0mm, Ext. Cl>=8.0mm, thr. Distance>=0.4mm, 115°C	IEC/EN 60747-5-5 UL1577	UL VDE
(alternative)	EVERLIGHT ELECTRONICS CO LTD	EL101 for UL, EL101(0;1;2;3;4;5;6;7;8;9) V for VDE	Reinforced insulation, Ext. Cr>=8.1mm, Ext. Cl>=8.1mm, thr. Distance>=0.4mm, min.110°C	IEC/EN 60747-5-5 UL1577	UL VDE

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict
(alternative)	COSMO ELECTRONICS CORP	K1010	Reinforced insulation, Ext. Cr>=6.5mm, Ext. Cl>=6.5mm, thr. Distance>= 0.4mm, 115°C	IEC/EN 60747-5-5 UL1577	UL VDE
(alternative)	TOSHIBA ELECTRONIC DEVICES & STORAGE CORPORATION	TLP383	Reinforced insulation, Ext. Cr>=7.0mm, Ext. Cl>=7.0mm, thr. Distance>= 0.4mm, 125°C	IEC/EN 60747-5-5 UL1577	UL VDE
(alternative)	VISHAY Semiconductor GmbH	VOL617A	Reinforced insulation, Ext. Cr>=7.0mm, Ext. Cl>=7.0mm, thr. Distance>= 0.4mm, 110°C	IEC/EN 60747-5-5 UL1577	UL VDE
(alternative)	TOSHIBA ELECTRONIC DEVICES & STORAGE CORPORATION	TLP781K, TLP781KF, TLP785, TLP785F, TLP781, TLP781F	Reinforced insulation, Ext. Cr>=7.0mm, Ext. Cl>=7.0mm, thr. Distance>= 0.4mm, 115°C	IEC/EN 60747-5-5 UL1577	UL VDE
Mylar sheet between PCB and metal enclosure	SABIC JAPAN L L C	FR700	Rated V-0, 130°C, minimum 0.25 mm thickness	UL 94, UL 746B	UL
(alternative)	ITW ELECTRONICS COMPONENTS/ PRODUCTS (SHANGHAI) CO LTD	FORMEX GK-10	Rated VTM-0, 115°C, minimum 0.25 mm thickness	UL 94, UL 746B	UL
(alternative)	SABIC INNOVATIVE PLASTICS US L L C	FR700 (GG)	Rated V-0, 130°C, minimum 0.25 mm thickness	UL 94, UL 746B	UL
(alternative)	SABIC JAPAN L L C	FR1(E1) (GG1)	Rated VTM-0, 125°C, minimum 0.25 mm thickness	UL 94, UL 746B	UL
(alternative)	SABIC INNOVATIVE PLASTICS US L L C	FR1(GG)	Rated V-0, 125°C, minimum 0.25mm thickness	UL 94, UL 746B	UL

IEC 62368-1					
Clause	Requirement + Test			Result - Remark	Verdict
(alternative)	ITW ELECTRONICS COMPONENTS/ PRODUCTS (SHANGHAI) CO LTD	Formex EP-(a)(d)(f2)	VTM-0, 105°C, minimum 0.25 mm thickness	UL 94, UL 746B	UL
(alternative)	SICHUAN DONGFANG INSULATING MATERIAL CO LTD	DFR3716A	V-0, 110°C, minimum 0.25mm thickness	UL 94, UL 746B	UL
Mylar sheet used between secondary circuit and fuse F100,F101)	SABIC JAPAN L L C	FR700	Rated V-0, 130°C, minimum 0.40 mm thickness	UL 94, UL 746B	UL
(alternative)	ITW ELECTRONICS COMPONENTS/ PRODUCTS (SHANGHAI) CO LTD	FORMEX GK-10	Rated VTM-0, 115°C, minimum 0.40 mm thickness	UL 94, UL 746B	UL
(alternative)	SABIC INNOVATIVE PLASTICS US L L C	FR700 (GG)	Rated V-0, 130°C, minimum 0.40 mm thickness	UL 94, UL 746B	UL
(alternative)	SABIC JAPAN L L C	FR1(E1) (GG1)	Rated VTM-0, 125°C, minimum 0.40 mm thickness	UL 94, UL 746B	UL
(alternative)	SABIC INNOVATIVE PLASTICS US L L C	FR1(GG)	Rated V-0, 125°C, minimum 0.40 mm thickness	UL 94, UL 746B	UL
(alternative)	ITW ELECTRONICS COMPONENTS/ PRODUCTS (SHANGHAI) CO LTD	Formex EP-(a)(d)(f2)	VTM-0, 105°C, minimum 0.40 mm thickness	UL 94, UL 746B	UL
(alternative)	SICHUAN DONGFANG INSULATING MATERIAL CO LTD	DFR3716A	V-0, 110°C, minimum 0.40 mm thickness	UL 94, UL 746B	UL

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict
Heat shrinkable tube(used on DC fan Lead wire, R100, DX100, RV100, RV103)	SHENZHEN WOER HEAT-SHRINKABLE MATERIAL CO LTD	RSFR-HPF, RSFR-H, RSFR	VW-1, minimum 300V, 125 degree C, min 0.4mm thickness	UL 224	UL
(alternative)	CHANGYUAN ELECTRONICS (SHENZHEN) CO LTD	CYG-ZHP, CB-HFT CB-HFT(XY), CYG-MT, CB-TT-T	VW-1, minimum 300V, 125 degree C, min 0.4mm thickness	UL 224	UL
(alternative)	DONGGUAN SALIPT CO LTD	SALIPT S-901-300, SALIPT S-901-600	VW-1, minimum 300V, 125 degree C, min 0.4mm thickness	UL 224	UL
(alternative)	DONG GUAN HUANG FENG INSULATION MATERIAL CO LTD	HFT-01, HFT-2	VW-1, minimum 300V, 125 degree C, min 0.4mm thickness	UL 224	UL
Internal wire (DC fan Lead wire)	Interchangeable	Interchangeable	VW-1, min. 60Vac min. 28AWG, min. 80°C.	UL 758	UL
Heat shrinkable (used on fuse, RV101)	Interchangeable	Interchangeable	Min. 125°C, VW-1, Min. 300Vac	UL 224	UL
Thermal conductivity of mud on T400	Interchangeable	Interchangeable	Min.V-2	UL 94	UL
Thermal sheet for on T400	Shenzhen Bornsun Industrial Co.,Ltd	BNG400	V-0, 150°C, minimum 0.23 mm thickness	UL 94, UL 746B	UL
(alternative)	AOCHUAN TECHNOLOGY(H EYUAN) CO LTD	TC900S	V-0, 150°C, minimum 0.23 mm thickness	UL 94, UL 746B	UL
Insulation tape used for EC901 (two layer)	XINYU SHENGDAFENG ELECTRIC MATERIAL CO LTD	SDF-312	Min.130°C	UL 510A	UL
(alternative)	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD	CT* (b)(g), WF* (c)(h)	Min.130°C	UL 510A	UL
(alternative)	DONGGUAN SHIN YAHUA ELECTRONIC MATERIAL CO LTD	WF* (c)(h), PF* (d)(g)	Min.130°C	UL 510A	UL

IEC 62368-1					
Clause	Requirement + Test		Result - Remark		Verdict
(alternative)	XINFENG GHILLIE ELECTRICAL MATERIALS CO LTD	JL-N408	Min.180°C	UL 510A	UL
NTC(RT200, RT202, RT201, RT400)	THINKING ELECTRONIC INDUSTRIAL CO LTD	TSM1A473	Each Max. 47K	UL 1434	UL
(Alternative)	Interchange	Interchange	Each Max. 47K	UL 1434	UL
Glue	Interchangeable	Interchangeable	V-2	UL94	UL
Supplementary information:					
1) Provided evidence ensures the agreed level of compliance. See OD-CB2039.					
2) License available upon request					

Enclosures

<u>Type</u>	<u>Supplement ID</u>	<u>Description</u>
Photographs	3-01	Overall View
Photographs	3-02	Overall View
Photographs	3-03	Internal view for models MR484000SG1XXXXXX
Photographs	3-04	Internal view for models MR484000SG1XXXXXX
Photographs	3-05	Internal view for models MR484000SG1XXXXXX
Photographs	3-06	Top view for PCB layout for models MR484000SG1XXXXXX
Photographs	3-07	Bottom view for PCB layout for models MR484000SG1XXXXXX
Photographs	3-08	Internal view for models MR484000SG2XXXXXX
Photographs	3-09	Internal view for models MR484000SG2XXXXXX
Photographs	3-10	Top view for PCB layout for models MR484000SG2XXXXXX
Photographs	3-11	Bottom view for PCB layout for models MR484000SG2XXXXXX
Photographs	3-12	DC Fan view
Photographs	3-13	Top view for Control board
Photographs	3-14	Bottom view for Control board
Photographs	3-15	Top view for little board
Photographs	3-16	Bottom view for little board
Photographs	3-17	Top view for little board
Photographs	3-18	Bottom view for little board
Photographs	3-19	Transformer T100 view
Photographs	3-20	Transformer T100 view
Photographs	3-21	Transformer T100 view
Photographs	3-22	Transformer T100 view
Photographs	3-23	Transformer T100 view
Photographs	3-24	Transformer T100 view
Photographs	3-25	Transformer T100 view
Photographs	3-26	Transformer T100 view
Photographs	3-27	Transformer T100 view
Photographs	3-28	Transformer T400 view
Photographs	3-29	Transformer T400 view
Photographs	3-30	Transformer TD700 view
Photographs	3-31	Transformer TD700 view

Diagrams	4-01	Transformer T100 spec
Diagrams	4-02	Transformer TD700 spec
Diagrams	4-03	Transformer T400 spec
PCB	5-01	PCB Layout (power board) for models MR484000SG1XXXXXX
PCB	5-02	PCB Layout (power board) for models MR484000SG2XXXXXX
PCB	5-03	PCB Layout (Control board)
PCB	5-04	PCB Layout (little board)

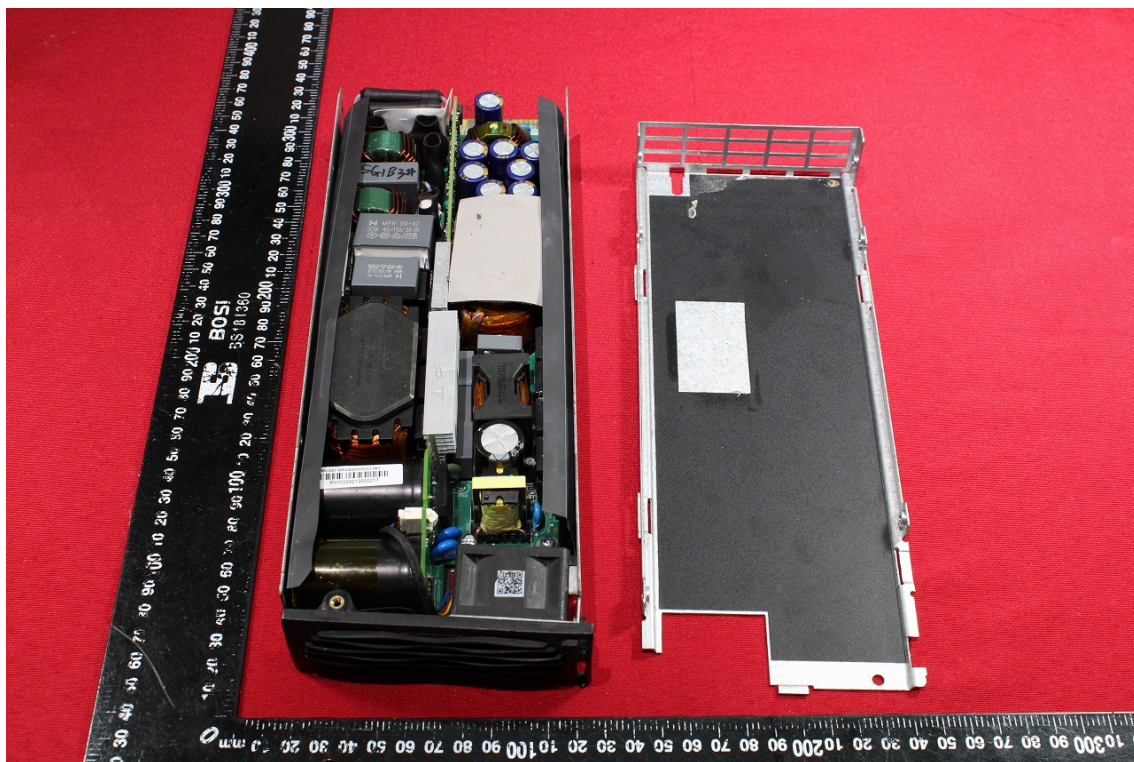
Photographs 3-01



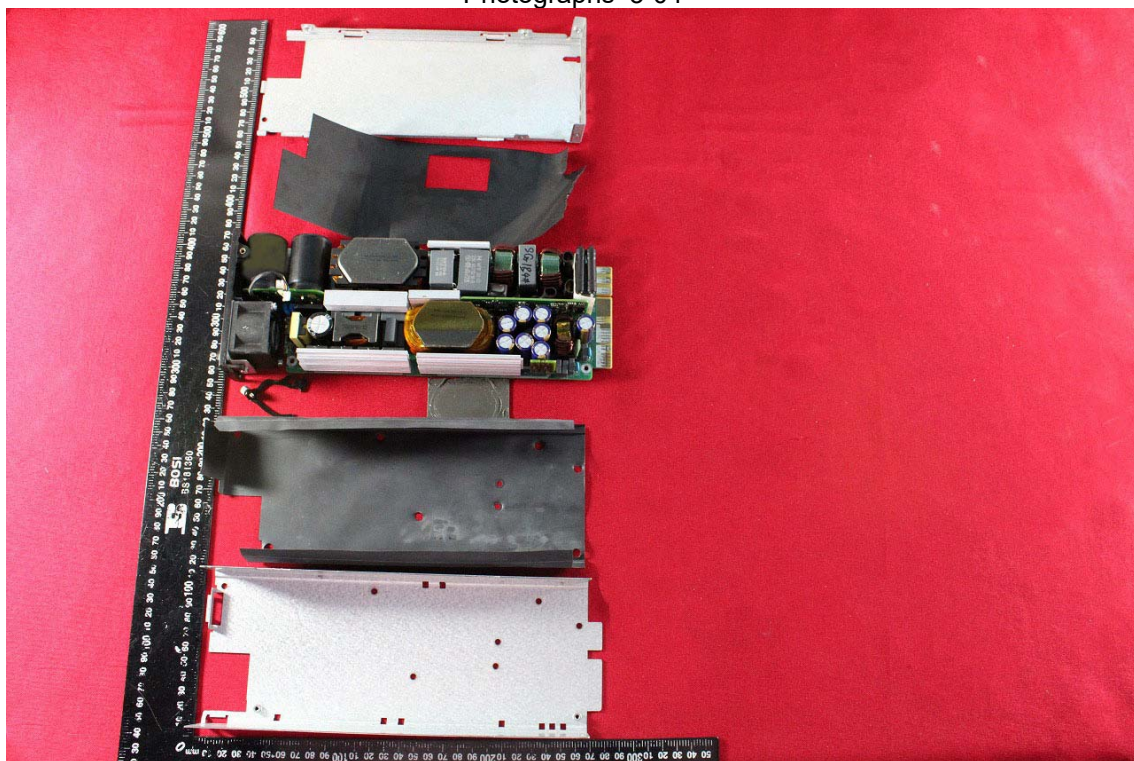
Photographs 3-02



Photographs 3-03



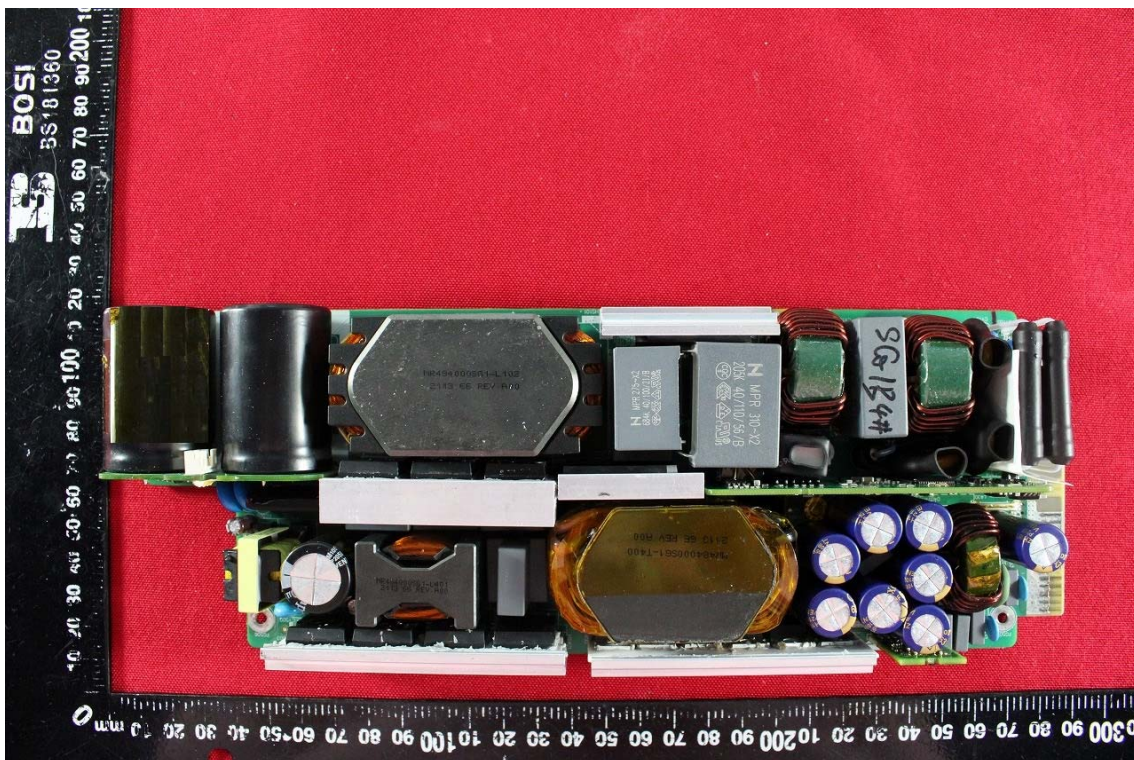
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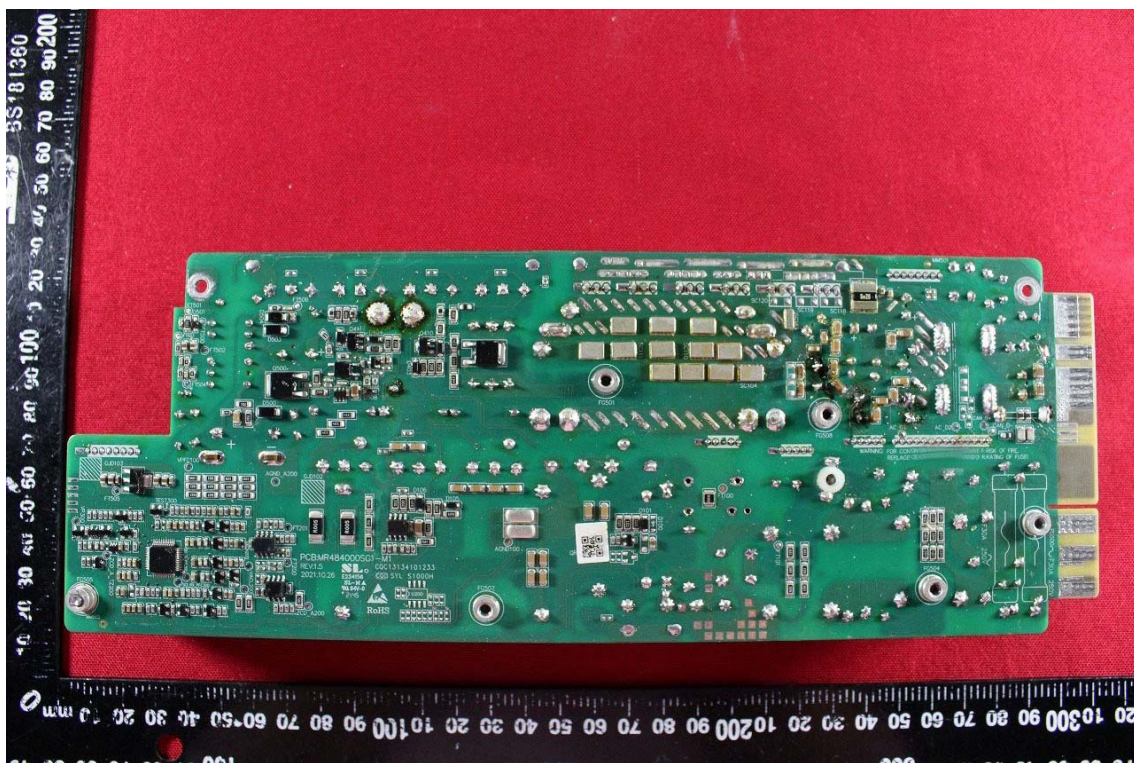
Photographs 3-05



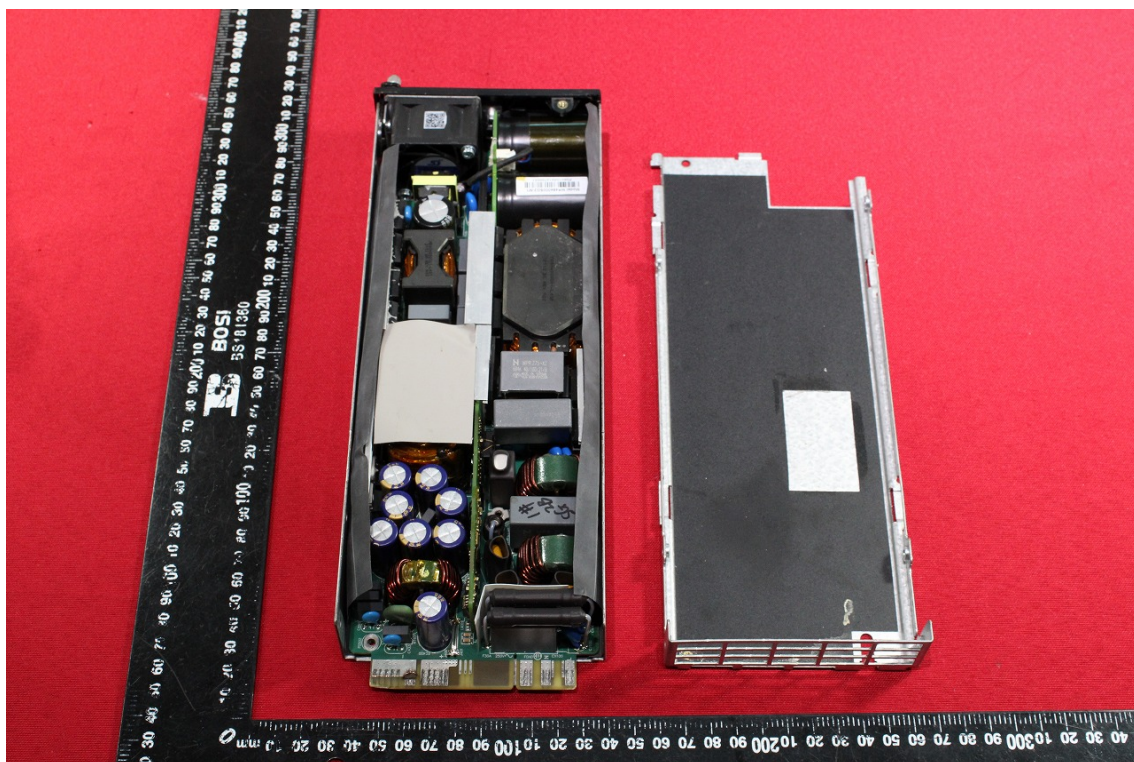
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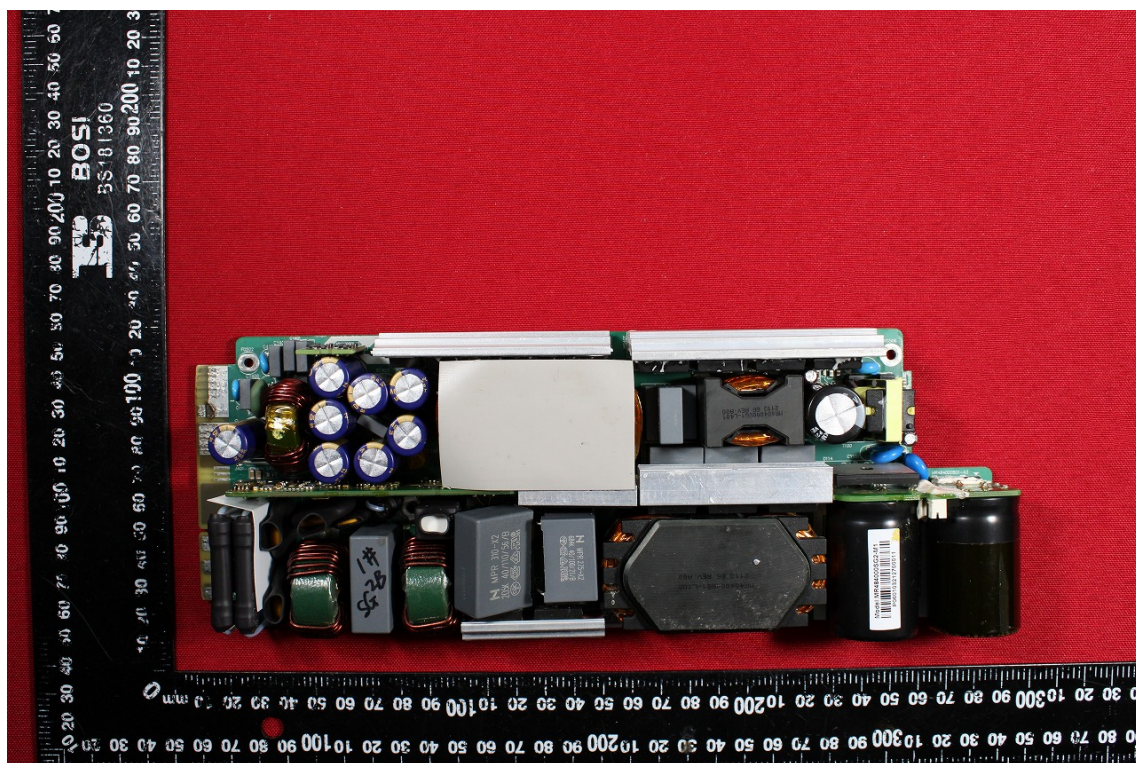
Photographs 3-07



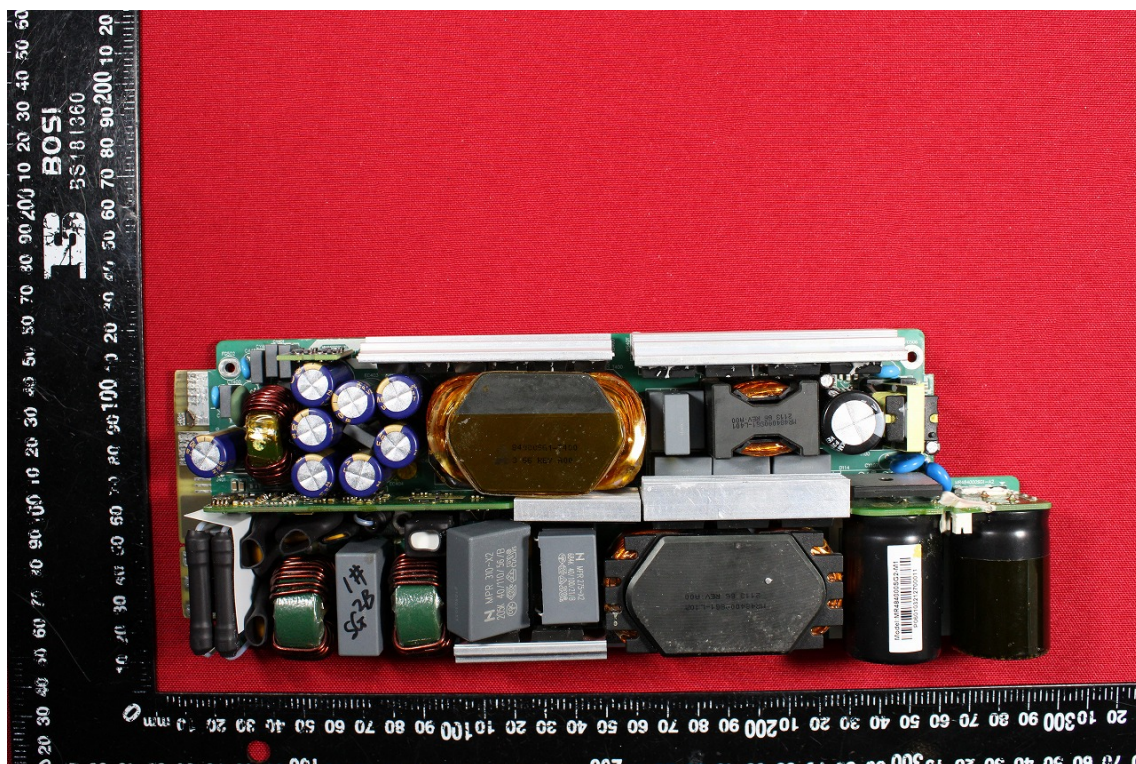
Photographs 3-08



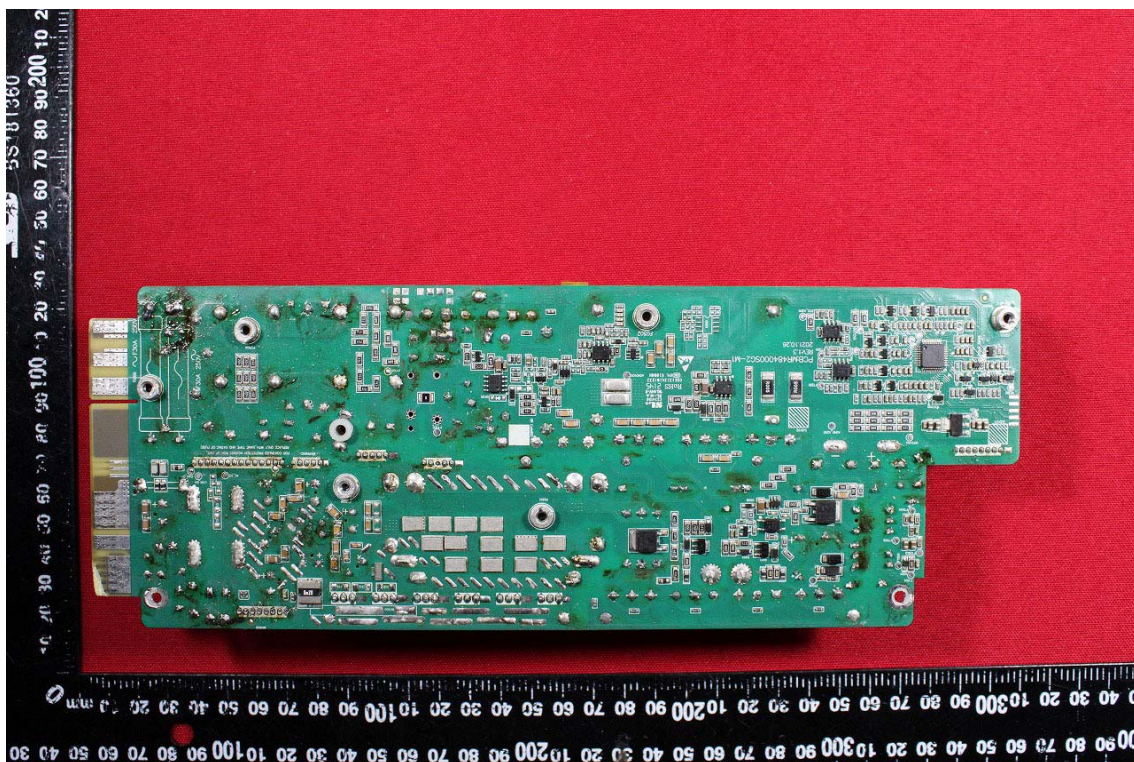
Photographs 3-09



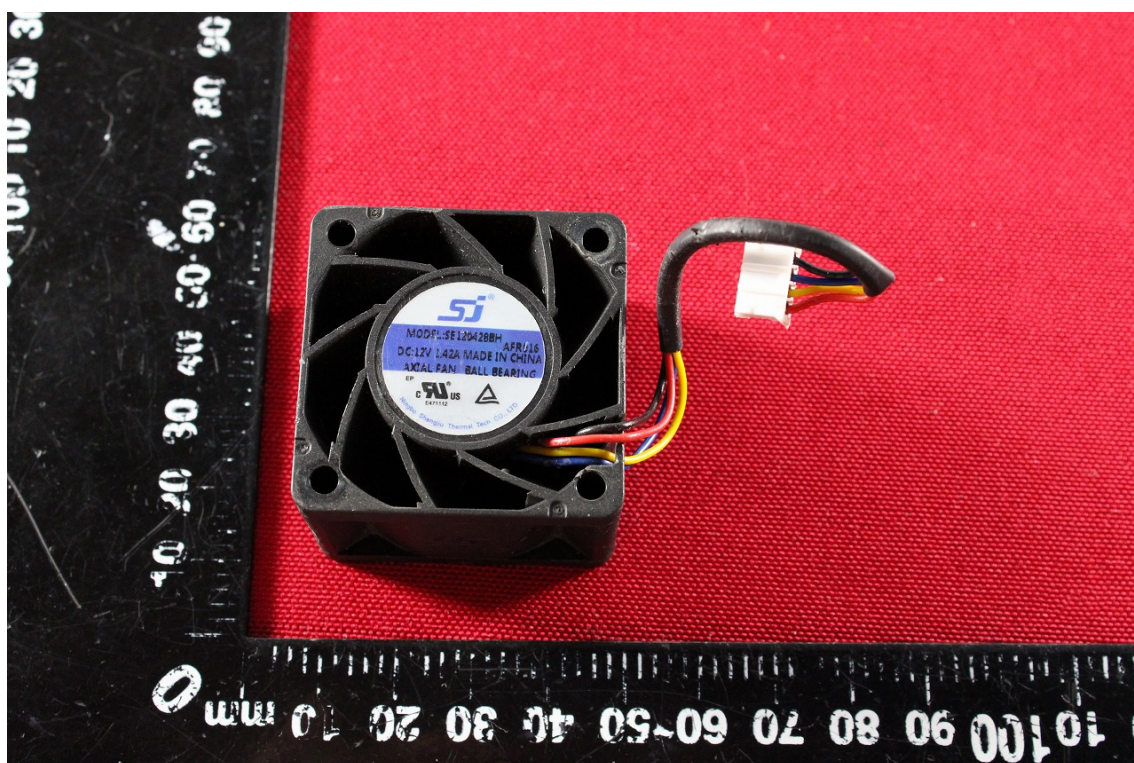
Photographs 3-10



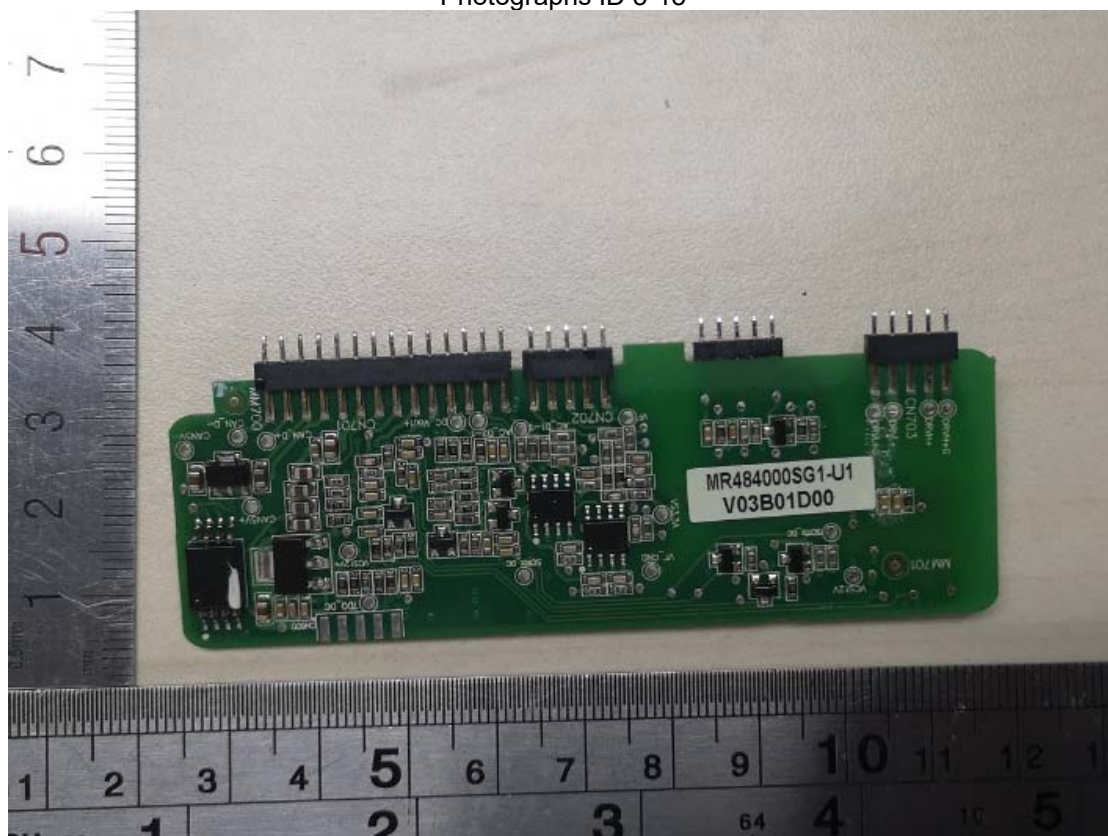
Photographs 3-11



Photographs 3-12



Photographs ID 3-13



Photographs ID 3-14



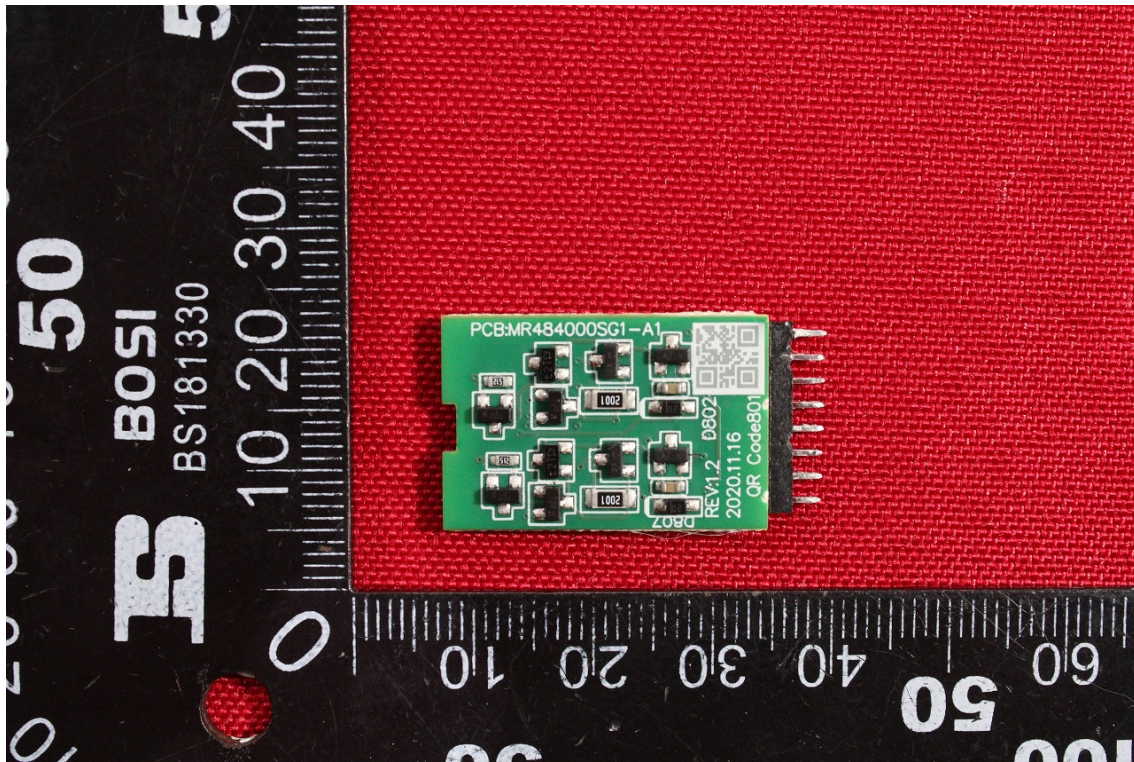
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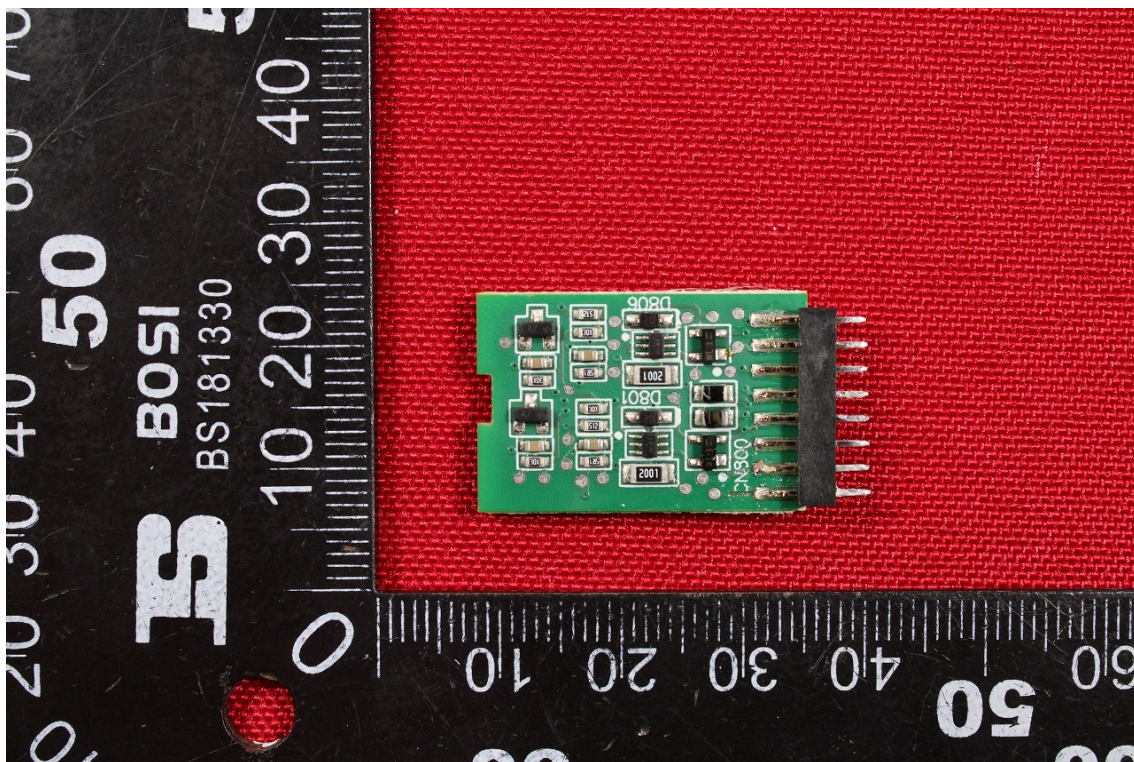
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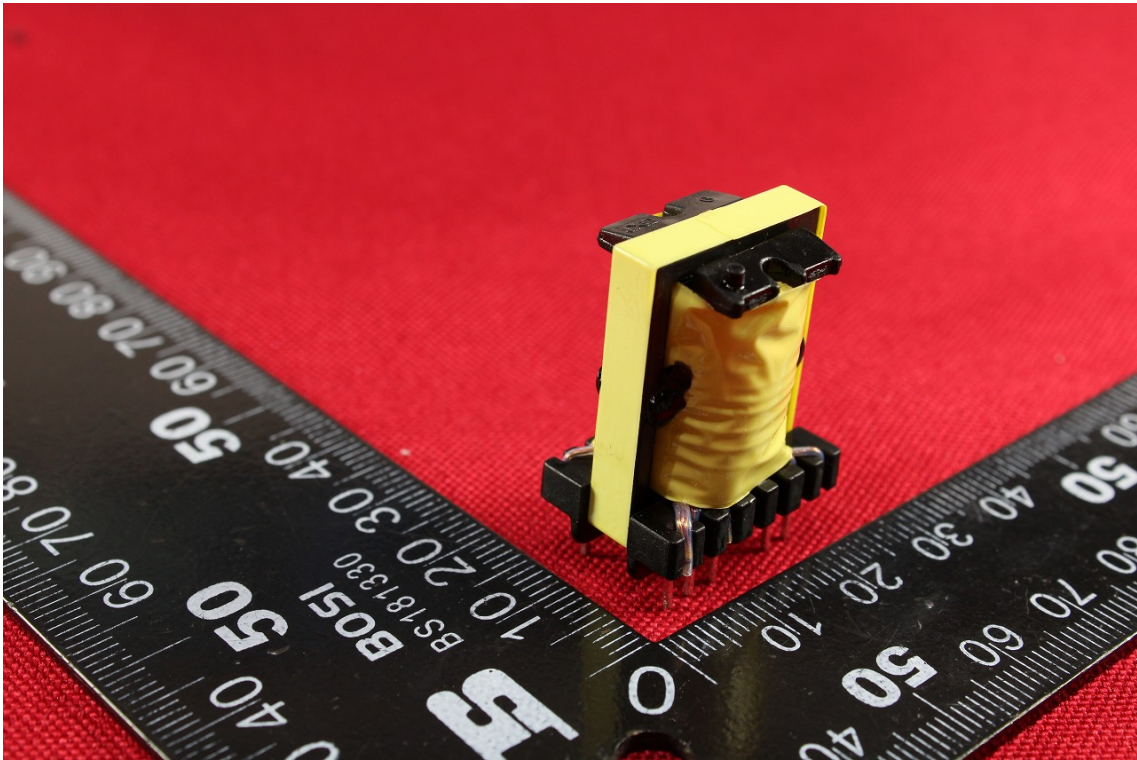
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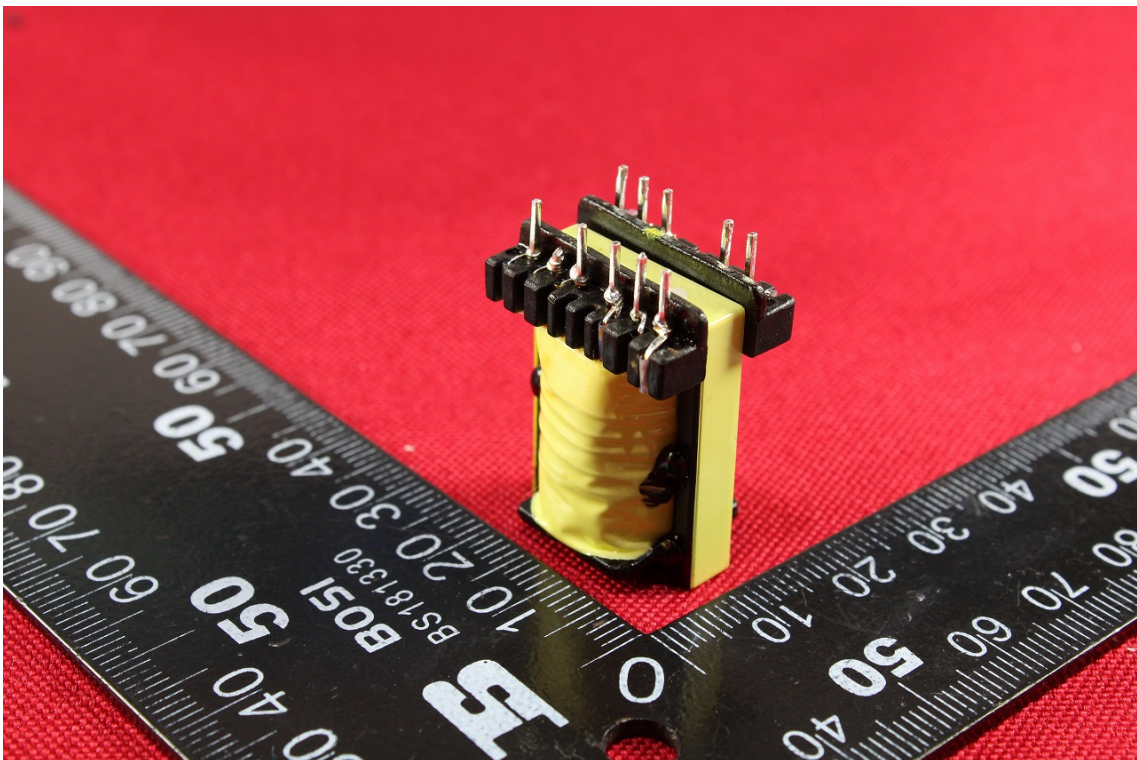
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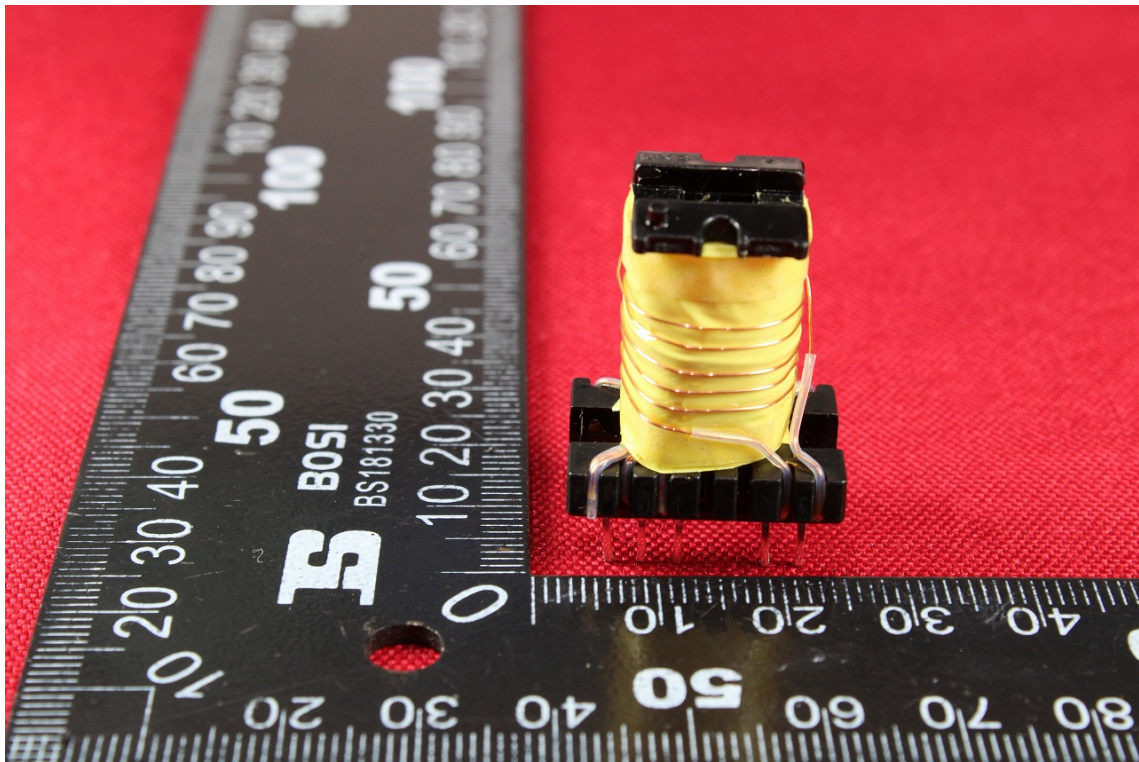
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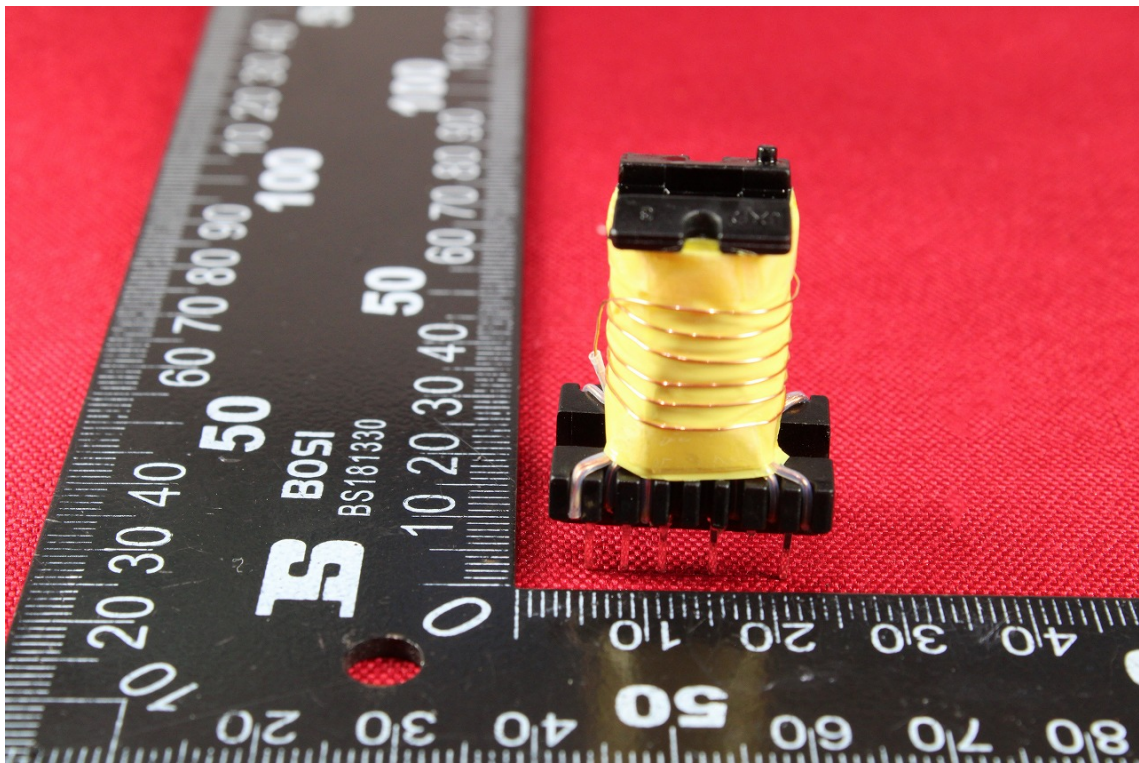
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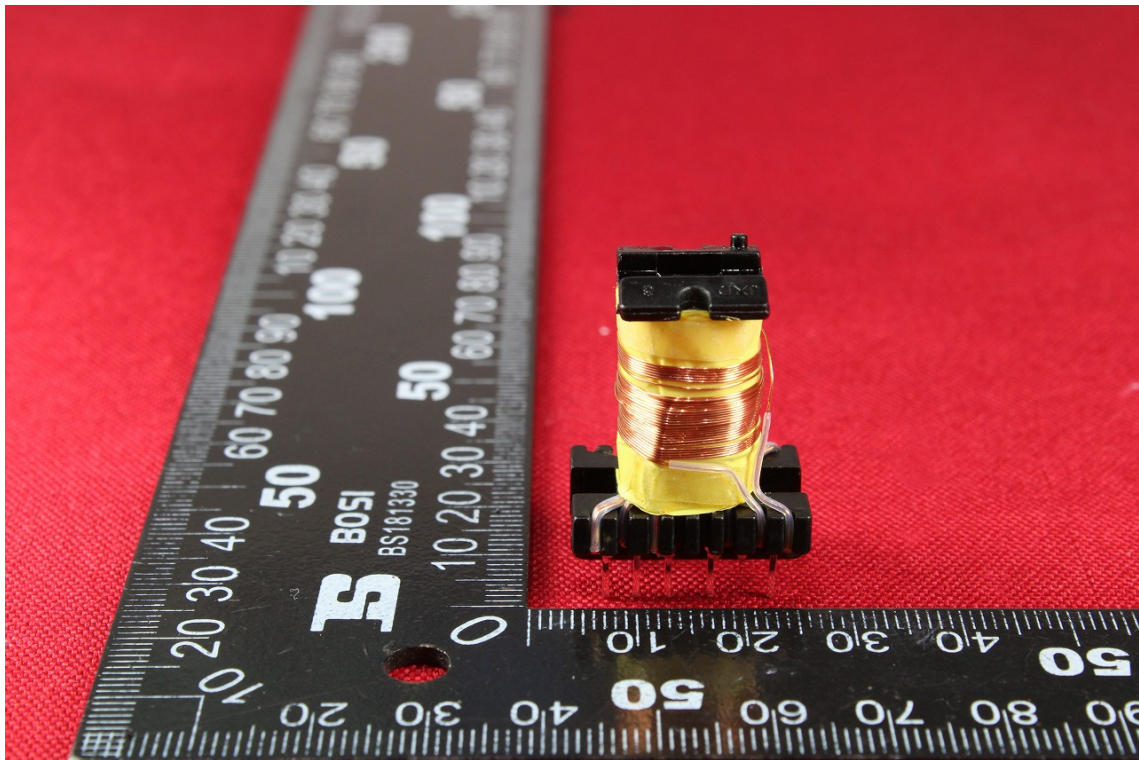
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Photographs ID 3-22



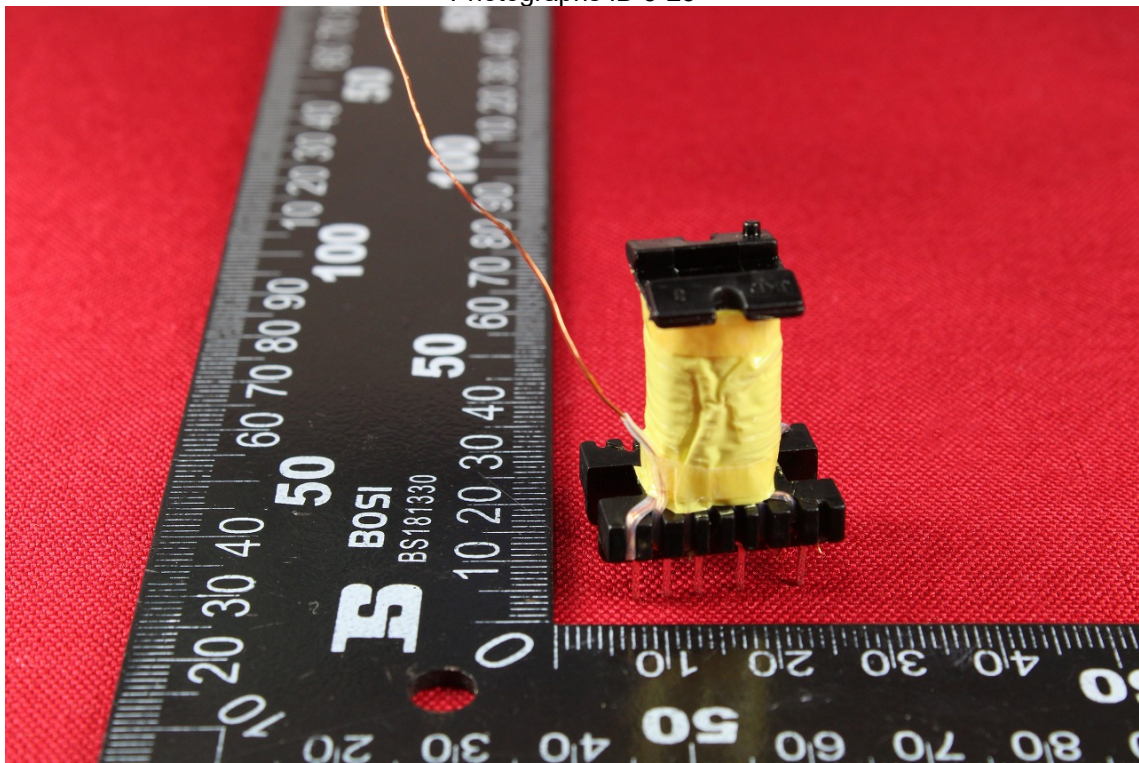
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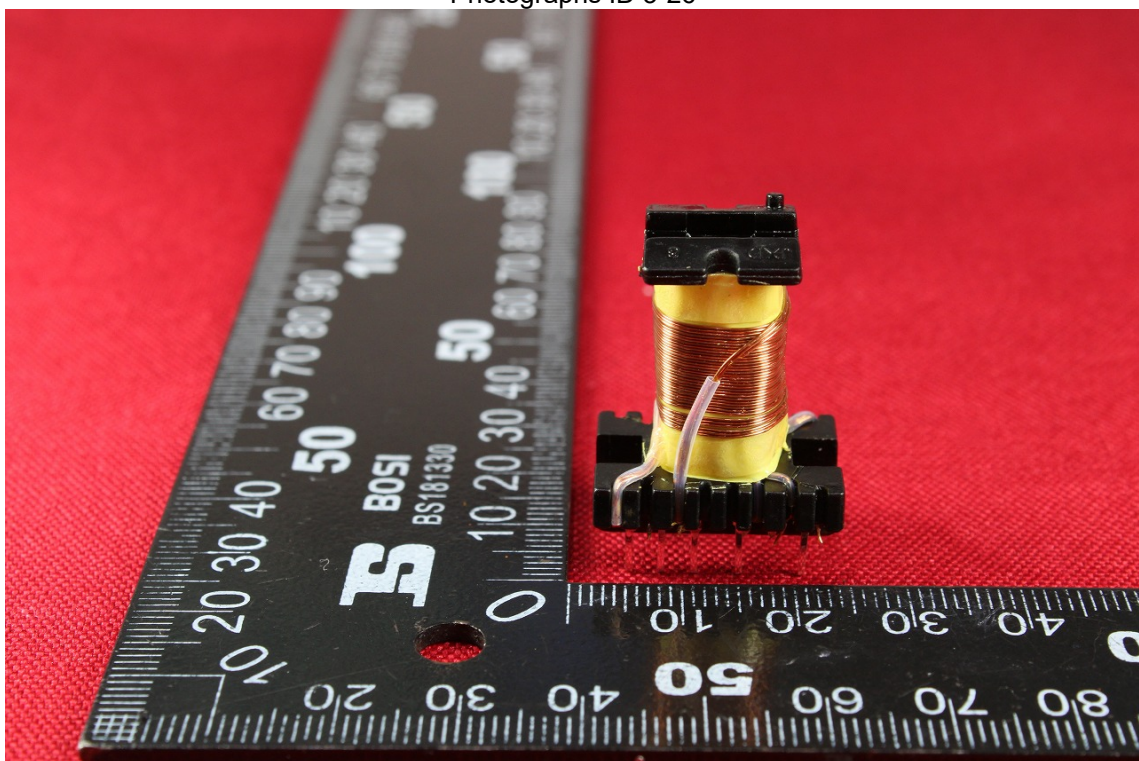
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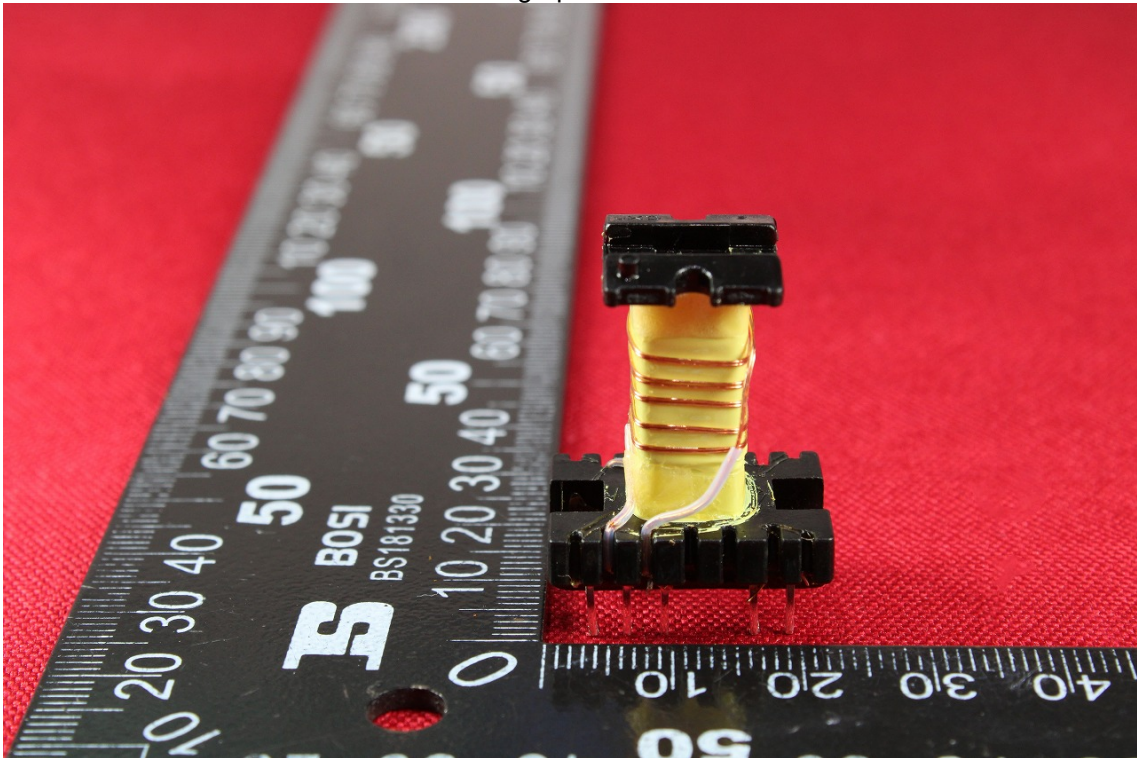
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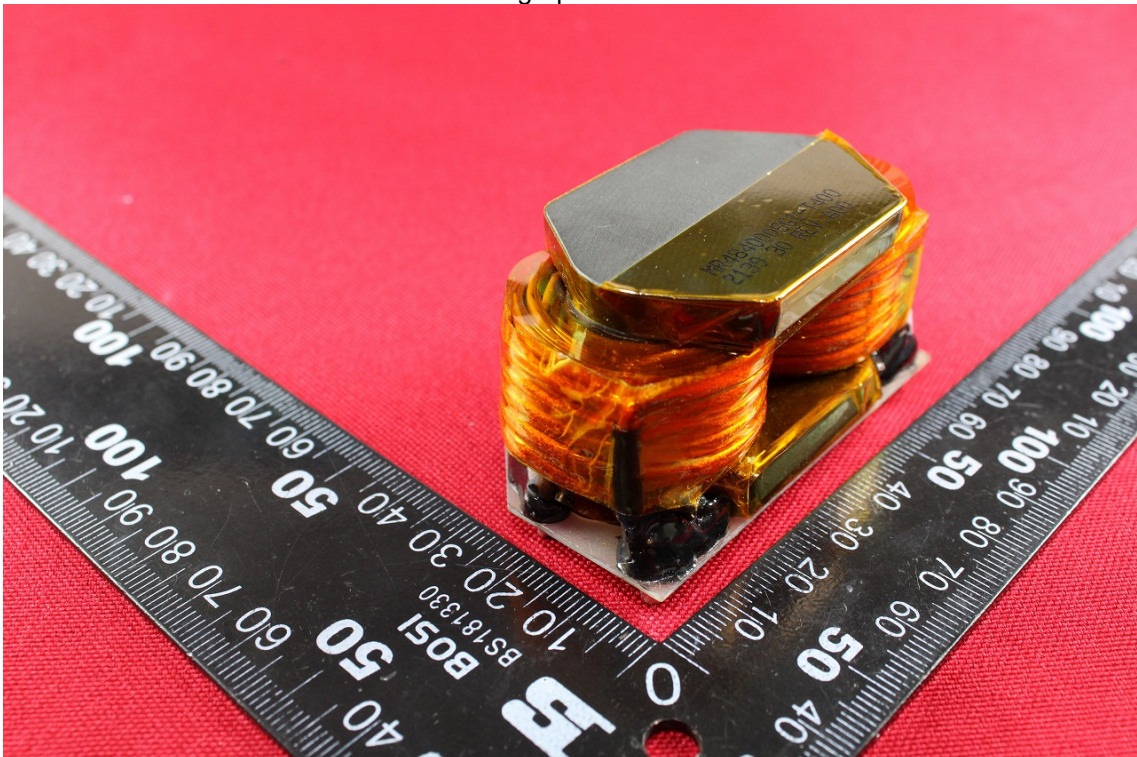
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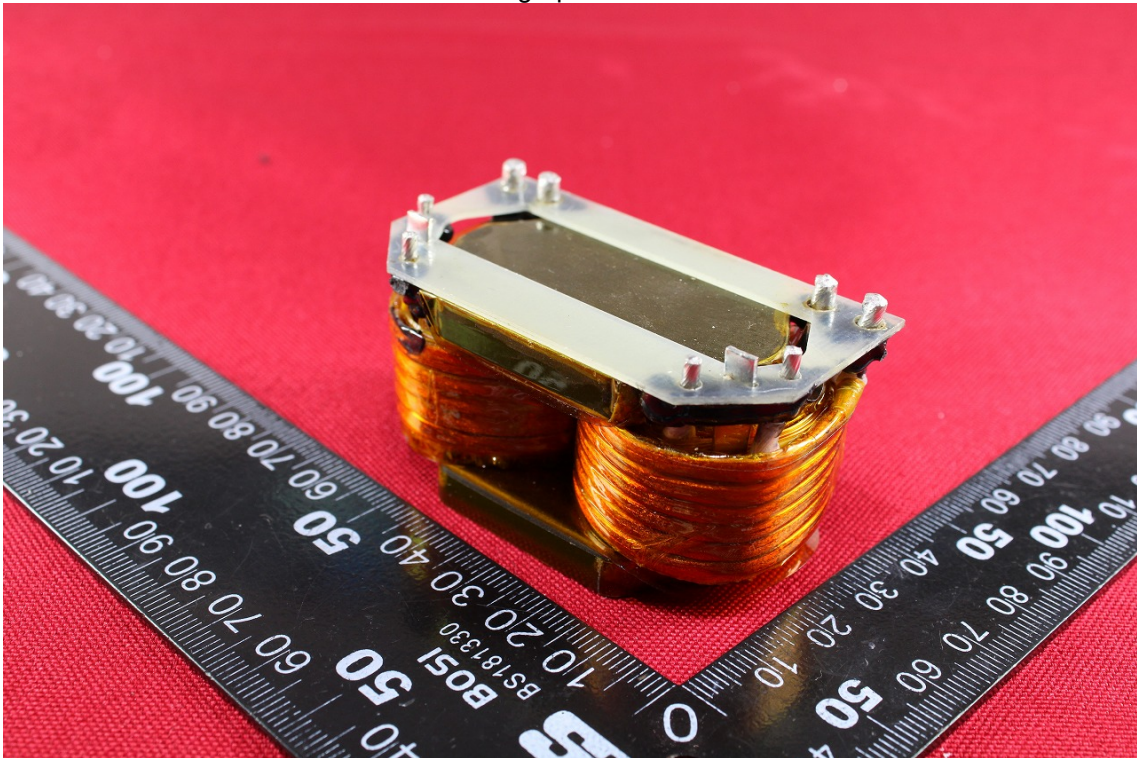
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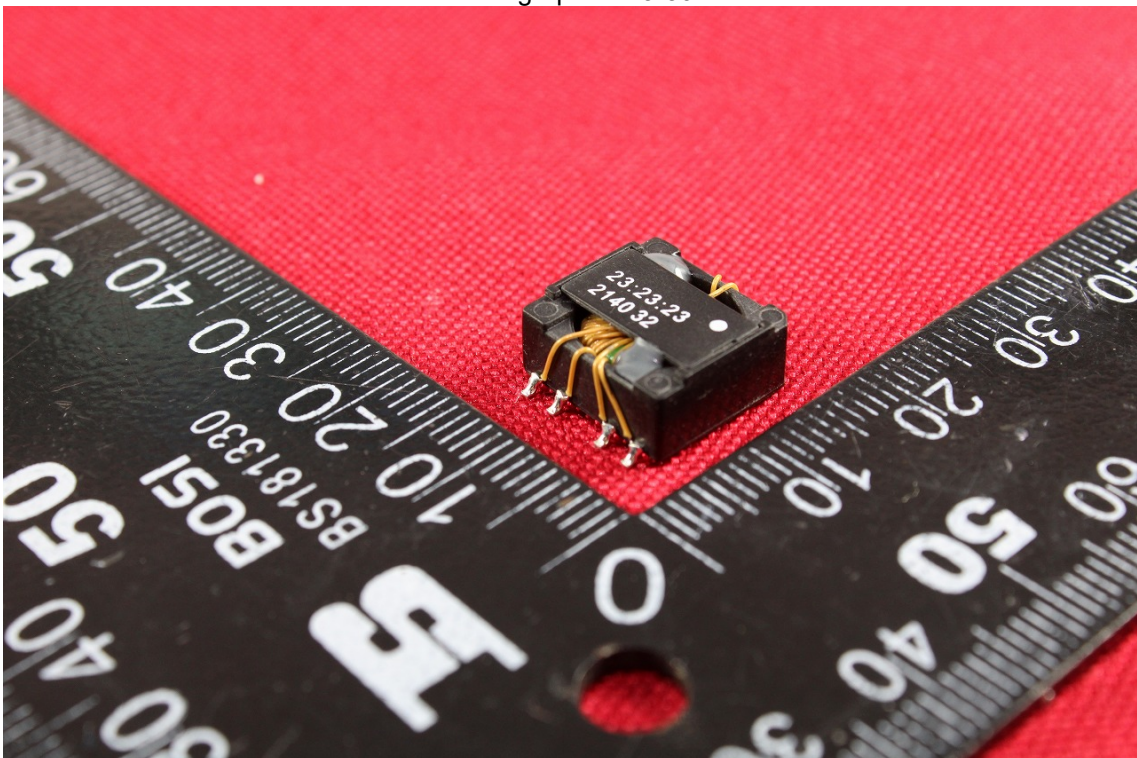
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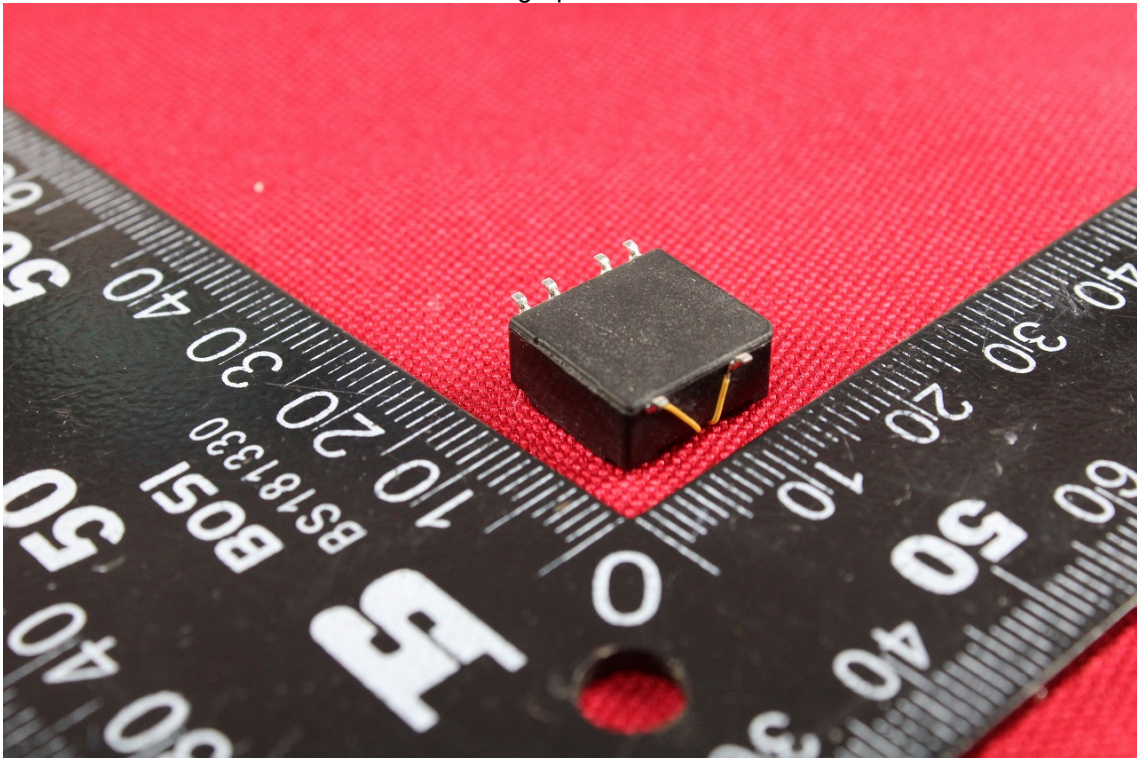
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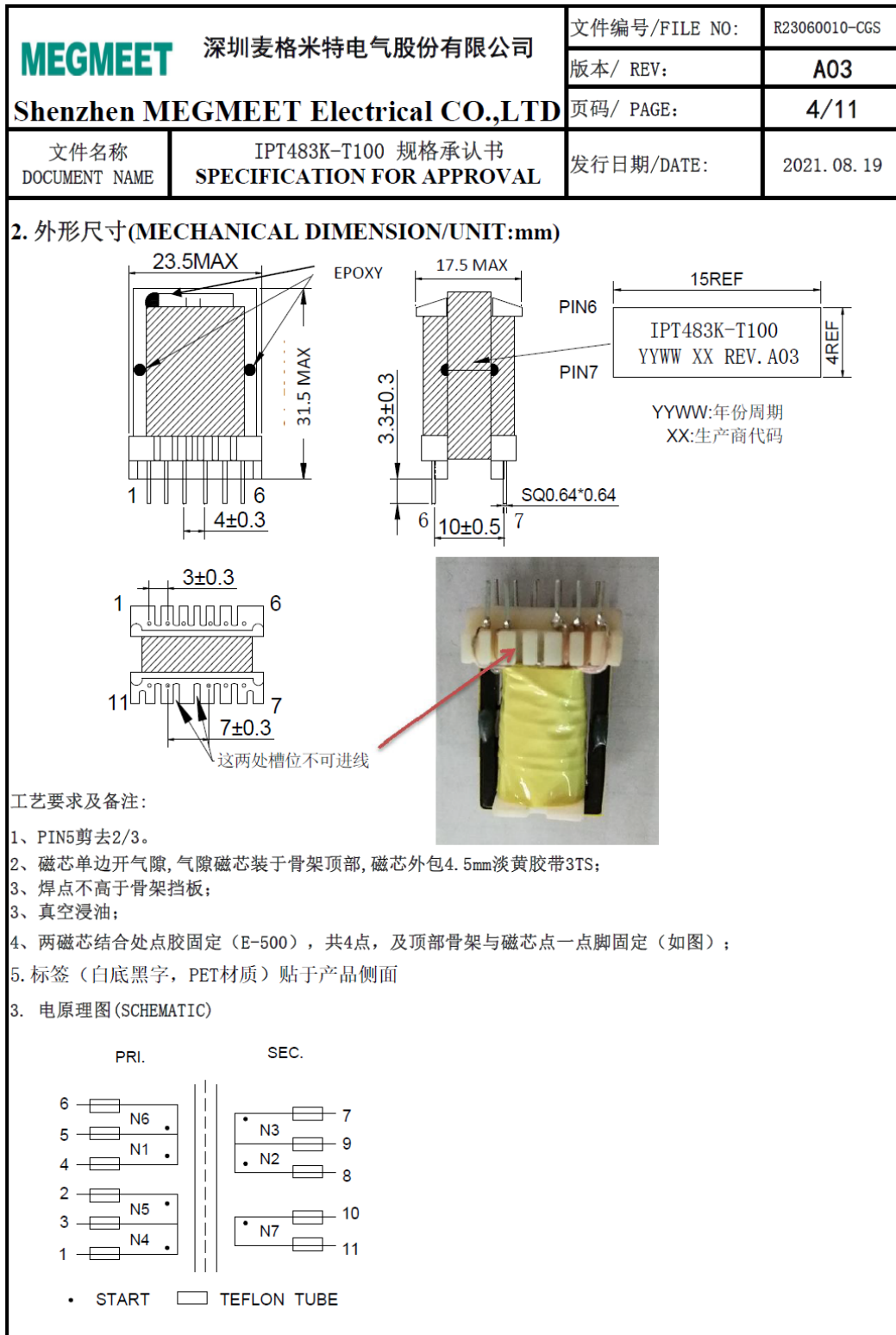
Photographs ID 3-30



Photographs ID 3-31



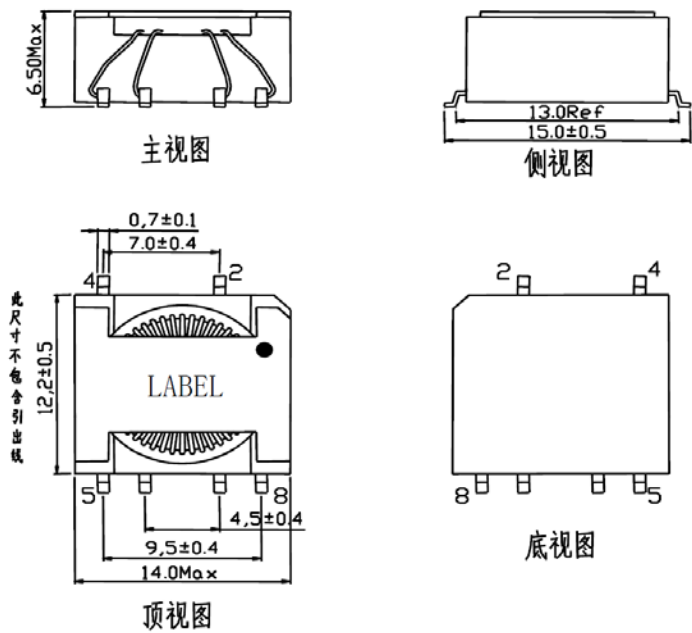
Diagrams 4-01



MEGMEET 深圳麦格米特电气股份有限公司 Shenzhen MEGMEET Electrical CO.,LTD				文件编号/FILE NO:		R23060010-CGS				
				版本/ REV:		A03				
				页码/ PAGE:		5/11				
文件名称 DOCUMENT NAME		IPT483K-T100 规格承认书 SPECIFICATION FOR APPROVAL		发行日期/DATE:		2021. 08. 19				
4. 绕线明细表(WINDING STEPS)										
Winding No (组别)	Margin Tape (挡墙胶带)		Wire 线材	Turns 圈数	端子/PIN		绝缘胶带 TAPE (黄色)	套管/Tube		备注NOTE
	PIN 侧	TOP 侧			入IN	出OUT		入IN	出OUT	
							1TS			1
N1	3.5mm	3.5mm	2UEW-F Φ 0.22mm*1P	40TS	4	5	3TS	TFT	TFT	2, 3
N2	3.5mm	3.5mm	2UEW-F Φ 0.22mm*2P	6TS	8	9	1TS	TFT	TFT	2, 3
N3	3.5mm	3.5mm	2UEW-F Φ 0.25mm*3P	13TS	7	9	3TS	TFT	TFT	2, 4
N4	3.5mm	3.5mm	2UEW-F Φ 0.25mm*3P	13TS	1	3	1TS	TFT	TFT	2, 4
N5	3.5mm	3.5mm	2UEW-F Φ 0.22mm*2P	6TS	2	3	2TS	TFT	TFT	2, 3
N6	3.5mm	3.5mm	2UEW-F Φ 0.22mm*1P	40TS	5	6	3TS	TFT	TFT	2, 3
N7	3.5mm	3.5mm	2UEW-F Φ 0.22mm*1P	6TS	10	11	3TS	TFT	TFT	2, 3
注解: 1、骨架打底一层胶带; 2、使用漆包线; 3、均绕一层; 4、密绕一层;										
5. 内部结构图(WINDING CONSTRUCTION)										
绕线方向: 所有绕组从PIN侧看为逆时针绕制 										

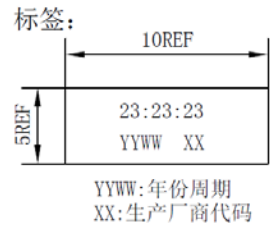
MEGMEET 深圳麦格米特电气股份有限公司 Shenzhen MEGMEET Electrical CO. LTD		文件编号/FILE NO:	M-3-WI-ENG-175
		版本/ REV: A01	A00
		页码/ PAGE:	4/11
文件名称 DOCUMENT NAME	MR483000HG-TD700 规格承认书 SPECIFICATION FOR APPROVAL	发行日期/DATE:	2018. 7. 13

2. 外形尺寸(MECHANICAL DIMENSION/UNIT:mm)

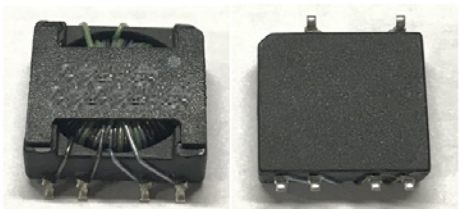
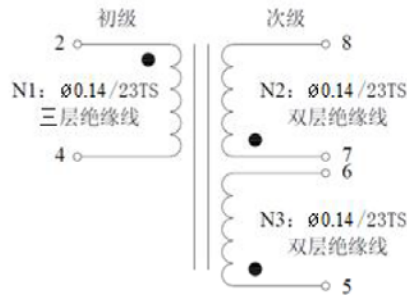


工艺要求及备注:

- (1) PIN1对应骨架顶部打极性点标示, 顶部视图逆时针数针脚;
- (2) 针脚共面度不大于0.15mm, 且上锡光亮, 可焊性好;
- (3) 产品不浸漆, 无铅制程;
- (4) N1, N2, N3绕组为并绕;
- (5) 顶盖与骨架之间点胶固定;
- (6) 标签内容喷印于顶部(如图所示)

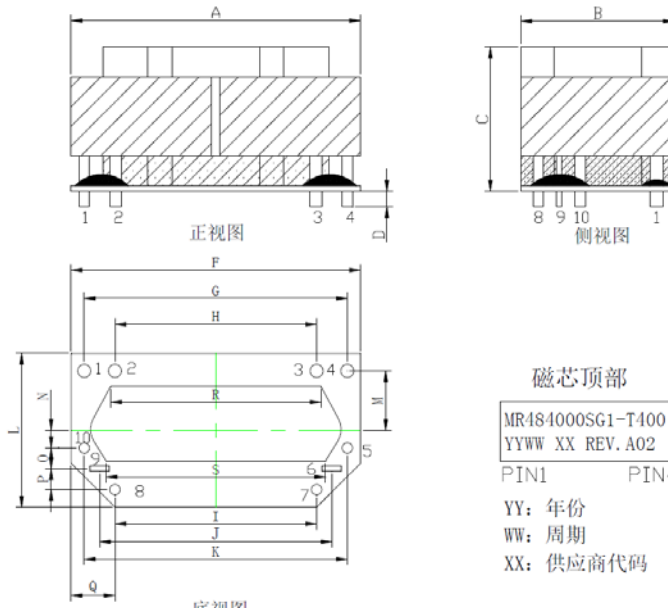
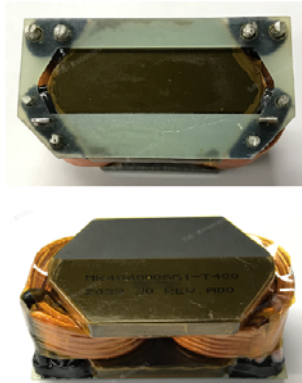
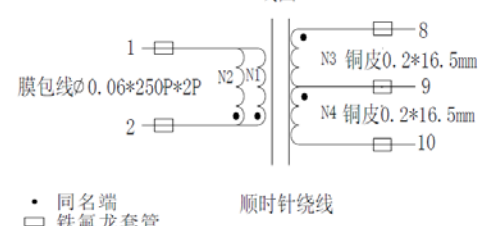
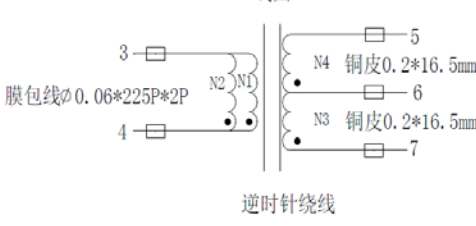


3. 电原理图(SCHEMATIC)



MEGMEET 深圳麦格米特电气股份有限公司 Shenzhen MEGMEET Electrical CO. LTD					文件编号/FILE NO:		M-3-WI-ENG-175			
					版本/ REV:		A00			
					页码/ PAGE:		5/11			
文件名称 DOCUMENT NAME		MR483000HG-TD700 规格承认书 SPECIFICATION FOR APPROVAL			发行日期/DATE:		2018. 7. 13			
4. 绕线明细表(WINDING STEPS)										
Winding No (组别)	Margin Tape (挡墙胶带)		Wire 线材	Turns 圈数	端子/PIN		绝缘胶带 TAPE	套管/Tube		备注NOTE
					入IN	出OUT		入IN	出OUT	
N1	/	/	W35T1.5PXXTC1A Φ0.14*1P	23	2	4	/	无	无	1. 2. 3. 4
N2	/	/	W35D1.5PXXTC1A Φ0.14*1P	23	7	8	/	无	无	
N3	/	/	W35D1.5PXXTC1A Φ0.14*1P	23	5	6	/	无	无	
注解: 1、N1使用三层绝缘线; 2、N2, N3使用二层绝缘线; 3、线不可交叉打结或损伤。 4、N1, N2, N3并绕。										
5. 内部结构图(WINDING CONSTRUCTION)										
NA										

Diagrams 4-03

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		版本/ REV:	A02																																																																
		页码/ PAGE:	4/11																																																																
文件名称 DOCUMENT NAME	MR484000SG1-T400 规格承认书 SPECIFICATION FOR APPROVAL	发行日期/DATE:	2021. 12. 15																																																																
2. 外形尺寸(MECHANICAL DIMENSION/UNIT:mm)																																																																			
 磁芯顶部 MR484000SG1-T400 YYWW XX REV. A02 PIN1 PIN4 YY: 年份 WW: 周期 XX: 供应商代码 <table border="1"> <tbody> <tr><td>A</td><td>67.5MAX</td><td>1</td><td>Φ3.0+0.5/-0.3</td></tr> <tr><td>B</td><td>35.5MAX</td><td>2</td><td>Φ3.0+0.5/-0.3</td></tr> <tr><td>C</td><td>33.7MAX</td><td>3</td><td>Φ2.8+0.5/-0.3</td></tr> <tr><td>D</td><td>3.0±0.5</td><td>4</td><td>Φ2.8+0.5/-0.3</td></tr> <tr><td>F</td><td>66±0.5</td><td>5</td><td>Φ2.1±0.3</td></tr> <tr><td>G</td><td>60±0.5</td><td>6</td><td>14.2±0.3MS 9+0.3/-0.2</td></tr> <tr><td>H</td><td>46±0.5</td><td>7</td><td>Φ2.1±0.3</td></tr> <tr><td>I</td><td>46±0.5</td><td>8</td><td>Φ2.1±0.3</td></tr> <tr><td>J</td><td>53±0.5</td><td>9</td><td>14.2±0.3MS 9+0.3/-0.2</td></tr> <tr><td>K</td><td>60±0.5</td><td>10</td><td>Φ2.1±0.3</td></tr> <tr><td>L</td><td>35±0.5</td><td></td><td></td></tr> <tr><td>M</td><td>13.5±0.5</td><td>R</td><td>48±0.5</td></tr> <tr><td>N</td><td>4±0.5</td><td>S</td><td>50±0.5</td></tr> <tr><td>O</td><td>4.75±0.5</td><td></td><td></td></tr> <tr><td>P</td><td>4.75±0.5</td><td></td><td></td></tr> <tr><td>Q</td><td>10±0.5</td><td></td><td></td></tr> </tbody> </table>				A	67.5MAX	1	Φ3.0+0.5/-0.3	B	35.5MAX	2	Φ3.0+0.5/-0.3	C	33.7MAX	3	Φ2.8+0.5/-0.3	D	3.0±0.5	4	Φ2.8+0.5/-0.3	F	66±0.5	5	Φ2.1±0.3	G	60±0.5	6	14.2±0.3MS 9+0.3/-0.2	H	46±0.5	7	Φ2.1±0.3	I	46±0.5	8	Φ2.1±0.3	J	53±0.5	9	14.2±0.3MS 9+0.3/-0.2	K	60±0.5	10	Φ2.1±0.3	L	35±0.5			M	13.5±0.5	R	48±0.5	N	4±0.5	S	50±0.5	O	4.75±0.5			P	4.75±0.5			Q	10±0.5		
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2. 工艺要求及备注: 1、用一层0.18mmNomexx纸1层+0.025mm高温胶带1层做骨架; 2、磁芯垫气隙调节电感,磁芯端面点环氧胶; 3、底部磁芯用0.025mm的高温胶带背胶二层,反包高度5mm 以上; 4、顶部磁芯PIN1-4脚侧磁芯用0.025mm高温胶带反包二层; 5、一个产品两个线包,两个产品组装成一个成品; 6、产品立式安装在底板上,磁芯与底板、引脚点黑色环氧胶固定; 7、产品不浸漆,标签贴或喷印在磁芯顶部; 8、磁芯和线包不超过底板。																																																																			
																																																																			
3. 电原理图 (SCHEMATIC)																																																																			
PRI 线圈1 SEC  膜包线0.06*250P*2P • 同名端 □ 铁氟龙套管 顺时针绕线 PRI 线圈2 SEC  膜包线0.06*225P*2P 逆时针绕线																																																																			

MEGMEET 深圳麦格米特电气股份有限公司 Shenzhen MEGMEET Electrical CO.,LTD		文件编号/FILE NO:	R23012319-CGS
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4. 绕线明细表(WINDING STEPS)

线圈一

Winding No (组别)	Margin Tape (擋牆膠帶)		Wire 线材	Turns 圈數	端子/PIN		绝缘胶带 TAPE	套管/Tube		备注 NOTE
					入IN	出OUT		入IN	出OUT	
N1			膜包线 2UEWH $\phi 0.06 \times 250 \times 2P$	5	2	1	2	TFL	TFL	1. 2. 3
N2			铜皮0.2*16.5mm	3	8	9	2	TFL		4.5
N3			铜皮0.2*16.5mm	3	9	10			TFL	4.5
N4			膜包线 2UEWH $\phi 0.06 \times 250 \times 2P$	5	2	1	2	TFL	TFL	1. 2. 3

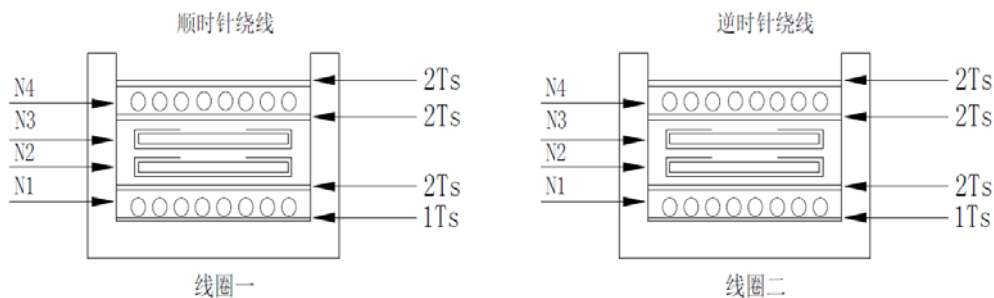
线圈二

Winding No (组别)	Margin Tape (擋牆膠帶)		Wire 线材	Turns 圈數	端子/PIN		绝缘胶带 TAPE	套管/Tube		备注 NOTE
					入IN	出OUT		入IN	出OUT	
N1			膜包线 2UEWH $\phi 0.06 \times 225 \times 2P$	6	4	3	2	TFL	TFL	1. 2. 3
N2			铜皮0.2*16.5mm	3	7	6	2	TFL		4.5
N3			铜皮0.2*16.5mm	3	6	5			TFL	4.5
N4			膜包线 2UEWH $\phi 0.06 \times 225 \times 2P$	6	4	3	2	TFL	TFL	1. 2. 3

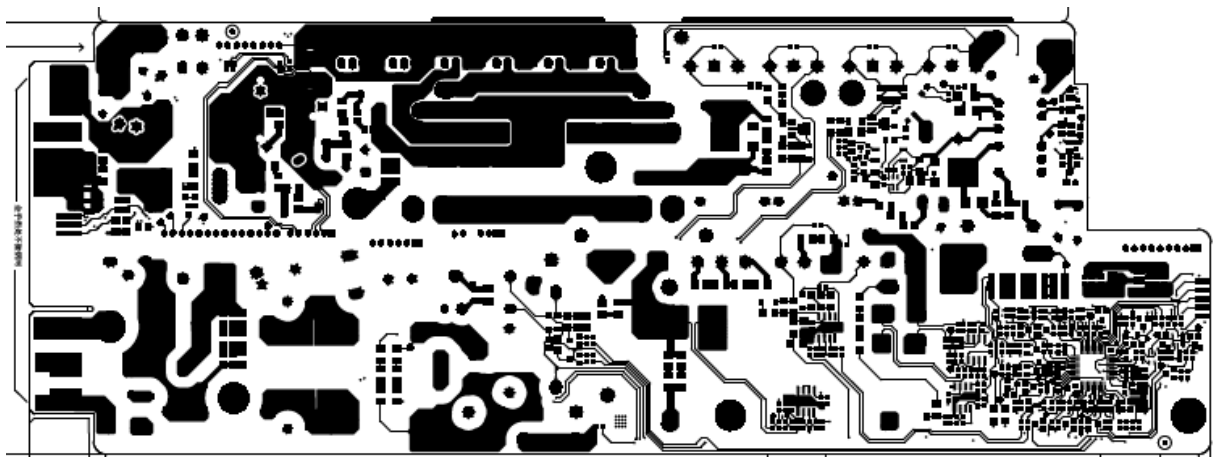
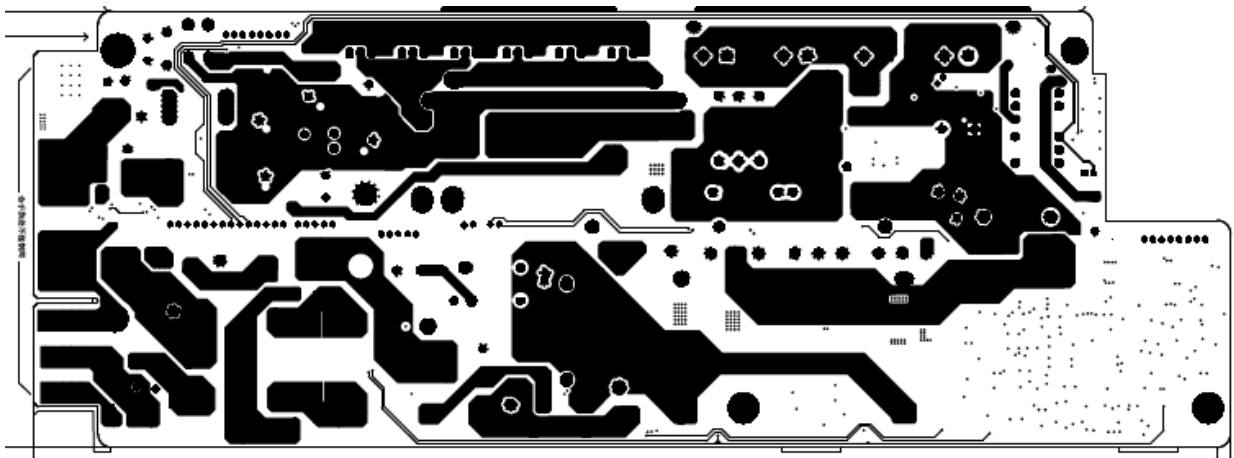
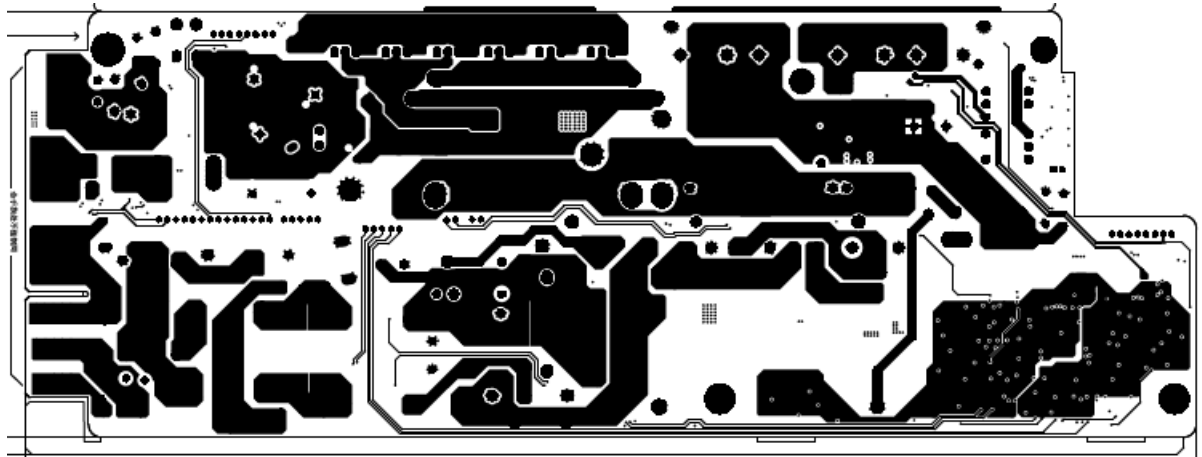
注解:

1. 产品无骨架绕制，采用nomex打底一层+外包一层胶带为骨架；
2. 使用膜包线绕制，膜包线外所包的高温胶带（胶带宽度 $\geq 8\text{mm}$ ），层与层之间须重叠1/2以上；
膜包线厂商：永志诚/ 嘉正欣 /言成；膜包线外包高温胶带厂商：亚华 杰力
3. 进出线穿铁氟龙套管
4. 铜皮用0.025mm高温胶带反包一层，反折两边5mm MIN
5. 铜皮起尾头用0.21*30*2P绞合线焊接穿铁氟龙套管引出，中心抽头焊接0.4*4*2P
铜皮并用高温胶带包二层引出；

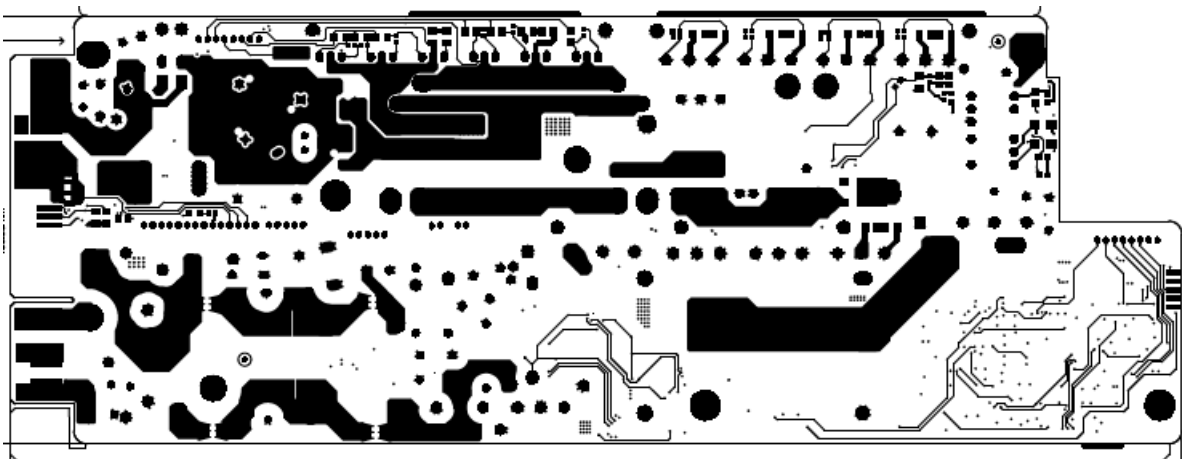
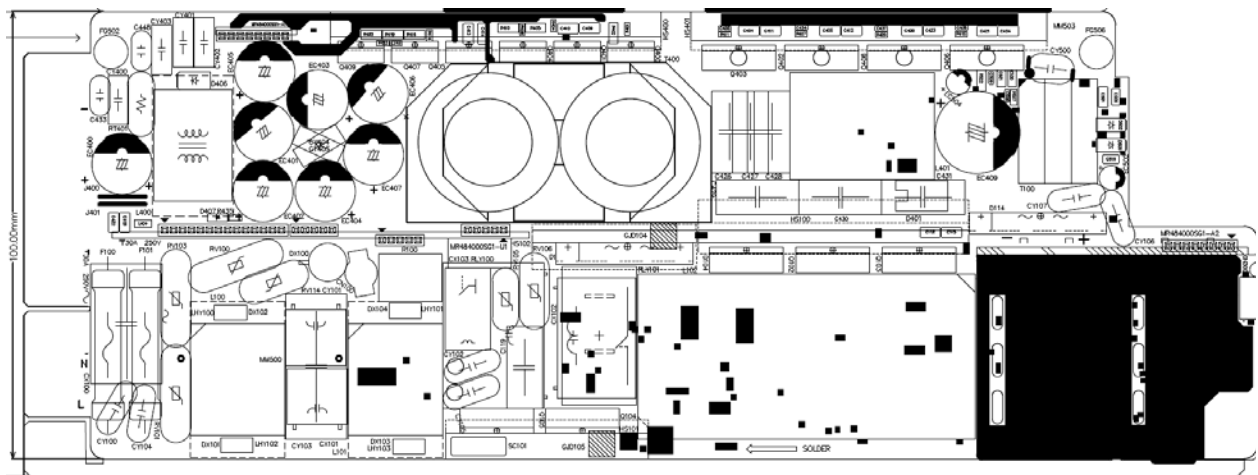
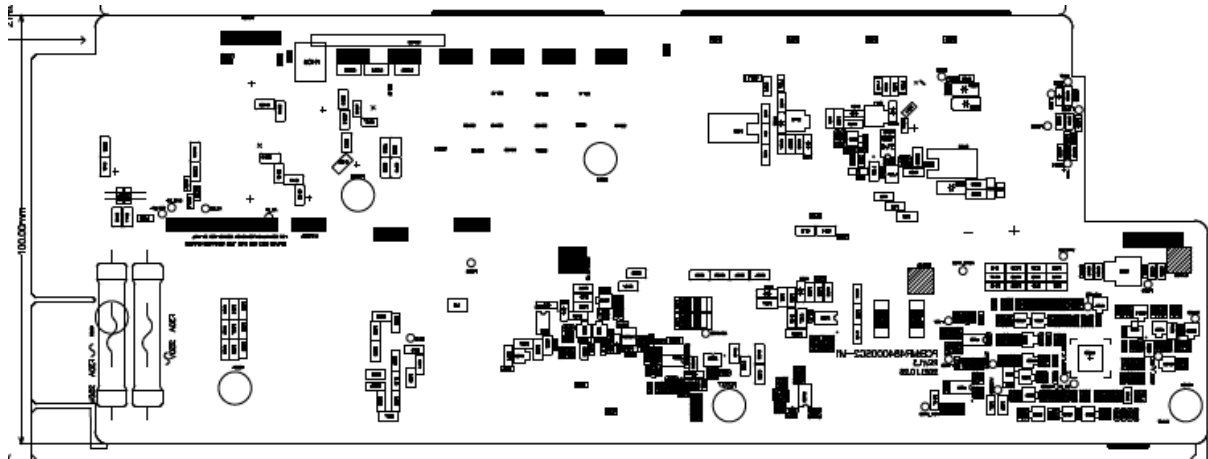
5. 内部结构图(WINDING CONSTRUCTION)



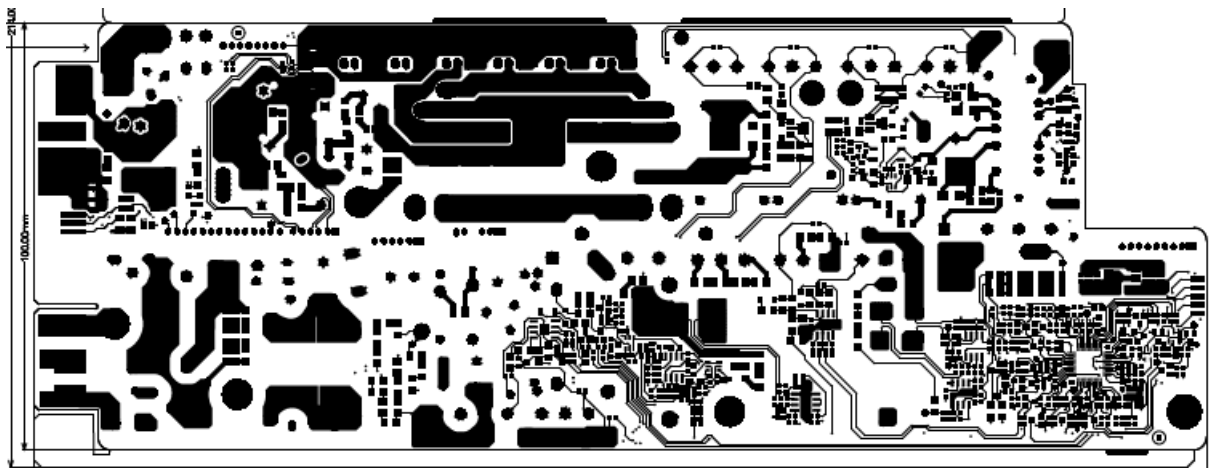
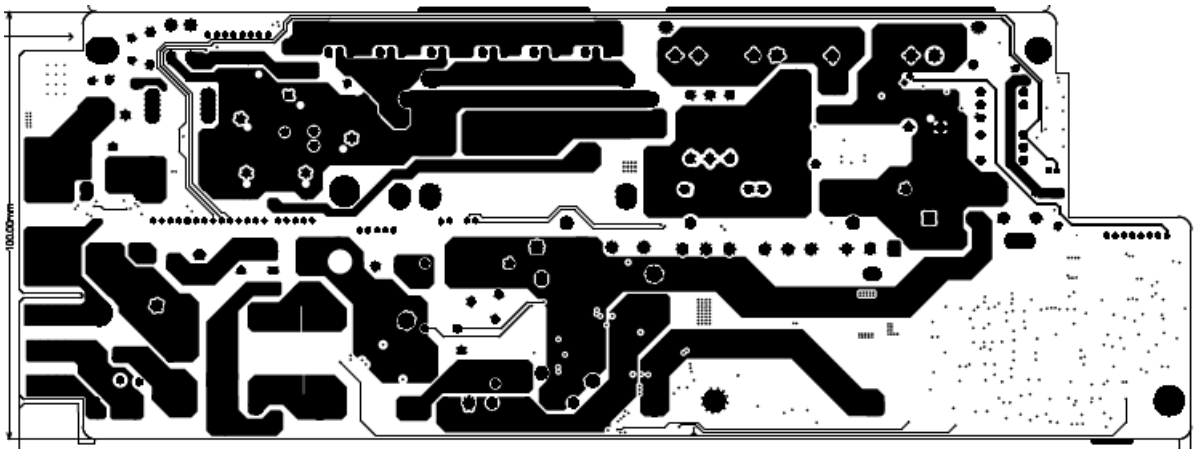
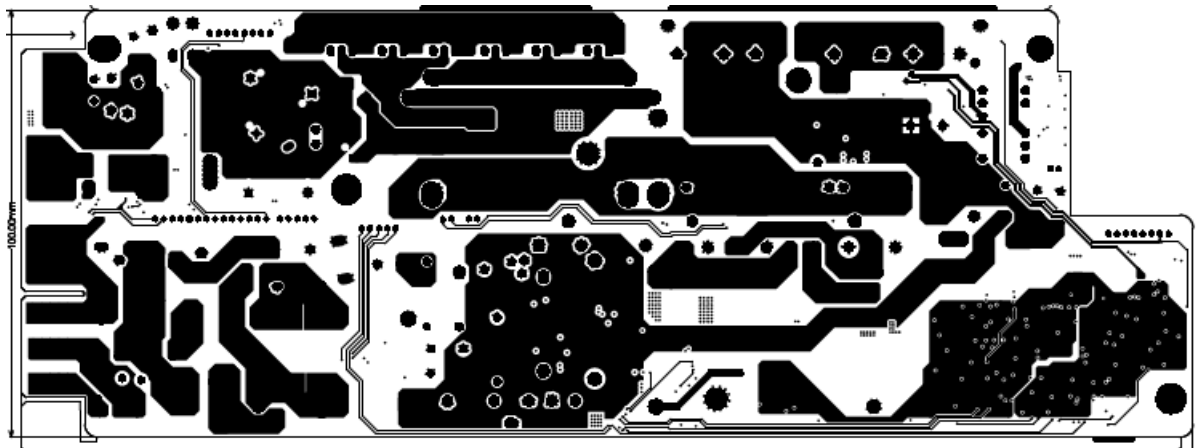
PCB 5-01



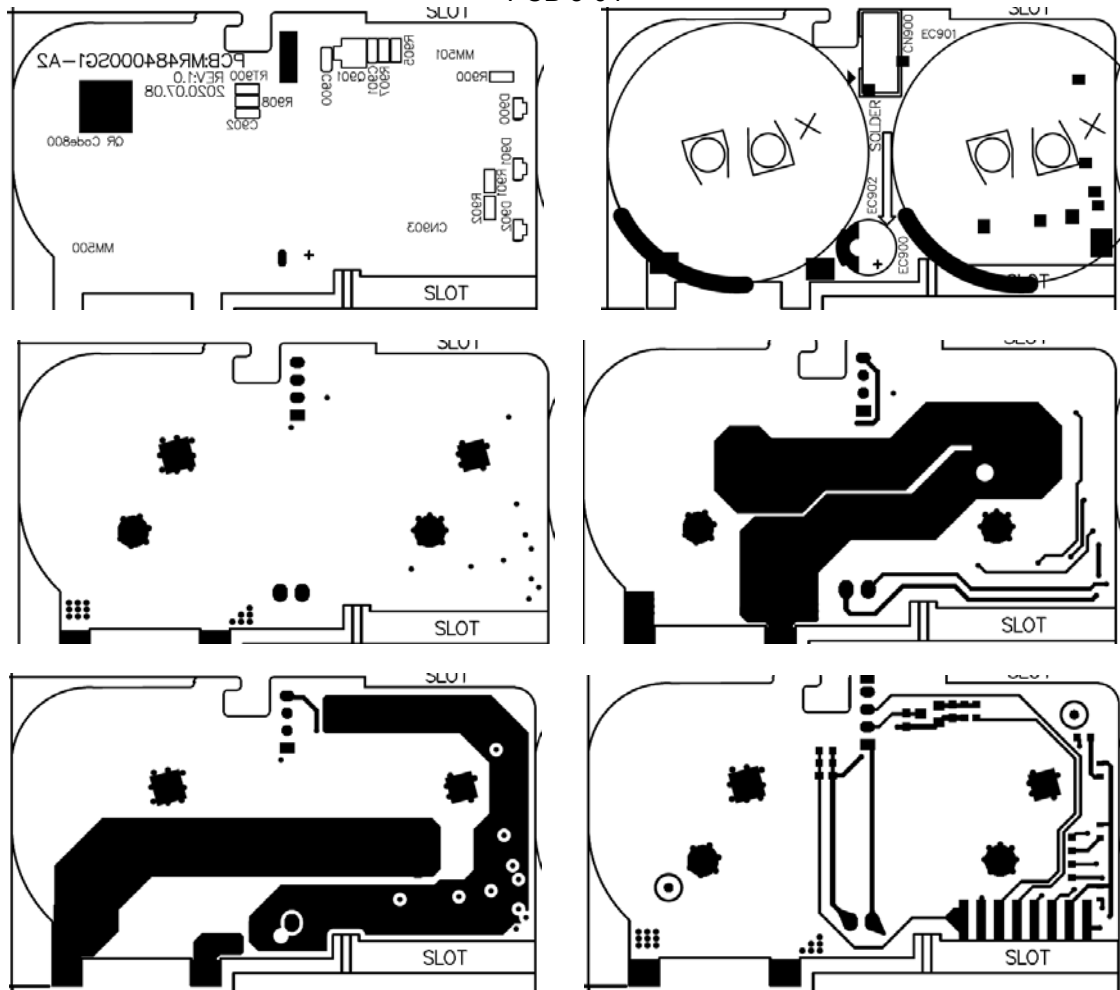
PCB 5-02



PCB 5-02



PCB 5-04



IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict

ATTACHMENT TO TEST REPORT		
IEC 62368-1		
EUROPEAN GROUP DIFFERENCES AND NATIONAL DIFFERENCES		
(Audio/video, information and communication technology equipment - Part 1: Safety requirements)		
Differences according to..... EN IEC 62368-1:2020+A11:2020		
Attachment Form No..... EU_GD_IEC62368_1E		
Attachment Originator UL(Demko)		
Master Attachment 2021-02-04		
Copyright © 2021 IEC System for Conformity Testing and Certification of Electrical Equipment (IECEE), Geneva, Switzerland. All rights reserved.		
	CENELEC COMMON MODIFICATIONS (EN)	P
	Clause numbers in the cells that are shaded light grey are clause references in EN IEC 62368-1:2020+A11:2020. All other clause numbers in that column, except for those in the paragraph below, refers to IEC 62368-1:2018. Clauses, subclauses, notes, tables, figures and annexes which are additional to those in IEC 62368-1:2018 are prefixed “Z”.	P
	Add the following annexes: Annex ZA (normative) Normative references to international publications with their corresponding European publications Annex ZB (normative) Special national conditions Annex ZC (informative) A-deviations Annex ZD (informative) IEC and CENELEC code designations for flexible cords	P
1	Modification to Clause 3 .	
3.3.19	Sound exposure <i>Replace 3.3.19 of IEC 62368-1 with the following definitions:</i>	N/A
3.3.19.1	momentary exposure level, MEL metric for estimating 1 s sound exposure level from the HD 483-1 S2 test signal applied to both channels, based on EN 50332-1:2013, 4.2. Note 1 to entry: MEL is measured as A-weighted levels in dB. Note 2 to entry: See B.3 of EN 50332-3:2017 for additional information.	N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
3.3.19.3	sound exposure, E A-weighted sound pressure (p) squared and integrated over a stated period of time, T Note 1 to entry: The SI unit is $\text{Pa}^2 \text{ s}$. $E = \int_0^T p(t)^2 dt$		N/A
3.3.19.4	sound exposure level, SEL logarithmic measure of sound exposure relative to a reference value, E_0, typically the 1 kHz threshold of hearing in humans. Note 1 to entry: SEL is measured as A-weighted levels in dB. $SEL = 10 \lg \left(\frac{E}{E_0} \right) \text{ dB}$ Note 2 to entry: See B.4 of EN 50332-3:2017 for additional information.		N/A
3.3.19.5	digital signal level relative to full scale, dBFS levels reported in dBFS are always r.m.s. Full scale level, 0 dBFS, is the level of a dc-free 997-Hz sine wave whose undithered positive peak value is positive digital full scale, leaving the code corresponding to negative digital full scale unused Note 1 to entry: It is invalid to use dBFS for non-r.m.s. levels. Because the definition of full scale is based on a sine wave, the level of signals with a crest factor lower than that of a sine wave may exceed 0 dBFS. In particular, square wave signals may reach +3,01 dBFS.		N/A
2	Modification to Clause 10		
10.6	Safeguards against acoustic energy sources Replace 10.6 of IEC 62368-1 with the following:		N/A
10.6.1.1	Introduction Safeguard requirements for protection against long-term exposure to excessive sound pressure levels from personal music players closely coupled to the ear are specified below. Requirements for earphones and headphones intended for use		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	with personal music players are also covered. A personal music player is a portable equipment intended for use by an ordinary person, that: – is designed to allow the user to listen to audio or audiovisual content / material; and – uses a listening device, such as headphones or earphones that can be worn in or on or around the ears; and – has a player that can be body worn (of a size suitable to be carried in a clothing pocket) and is intended for the user to walk around with while in continuous use (for example, on a street, in a subway, at an airport, etc.). EXAMPLES Portable CD players, MP3 audio players, mobile phones with MP3 type features, PDAs or similar equipment. Personal music players shall comply with the requirements of either 10.6.2 or 10.6.3. NOTE 1 Protection against acoustic energy sources from telecom applications is referenced to ITU-T P.360. NOTE 2 It is the intention of the Committee to allow the alternative methods for now, but to only use the dose measurement method as given in 10.6.5 in future. Therefore, manufacturers are encouraged to implement 10.6.5 as soon as possible. Listening devices sold separately shall comply with the requirements of 10.6.6. These requirements are valid for music or video mode only. The requirements do not apply to: – professional equipment; NOTE 3 Professional equipment is equipment sold through special sales channels. All products sold through normal electronics stores are considered not to be professional equipment. – hearing aid equipment and other devices for assistive listening; – the following type of analogue personal music players: • long distance radio receiver (for example, a multiband radio receiver or world band radio receiver, an AM radio receiver), and • cassette player/recorder;		

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	NOTE 4 This exemption has been allowed because this technology is falling out of use and it is expected that within a few years it will no longer exist. This exemption will not be extended to other technologies. – a player while connected to an external amplifier that does not allow the user to walk around while in use. For equipment that is clearly designed or intended primarily for use by children, the limits of the relevant toy standards may apply. The relevant requirements are given in EN 71-1:2011, 4.20 and the related tests methods and measurement distances apply.		
10.6.1.2	Non-ionizing radiation from radio frequencies in the range 0 to 300 GHz The amount of non-ionizing radiation is regulated by European Council Recommendation 1999/519/EC of 12 July 1999 on the limitation of exposure of the general public to electromagnetic fields (0 Hz to 300 GHz). For intentional radiators, ICNIRP guidelines should be taken into account for Limiting Exposure to Time-Varying Electric, Magnetic, and Electromagnetic Fields (up to 300 GHz). For hand-held and body mounted devices, attention is drawn to EN 50360 and EN 50566.		N/A
10.6.2	Classification of devices without the capacity to estimate sound dose		N/A
10.6.2.1	General This standard is transitioning from short-term based (30 s) requirements to long-term based (40 hour) requirements. These clauses remain in effect only for devices that do not comply with sound dose estimation as stipulated in EN 50332-3. For classifying the acoustic output $L_{Aeq,T}$, measurements are based on the A-weighted equivalent sound pressure level over a 30 s period. For music where the average sound pressure (long term $L_{Aeq,T}$) measured over the duration of the song is lower than the average produced by the programme simulation noise, measurements may be done over the duration of the complete song. In this case, T becomes the duration of the song.		N/A

IEC62368_1E - ATTACHMENT			
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	NOTE Classical music, acoustic music and broadcast typically has an average sound pressure (long term $L_{Aeq,T}$) which is much lower than the average programme simulation noise. Therefore, if the player is capable to analyse the content and compare it with the programme simulation noise, the warning does not need to be given as long as the average sound pressure of the song does not exceed the required limit. For example, if the player is set with the programme simulation noise to 85 dB, but the average music level of the song is only 65 dB, there is no need to give a warning or ask an acknowledgement as long as the average sound level of the song is not above the basic limit of 85 dB.		
10.6.2.2	RS1 limits (to be superseded, see 10.6.3.2) RS1 is a class 1 acoustic energy source that does not exceed the following: – for equipment provided as a package (player with its listening device), and with a proprietary connector between the player and its listening device, or where the combination of player and listening device is known by other means such as setting or automatic detection, the $L_{Aeq,T}$ acoustic output shall be ≤ 85 dB when playing the fixed “programme simulation noise” described in EN 50332-1. – for equipment provided with a standardized connector (for example, a 3,5 phone jack) that allows connection to a listening device for general use, the unweighted r.m.s. output voltage shall be ≤ 27 mV (analogue interface) or -25 dBFS (digital interface) when playing the fixed “programme simulation noise” described in EN 50332-1. – The RS1 limits will be updated for all devices as per 10.6.3.2.		N/A
10.6.2.3	RS2 limits (to be superseded, see 10.6.3.3) RS2 is a class 2 acoustic energy source that does not exceed the following: – for equipment provided as a package (player with its listening device), and with a proprietary connector between the player and its listening device, or when the combination of player and listening device is known by other means such as setting or automatic 130 detection, the $L_{Aeq,T}$ acoustic output shall be ≤ 100 dB(A) when playing the fixed “programme simulation noise” as described in EN 50332-1. – for equipment provided with a standardized connector (for example, a 3,5 phone jack) that allows connection to a listening device for general use, the unweighted r.m.s. output voltage shall be ≤ 150 mV (analogue interface) or -10 dBFS (digital interface) when playing the fixed “programme simulation noise” as described in EN 50332-1.		N/A
10.6.2.4	RS3 limits		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	RS3 is a class 3 acoustic energy source that exceeds RS2 limits.		
10.6.3	Classification of devices (new)		N/A
10.6.3.1	General Previous limits (10.6.2) created abundant false negative and false positive PMP sound level warnings. New limits, compliant with The Commission Decision of 23 June 2009, are given below.		N/A
10.6.3.2	RS1 limits (new) RS1 is a class 1 acoustic energy source that does not exceed the following: – for equipment provided as a package (player with its listening device), and with a proprietary connector between the player and its listening device, or where the combination of player and listening device is known by other means such as setting or automatic detection, the $L_{Aeq,T}$ acoustic output shall be ≤ 80 dB when playing the fixed “programme simulation noise” described in EN 50332-1. – for equipment provided with a standardized connector (for example, a 3,5 phone jack) that allows connection to a listening device for general use, the unweighted r.m.s. output voltage shall be ≤ 15 mV (analogue interface) or -30 dBFS (digital interface) when playing the fixed “programme simulation noise” described in EN 50332-1.		N/A
10.6.3.3	RS2 limits (new) RS2 is a class 2 acoustic energy source that does not exceed the following: – for equipment provided as a package (player with its listening device), and with a proprietary connector between the player and its listening device, or where the combination of player and listening device is known by other means such as setting or automatic detection, the weekly sound exposure level, as described in EN 50332-3, shall be ≤ 80 dB when playing the fixed “programme simulation noise” described in EN 50332-1. – for equipment provided with a standardized connector (for example, a 3,5 phone jack) that allows connection to a listening device for general use, the unweighted r.m.s. output level, integrated over one week, as described in EN50332-3, shall be ≤ 15 mV (analogue interface) or -30 dBFS (digital interface) when playing the fixed “programme simulation noise” described in EN 50332-1.		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
10.6.4	Requirements for maximum sound exposure		N/A
10.6.4.1	Measurement methods All volume controls shall be turned to maximum during tests. Measurements shall be made in accordance with EN 50332-1 or EN 50332-2 as applicable.		N/A
10.6.4.2	Protection of persons Except as given below, protection requirements for parts accessible to ordinary persons, instructed persons and skilled persons are given in 4.3. NOTE 1 Volume control is not considered a safeguard . Between RS2 and an ordinary person , the basic safeguard may be replaced by an instructional safeguard in accordance with Clause F.5, except that the instructional safeguard shall be placed on the equipment, or on the packaging, or in the instruction manual. Alternatively, the instructional safeguard may be given through the equipment display during use. The elements of the instructional safeguard shall be as follows:  – element 1a: the symbol , IEC 60417-6044 (2011-01) – element 2: “High sound pressure” or equivalent wording – element 3: “Hearing damage risk” or equivalent wording – element 4: “Do not listen at high volume levels for long periods.” or equivalent wording An equipment safeguard shall prevent exposure of an ordinary person to an RS2 source without intentional physical action from the ordinary person and shall automatically return to an output level not exceeding what is specified for an RS1 source when the power is switched off. The equipment shall provide a means to actively inform the user of the increased sound level when the equipment is operated with an output exceeding RS1. Any means used shall be acknowledged by the user before activating a mode of operation which		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	allows for an output exceeding RS1. The acknowledgement does not need to be repeated more than once every 20 h of cumulative listening time. NOTE 2 Examples of means include visual or audible signals. Action from the user is always needed. NOTE 3 The 20 h listening time is the accumulative listening time, independent of how often and how long the personal music player has been switched off. A skilled person shall not be unintentionally exposed to RS3.		
10.6.5	Requirements for dose-based systems		N/A
10.6.5.1	General requirements Personal music players shall give the warnings as provided below when tested according to EN 50332-3, using the limits from this clause. The manufacturer may offer optional settings to allow the users to modify when and how they wish to receive the notifications and warnings to promote a better user experience without defeating the safeguards. This allows the users to be informed in a method that best meets their physical capabilities and device usage needs. If such optional settings are offered, an administrator (for example, parental restrictions, business/educational administrators, etc.) shall be able to lock any optional settings into a specific configuration. The personal music player shall be supplied with easy to understand explanation to the user of the dose management system, the risks involved, and how to use the system safely. The user shall be made aware that other sources may significantly contribute to their sound exposure, for example work, transportation, concerts, clubs, cinema, car races, etc.		N/A
10.6.5.2	Dose-based warning and requirements When a dose of 100 % CSD is reached, and at least at every 100 % further increase of CSD, the device shall warn the user and require an acknowledgement. In case the user does not acknowledge, the output level shall automatically decrease to compliance with class RS1. The warning shall at least clearly indicate that listening above 100 % CSD leads to the risk of		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	hearing damage or loss.		
10.6.5.3	Exposure-based requirements With only dose-based requirements, cause and effect could be far separated in time, defying the purpose of educating users about safe listening practice. In addition to dose-based requirements, a PMP shall therefore also put a limit to the short-term sound level a user can listen at. The exposure-based limiter (EL) shall automatically reduce the sound level not to exceed 100 dB(A) or 150 mV integrated over the past 180 s, based on methodology defined in EN 50332-3. The EL settling time (time from starting level reduction to reaching target output) shall be 10 s or faster. Test of EL functionality is conducted according to EN 50332-3, using the limits from this clause. For equipment provided as a package (player with its listening device), the level integrated over 180 s shall be 100 dB or lower. For equipment provided with a standardized connector, the unweighted level integrated over 180 s shall be no more than 150 mV for an analogue interface and no more than -10 dBFS for a digital interface. NOTE In case the source is known not to be music (or test signal), the EL may be disabled.		N/A
10.6.6	Requirements for listening devices (headphones, earphones, etc.)		N/A
10.6.6.1	Corded listening devices with analogue input With 94 dB L_{Aeq} acoustic pressure output of the listening device, and with the volume and sound settings in the listening device (for example, built-in volume level control, additional sound features like equalization, etc.) set to the combination of positions that maximize the measured acoustic output, the input voltage of the listening device when playing the fixed “programme simulation noise” as described in EN 50332-1 shall be ≥ 75 mV. NOTE The values of 94 dB and 75 mV correspond with 85 dB and 27 mV or 100 dB and 150 mV.		N/A
10.6.6.2	Corded listening devices with digital input With any playing device playing the fixed “programme simulation noise” described in EN 50332-1, and with the volume and sound settings in		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	the listening device (for example, built-in volume level control, additional sound features like equalization, etc.) set to the combination of positions that maximize the measured acoustic output, the $L_{Aeq,T}$ acoustic output of the listening device shall be ≤ 100 dB with an input signal of -10 dBFS.		
10.6.6.3	Cordless listening devices In cordless mode, – with any playing and transmitting device playing the fixed programme simulation noise described in EN 50332-1; and – respecting the cordless transmission standards, where an air interface standard exists that specifies the equivalent acoustic level; and – with volume and sound settings in the receiving device (for example, built-in volume level control, additional sound features like equalization, etc.) set to the combination of positions that maximize the measured acoustic output for the above mentioned programme simulation noise, the $L_{Aeq,T}$ acoustic output of the listening device shall be ≤ 100 dB with an input signal of -10 dBFS.		N/A
10.6.6.4	Measurement method <i>Measurements shall be made in accordance with EN 50332-2 as applicable.</i>		N/A
3	Modification to the whole document		

IEC62368_1E - ATTACHMENT						
Clause	Requirement + Test			Result - Remark		Verdict
	Delete all the “country” notes in the reference document according to the following list:					N/A
	0.2.1	Note 1 and 2	1	Note 4 and 5	3.3.8.1	Note 2
	3.3.8.3	Note 1	4.1.15	Note	4.7.3	Note 1 and 2
	5.2.2.2	Note	5.4.2.3.2.2 Table 12	Note c	5.4.2.3.2.4	Note 1 and 3
	5.4.2.3.2.4 Table 13	Note 2	5.4.2.5	Note 2	5.4.5.1	Note
	5.4.10.2.1	Note	5.4.10.2.2	Note	5.4.10.2.3	Note
	5.5.2.1	Note	5.5.6	Note	5.6.4.2.1	Note 2 and 3 and 4
	5.6.8	Note 2	5.7.6	Note	5.7.7.1	Note 1 and Note 2
	8.5.4.2.3	Note	10.2.1 Table 39	Note 3 and 4 and 5	10.5.3	Note 2
	10.6.1	Note 3	F.3.3.6	Note 3	Y.4.1	Note
	Y.4.5	Note				
4	Modification to Clause 1					--
1	Add the following note: NOTE Z1 The use of certain substances in electrical and electronic equipment is restricted within the EU: see Directive 2011/65/EU.					P
5	Modification to 4.Z1					--

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
4.21	Add the following new subclause after 4.9: To protect against excessive current, short-circuits and earth faults in circuits connected to an a.c. mains, protective devices shall be included either as integral parts of the equipment or as parts of the building installation, subject to the following, a), b) and c): a) except as detailed in b) and c), protective devices necessary to comply with the requirements of B.3.1 and B.4 shall be included as parts of the equipment; b) for components in series with the mains input to the equipment such as the supply cord, appliance coupler, r.f.i. filter and switch, short-circuit and earth fault protection may be provided by protective devices in the building installation; c) it is permitted for pluggable equipment type B or permanently connected equipment, to rely on dedicated overcurrent and short-circuit protection in the building installation, provided that the means of protection, e.g. fuses or circuit breakers, is fully specified in the installation instructions. If reliance is placed on protection in the building installation, the installation instructions shall so state, except that for pluggable equipment type A the building installation shall be regarded as providing protection in accordance with the rating of the wall socket outlet.		P
6	Modification to 5.4.2.3.2.4		--
5.4.2.3.2.4	Add the following to the end of this subclause: The requirement for interconnection with external circuit is in addition given in EN 50491-3:2009.		N/A
7	Modification to 10.2.1		--
10.2.1	Add the following to ^{c)} and ^{d)} in table 39: For additional requirements, see 10.5.1.		N/A
8	Modification to 10.5.1		--

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
10.5.1	Add the following after the first paragraph: For RS 1 compliance is checked by measurement under the following conditions: In addition to the normal operating conditions, all controls adjustable from the outside by hand, by any object such as a tool or a coin, and those internal adjustments or pre-sets which are not locked in a reliable manner, are adjusted so as to give maximum radiation whilst maintaining an intelligible picture for 1 h, at the end of which the measurement is made. NOTE Z1 Soldered joints and paint lockings are examples of adequate locking. The dose-rate is determined by means of a radiation monitor with an effective area of 10 cm², at any point 10 cm from the outer surface of the apparatus. Moreover, the measurement shall be made under fault conditions causing an increase of the high voltage, provided an intelligible picture is maintained for 1 h, at the end of which the measurement is made. For RS1, the dose-rate shall not exceed 1 µSv/h taking account of the background level. NOTE Z2 These values appear in Directive 96/29/Euratom of 13 May 1996.		N/A
9	Modification to G.7.1		--
G.7.1	Add the following note: NOTE Z1 The harmonized code designations corresponding to the IEC cord types are given in Annex ZD.		N/A
10	Modification to Bibliography		--

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	Add the following notes for the standards indicated: IEC 60130-9 NOTE Harmonized as EN 60130-9. IEC 60269-2 NOTE Harmonized as HD 60269-2. IEC 60309-1 NOTE Harmonized as EN 60309-1. IEC 60364 NOTE some parts harmonized in HD 384/HD 60364 series. IEC 60601-2-4 NOTE Harmonized as EN 60601-2-4. IEC 60664-5 NOTE Harmonized as EN 60664-5. IEC 61032:1997 NOTE Harmonized as EN 61032:1998 (not modified). IEC 61508-1 NOTE Harmonized as EN 61508-1. IEC 61558-2-1 NOTE Harmonized as EN 61558-2-1. IEC 61558-2-4 NOTE Harmonized as EN 61558-2-4. IEC 61558-2-6 NOTE Harmonized as EN 61558-2-6. IEC 61643-1 NOTE Harmonized as EN 61643-1. IEC 61643-21 NOTE Harmonized as EN 61643-21. IEC 61643-311 NOTE Harmonized as EN 61643-311. IEC 61643-321 NOTE Harmonized as EN 61643-321. IEC 61643-331 NOTE Harmonized as EN 61643-331.		N/A
11	ADDITION OF ANNEXES		--
ZB	ANNEX ZB, SPECIAL NATIONAL CONDITIONS (EN)		N/A
4.1.15	Denmark, Finland, Norway and Sweden To the end of the subclause the following is added: Class I pluggable equipment type A intended for connection to other equipment or a network shall, if safety relies on connection to reliable earthing or if surge suppressors are connected between the network terminals and accessible parts, have a marking stating that the equipment shall be connected to an earthed mains socket-outlet. The marking text in the applicable countries shall be as follows: In Denmark: "Apparatets stikprop skal tilsluttes en stikkontakt med jord som giver forbindelse til stikproppens jord." In Finland: "Laitte on liitettävä suojakoskettimilla varustettuun pistorasiaan" In Norway: "Apparatet må tilkoples jordet stikkontakt" In Sweden: "Apparaten skall anslutas till jordat uttag"		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
4.7.3	United Kingdom To the end of the subclause the following is added: The torque test is performed using a socket-outlet complying with BS 1363, and the plug part shall be assessed to the relevant clauses of BS 1363. Also see Annex G.4.2 of this annex		N/A
5.2.2.2	Denmark After the 2nd paragraph add the following: A warning (marking safeguard) for high touch current is required if the touch current exceeds the limits of 3,5 mA a.c. or 10 mA d.c.		N/A
5.4.11.1 and Annex G	Finland and Sweden To the end of the subclause the following is added: For separation of the telecommunication network from earth the following is applicable: If this insulation is solid, including insulation forming part of a component, it shall at least consist of either • two layers of thin sheet material, each of which shall pass the electric strength test below, or • one layer having a distance through insulation of at least 0,4 mm, which shall pass the electric strength test below. If this insulation forms part of a semiconductor component (e.g. an optocoupler), there is no distance through insulation requirement for the insulation consisting of an insulating compound completely filling the casing, so that clearances and creepage distances do not exist, if the component passes the electric strength test in accordance with the compliance clause below and in addition • passes the tests and inspection criteria of 5.4.8 with an electric strength test of 1,5 kV multiplied by 1,6 (the electric strength test of 5.4.9 shall be performed using 1,5 kV),		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	and • is subject to routine testing for electric strength during manufacturing, using a test voltage of 1,5 kV. It is permitted to bridge this insulation with a capacitor complying with EN 60384-14:2005, subclass Y2. A capacitor classified Y3 according to EN 60384-14:2005, may bridge this insulation under the following conditions: • the insulation requirements are satisfied by having a capacitor classified Y3 as defined by EN 60384-14, which in addition to the Y3 testing, is tested with an impulse test of 2,5 kV defined in 5.4.11; • the additional testing shall be performed on all the test specimens as described in EN 60384-14; the impulse test of 2,5 kV is to be performed before the endurance test in EN 60384-14, in the sequence of tests as described in EN 60384-14.		
5.5.2.1	Norway After the 3rd paragraph the following is added: Due to the IT power system used, capacitors are required to be rated for the applicable line-to-line voltage (230 V).		N/A
5.5.6	Finland, Norway and Sweden To the end of the subclause the following is added: Resistors used as basic safeguard or bridging basic insulation in class I pluggable equipment type A shall comply with G.10.1 and the test of G.10.2.		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
5.6.1	Denmark Add to the end of the subclause Due to many existing installations where the socket-outlets can be protected with fuses with higher rating than the rating of the socket-outlets the protection for pluggable equipment type A shall be an integral part of the equipment. Justification: In Denmark an existing 13 A socket outlet can be protected by a 20 A fuse.		N/A
5.6.4.2.1	Ireland and United Kingdom After the indent for pluggable equipment type A, the following is added: – the protective current rating is taken to be 13 A, this being the largest rating of fuse used in the mains plug.		N/A
5.6.4.2.1	France After the indent for pluggable equipment type A, the following is added: – in certain cases, the protective current rating of the circuit supplied from the mains is taken as 20 A instead of 16 A.		N/A
5.6.5.1	To the second paragraph the following is added: The range of conductor sizes of flexible cords to be accepted by terminals for equipment with a rated current over 10 A and up to and including 13 A is: 1,25 mm² to 1,5 mm² in cross-sectional area.		N/A
5.6.8	Norway To the end of the subclause the following is added: Equipment connected with an earthed mains plug is classified as class I equipment. See the Norway marking requirement in 4.1.15. The symbol IEC 60417-6092, as specified in F.3.6.2, is accepted.		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
5.7.6	Denmark To the end of the subclause the following is added: The installation instruction shall be affixed to the equipment if the protective conductor current exceeds the limits of 3,5 mA a.c. or 10 mA d.c.		N/A
5.7.6.2	Denmark To the end of the subclause the following is added: The warning (marking safeguard) for high touch current is required if the touch current or the protective current exceed the limits of 3,5 mA .		N/A
5.7.7.1	Norway and Sweden To the end of the subclause the following is added: The screen of the television distribution system is normally not earthed at the entrance of the building and there is normally no equipotential bonding system within the building. Therefore the protective earthing of the building installation needs to be isolated from the screen of a cable distribution system. It is however accepted to provide the insulation external to the equipment by an adapter or an interconnection cable with galvanic isolator, which may be provided by a retailer, for example. The user manual shall then have the following or similar information in Norwegian and Swedish language respectively, depending on in what country the equipment is intended to be used in: “Apparatus connected to the protective earthing of the building installation through the mains connection or through other apparatus with a connection to protective earthing – and to a television distribution system using coaxial cable, may in some circumstances create a fire hazard. Connection to a television distribution system therefore has to be provided through a device providing electrical isolation below a certain frequency range (galvanic isolator, see EN 60728-11)” NOTE In Norway, due to regulation for CATV-installations, and in Sweden, a galvanic isolator shall provide electrical insulation below 5 MHz. The insulation shall withstand a dielectric strength of 1,5 kV r.m.s., 50 Hz or 60 Hz, for 1 min.		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	Translation to Norwegian (the Swedish text will also be accepted in Norway): “Apparater som er koplet til beskyttelsesjord via nettplugg og/eller via annet jordtilkoplet utstyr – og er tilkoplet et koaksialbasert kabel-TV nett, kan forårsake brannfare. For å unngå dette skal det ved tilkopling av apparater til kabel-TV nett installeres en galvanisk isolator mellom apparatet og kabel-TV nettet.” Translation to Swedish: ”Apparater som är kopplad till skyddsjord via jordat vägguttag och/eller via annan utrustning och samtidigt är kopplad till kabel-TV nät kan i vissa fall medföra risk för brand. För att undvika detta skall vid anslutning av apparaten till kabel-TV nät galvanisk isolator finnas mellan apparaten och kabel-TV nätet.”.		
8.5.4.2.3	United Kingdom Add the following after the 2nd dash bullet in 3rd paragraph: An emergency stop system complying with the requirements of IEC 60204-1 and ISO 13850 is required where there is a risk of personal injury.		N/A
B.3.1 and B.4	Ireland and United Kingdom The following is applicable: To protect against excessive currents and short-circuits in the primary circuit of direct plug-in equipment, tests according to Annexes B.3.1 and B.4 shall be conducted using an external miniature circuit breaker complying with EN 60898-1, Type B, rated 32A. If the equipment does not pass these tests, suitable protective devices shall be included as an integral part of the direct plug-in equipment, until the requirements of Annexes B.3.1 and B.4 are met		N/A
G.4.2	Denmark To the end of the subclause the following is added: Supply cords of single phase appliances having a rated current not exceeding 13 A shall be provided		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	with a plug according to DS 60884-2-D1:2011. CLASS I EQUIPMENT provided with socket-outlets with earth contacts or which are intended to be used in locations where protection against indirect contact is required according to the wiring rules shall be provided with a plug in accordance with standard sheet DK 2-1a or DK 2-5a. If a single-phase equipment having a RATED CURRENT exceeding 13 A or if a polyphase equipment is provided with a supply cord with a plug, this plug shall be in accordance with the standard sheets DK 6-1a in DS 60884-2-D1 or EN 60309-2. Mains socket outlets intended for providing power to Class II apparatus with a rated current of 2,5 A shall be in accordance DS 60884-2-D1:2011 standard sheet DKA 1-4a. Other current rating socket outlets shall be in compliance with Standard Sheet DKA 1-3a or DKA 1-1c. Mains socket-outlets with earth shall be in compliance with DS 60884-2-D1:2011 Standard Sheet DK 1-3a, DK 1-1c, DK1-1d, DK 1-5a or DK 1-7a <i>Justification:</i> Heavy Current Regulations, Section 6c		
G.4.2	United Kingdom To the end of the subclause the following is added: The plug part of direct plug-in equipment shall be assessed to BS 1363: Part 1, 12.1, 12.2, 12.3, 12.9, 12.11, 12.12, 12.13, 12.16, and 12.17, except that the test of 12.17 is performed at not less than 125 °C. Where the metal earth pin is replaced by an Insulated Shutter Opening Device (ISOD), the requirements of clauses 22.2 and 23 also apply.		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
G.7.1	United Kingdom To the first paragraph the following is added: Equipment which is fitted with a flexible cable or cord and is designed to be connected to a mains socket conforming to BS 1363 by means of that flexible cable or cord shall be fitted with a 'standard plug' in accordance with the Plugs and Sockets etc. (Safety) Regulations 1994, Statutory Instrument 1994 No. 1768, unless exempted by those regulations. NOTE "Standard plug" is defined in SI 1768:1994 and essentially means an approved plug conforming to BS 1363 or an approved conversion plug.		N/A
G.7.1	Ireland To the first paragraph the following is added: Apparatus which is fitted with a flexible cable or cord shall be provided with a plug in accordance with Statutory Instrument 525: 1997, "13 A Plugs and Conversion Adapters for Domestic Use Regulations: 1997. S.I. 525 provides for the recognition of a standard of another Member State which is equivalent to the relevant Irish Standard		N/A
G.7.2	Ireland and United Kingdom To the first paragraph the following is added: A power supply cord with a conductor of 1,25 mm² is allowed for equipment which is rated over 10 A and up to and including 13 A.		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
ZC	ANNEX ZC, NATIONAL DEVIATIONS (EN)		N/A
10.5.2	Germany The following requirement applies: For the operation of any cathode ray tube intended for the display of visual images operating at an acceleration voltage exceeding 40 kV, authorization is required, or application of type approval (Bauartzulassung) and marking. <i>Justification:</i> German ministerial decree against ionizing radiation (Röntgenverordnung), in force since 2002-07-01, implementing the European Directive 96/29/EURATOM. NOTE Contact address: Physikalisch-Technische Bundesanstalt, Bundesallee 100, D-38116 Braunschweig. Tel.: Int+49-531-592-6320, Internet: http://www.ptb.de		N/A

IEC62368_1E - ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict

ZD	IEC and CENELEC CODE DESIGNATIONS FOR FLEXIBLE CORDS (EN)		N/A					
	<table><tr><th rowspan="2">Type of flexible cord</th><th colspan="2">Code designations</th></tr><tr><th>IEC</th><th>CENELEC</th></tr></table>		Type of flexible cord	Code designations		IEC	CENELEC	N/A
	Type of flexible cord	Code designations						
		IEC	CENELEC					
	PVC insulated cords							
	Flat twin tinsel cord	60227 IEC 41	H03VH-Y					
	Light polyvinyl chloride sheathed flexible cord	60227 IEC 52	H03VV-F H03VVH2-F					
	Ordinary polyvinyl chloride sheathed flexible cord	60227 IEC 53	H05VV-F H05VVH2-F					
	Rubber insulated cords							
	Braided cord	60245 IEC 51	H03RT-F					
	Ordinary tough rubber sheathed flexible cord	60245 IEC 53	H05RR-F					
	Ordinary polychloroprene sheathed flexible cord	60245 IEC 57	H05RN-F					
	Heavy polychloroprene sheathed flexible cord	60245 IEC 66	H07RN-F					
	Cords having high flexibility							
	Rubber insulated and sheathed cord	60245 IEC 86	H03RR-H					
	Rubber insulated, crosslinked PVC sheathed cord	60245 IEC 87	H03RV4-H					
Crosslinked PVC insulated and sheathed cord	60245 IEC 88	H03V4V4-H						
Cords insulated and sheathed with halogen-free thermoplastic compounds								
Light halogen-free thermoplastic insulated and sheathed flexible cords		H03Z1Z1-F H03Z1Z1H2-F						
Ordinary halogen-free thermoplastic insulated and sheathed flexible cords		H05Z1Z1-F H05Z1Z1H2-F						

IEC62368_1E ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
ATTACHMENT TO TEST REPORT IEC 62368-1 U.S.A. AND CANADA NATIONAL DIFFERENCES (Audio/video, information and communication technology equipment – Part 1: Safety requirements)			
Differences according to : CSA/UL 62368-1:2019			
TRF template used: : IECCE OD-2020-F3, Ed. 1.1			
Attachment Form No. : US_CA_ND_IEC62368_1E			
Attachment Originator : UL(US)			
Master Attachment : Dated 2022-03-04			
Copyright © 2022 IEC System for Conformity Testing and Certification of Electrical Equipment (IECEE), Geneva, Switzerland. All rights reserved.			
IEC 62368-1 - US and Canadian National Differences Special National Conditions based on Regulations and Other National Differences			
1 (1DV.1) (1.3)	All equipment is to be designed to allow installation in accordance with the National Electrical Code (NEC), ANSI/NFPA 70, the Canadian Electrical Code (CEC), Part 1, CAN/CSA C22.1, and when applicable, the National Electrical Safety Code, IEEE C2. Also, for such equipment marked or otherwise identified, installation is allowed per the Standard for the Protection of Information Technology Equipment, ANSI/NFPA 75.		P
1 (1DV.2.1)	This standard includes additional requirements for equipment used for entertainment purposes intended for installation in general patient care areas of health care facilities. See Annex DVB.		N/A
1 (1DV.2.2)	This standard includes additional requirements for equipment intended for mounting under cabinets. See Annex DVC.		N/A
1 (1DV.2.3)	IEC 62368-3 clause 5 for DC power transfer at ES1 or ES2 voltage levels is considered informative. IEC 62368-3 clause 6 for remote power feeding telecommunication (RFT) circuits is considered normative (see ITU K.50). Alternatively, equipment with RFT circuits are given in either UL 2391 or CSA/UL 60950-21. RFT-C circuits are not permitted unless the RFT-C circuit complies with RFT-V limits ($\leq 200V$ per conductor to earth).		N/A

IEC62368_1E ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
1 (1DV.3)	For protection against direct lightning strikes, reference is made to NFPA 780 and CAN/CSA-B72 for additional requirements.		N/A
1 (DV.5)	Additional requirements apply to some forms of power distribution equipment, including sub-assemblies.		P
4.1 (4.1.17)	For lengths exceeding 3.05 m, external interconnecting cable assemblies are required to be a suitable cable type (e.g., DP, CL2) specified in the NEC.		N/A
	For lengths 3.05 m or less, external interconnecting cable assemblies that are not types specified in the NEC generally are required to have special construction features and identification markings.		N/A
4.6 (4.6.2)	Wire-wrap terminals have special construction and performance requirements.		N/A
4.8 (4.8.3, 4.8.4.5, 4.8.5)	Coin / button cell batteries have modified special construction and performance requirements.		N/A
5.4.2.3.2 (5.4.2.3.2.1)	Surge Arrestors and Transient Voltage Surge Suppressors installed external to the equipment are required to comply with the appropriate NEC and CEC requirements.		N/A
5.5.9	Receptacles, rated 125-V, single phase, 15- or 20-A accessible to either ordinary, instructed, or skilled persons are required to be provided with GFCI Protection for Personnel if the equipment containing the receptacles is installed outdoors. The protection devices are required to comply with UL 943, and CAN/CSA C22.2 No.144.		N/A
5.6.3	Protective earthing conductors comply with the minimum conductor sizes in Table G.7, except as required by Table G.7ADV.1 for cord connected equipment, or Annex DVH for permanently connected equipment.		N/A
5.7.8 (5.7.8.1)	Equipment intended to receive telecommunication ringing signals is required to comply with a special touch current measurement tests.		N/A
6.5.1	PS3 wiring outside a fire enclosure is required to comply with single fault testing in B.4, or be current limited per one of the permitted methods.		N/A

IEC62368_1E ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
Annex F (F.3.3.9)	Output terminals provided for supply of other equipment, except mains supply, are required to be marked with a maximum rating or reference to equipment permitted to be connected.		N/A
Annex F (F.3.7)	Outdoor Enclosures are required to be classified and marked in accordance with UL 50 or 50E, or CAN/CSA C22.2 No. 94.1 or 94.2.		N/A
Annex G (G.7)	Permanent connection of equipment to the mains supply by a power supply cord is not permitted, except for certain equipment, such as ATMs.		N/A
	Power supply cords are required to have attachment plugs rated not less than 125 percent of the rated current of the equipment.		N/A
	Flexible power supply cords are required to be compatible with Article 400 of the NEC, and Tables 11 and 12 of the CEC.		N/A
	Minimum cord length is required to be 1.5 m, with certain constructions such as external power supplies allowed to consider both input and output cord lengths into the requirement. Power supply cords are required to be no longer than 4.5 m in length if used in ITE Rooms.		N/A
	Power supply cords for outdoor equipment are required to be suitable outdoor use type as required by Section 400.4 of the NEC and Rule 4-012 of the CEC, i.e., marked "W."		N/A
Annex H.2	Continuous ringing signals under normal operating conditions up to 16 mA only are permitted if the equipment is subjected to special installation and performance restrictions.		N/A
Annex H.4	For circuits with other than ringing signals and with voltages exceeding 42.4 V _{peak} or 60 V _{d.c.} , the maximum acceptable current through a 2000 ohm resistor (or greater) connected across the voltage source with other loads disconnected is 7.1 mA peak or 30 mA d.c. under normal operating conditions.		N/A
Annex Q (Q.3)	Equipment with paired conductor and/or coax communications cables/wiring connected to building wiring are required to have special voltage, current, power and marking requirements.		N/A

IEC62368_1E ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
Annex DVA (1)	Equipment that is designed such that it may be powered from a separate electrical service, is required to meet applicable requirements for service equipment for control and protection of services and their installation and complies with Article 230 of the National Electrical Code (NEC), NFPA 70 and Section 6 of the Canadian Electrical Code, Part I, CSA C22.1.		N/A
	Equipment intended for use in spaces used for environmental air (plenums) are subjected to special flammability requirements for heat and visible smoke release.		N/A
	For ITE room applications, automated information storage systems with combustible media greater than 0.76 m ³ (27 cu ft) are required to have a provision for connection of either automatic sprinklers or a gaseous agent extinguishing system with an extended discharge.		N/A
	Consumer products designed or intended primarily for children 12 years of age or younger are subject to additional requirements in accordance with U.S. and Canadian Regulations.		N/A
	Baby monitors are required to additionally comply with ASTM F2951, Consumer Safety Specification for Baby Monitors.		N/A
	Storage batteries and battery management equipment, other than associated with lead-acid batteries, and including battery backup systems that are not an integral part of stationary AV and ICT equipment, such as provided in separate cabinets, are required to be certified (listed) to the appropriate standard(s) for such storage batteries and equipment.		N/A
Annex DVA (5.6)	For Pluggable Equipment Type A, the protection in the installation is assumed to be 20A.		N/A
Annex DVA (6.3)	The maximum quantity of flammable liquid stored in equipment is required to comply with NFPA 30.		N/A

IEC62368_1E ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
Annex DVA (6.4.8)	For ITE room applications, enclosures with combustible material measuring greater than 0.9 m ² (10 sq ft) or a single dimension greater than 1.8 m (6 ft) are required to have a flame spread rating of 50 or less. For equipment with the same dimensions for other applications, an external surface that is not a fire enclosure requires a minimum flammability classification of V-1.		N/A
Annex DVA (10.3)	Equipment with lasers is required to meet the U.S. Code of Federal Regulations 21 CFR 1040 (and the Canadian Radiation Emitting Devices Act, REDR C1370).		N/A
Annex DVA (10.5)	Equipment that produces ionizing radiation is required to comply with the U.S. Code of Federal Regulations, 21 CFR 1020 (and the Canadian Radiation Emitting Devices Act, REDR C1370).		N/A
Annex DVA (F.3.3.4)	Equipment for use on a.c. mains supply systems with a neutral and more than one phase conductor (e.g. 120/240 V, 3-wire) require a special marking format for electrical ratings. Additional considerations apply for voltage ratings that exceed the attachment cap rating or that are lower than the "Normal Operating Condition" in Table 2 of CAN/CSA C22.2 No. 235."		N/A
Annex DVA (F.3.3.6)	Equipment identified for ITE (computer) room installation is required to be marked with the rated current.		N/A
Annex DVA (G.1)	Vertically-mounted disconnect switches and circuit breakers are required to have the "on" position indicated by the handle in the up position, where mounted in an enclosure, vertically mounted disconnect switches and circuit breakers with vertical operating means extending outside the enclosure are required to indicate in a location visible when accessing the external operating means whether the switch or circuit breaker is in the open (off) or closed (on) position.		N/A
Annex DVA (G.3.4)	Suitable NEC/CEC branch circuit protection rated at the maximum circuit rating is required for all standard supply outlets and receptacles (such as supplied in power distribution units) if the supply branch circuit protection is not suitable.		N/A
	Where a fuse is used to provide Class 2 or Class 3 current limiting, it is not operator-accessible unless it is non- interchangeable.		N/A

IEC62368_1E ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
Annex DVA (G.4.2)	Equipment with isolated ground (earthing) receptacles is required to comply with NEC 250.146(D) and CEC 10-400 and 10-612.		N/A
Annex DVA (G.4.3)	Interconnection of units by conductors supplied by a limited power source, or a Class 2 circuit defined in the NEC/CEC may have field wiring connections other than specified in DVH.3, such as wire-wrap and crimp-on types, if the limited power source and Class 2 circuits are separated from all other circuits by barriers, routing or fixing.		N/A
Annex DVA (G.5.3)	Power distribution transformers distributing power at 100 volts or more, and rated 10 kVA or more, require special transformer overcurrent protection.		N/A
Annex DVA (G.5.4)	Motor control devices are required for cord-connected equipment with a mains-connected motor if the equipment is rated more than 12 A, or if the equipment has a nominal voltage rating greater than 120 V, or if the motor is rated more than 1/3 hp (locked rotor current over 43 A).		N/A
Annex DVA (G.7)	Flexible cords used outdoors are required to have the suffix "W" marked on the flexible cord.		N/A
Annex DVA (M)	For ITE room applications, equipment with battery systems capable of supplying 750 VA for five minutes are required to have a battery disconnect means that may be connected to the ITE room remote power-off circuit.		N/A
Annex DVA (Q)	If applicable per NEC 725.121(C), some limited power sources supplied from AV/ICT equipment are required to have a label indicating the maximum voltage and rated current output for per conductor for each connection point. Where multiple connection points have the same rating, a single label is permitted to be used.		N/A
	Wiring terminals intended to supply Class 2 outputs in accordance with the NEC or CEC Part 1 are required to be marked with the voltage rating and "Class 2" or equivalent. The marking is located adjacent to the terminals and visible during wiring.		N/A
	Applicable parts of Chapter 8 of the NEC, and Rules 54 and 60 of the CEC, may be applicable to ITE installed outdoors with connections to communication systems.		N/A

IEC62368_1E ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
Annex DVB (1)	Additional requirements apply for equipment used for entertainment purposes intended for installation in general patient care areas of health care facilities.		N/A
Annex DVC (1)	Additional requirements apply for equipment intended for mounting under kitchen cabinets.		N/A
Annex DVE (4.1.1)	Some equipment, components, sub-assemblies and materials associated with the risk of fire, electric shock, or personal injury are required to have component or material ratings in accordance with the applicable national (U.S. and Canadian) component or material requirements. These equipment and components include: appliance couplers, attachment plugs, battery backup systems, circuit breakers, communication circuit accessories, connectors (used for current interruption of non-LPS circuits), direct plug-in equipment, electrochemical capacitor modules (energy storage modules with ultracapacitors), enclosures (outdoor), flexible cords and cables, fuses (branch circuit), ground-fault current interrupters, interconnecting cables, modular data centres, power supply cords, some power distribution equipment, printed wiring, protectors for communications circuits, receptacles, surge protective devices, vehicle battery adapters, wire connectors, and wire and cables.		P
Annex DVH	Equipment for permanent connection to the mains supply is subjected to additional requirements.		N/A
Annex DVH (DVH.1)	Wiring methods (terminals, leads, etc.) used for the connection of the equipment to the mains are required to be in accordance with the NEC/CEC.		N/A
Annex DVH (DVH.2.1)	For safe and reliable connection to a mains, permanently connected equipment is to be provided.		N/A
Annex DVH (DVH.2.2)	Additional considerations for D.C. mains.		N/A
Annex DVH (DVH.3.2.1)	Terminals for permanent wiring, including protective earthing terminals, are required to be suitable for U.S./Canadian wire gauge sizes, rated 125 percent of the equipment rating, and be specially marked when specified.		N/A
Annex DVH (DVH.3.2.3)	Wire binding screws are not permitted to attach conductors larger than 10 AWG (5.3 mm ²).		N/A

IEC62368_1E ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
Annex DVH (DVH.3.2.4)	All associated mains supply terminals are located in proximity to each other and to the main protective earthing terminal, if any.		N/A
Annex DVH (DVH.3.2.5)	Terminals are located, guarded or insulated so that, should a strand of a conductor escape when the conductor is fitted, there is no likelihood of accidental contact between such a strand and accessible conductive parts or unearthed conductive parts separated from accessible conductive parts by supplementary insulation only.		N/A
Annex DVH (DVH.3.3)	When field connection to an external circuit is via wires (example, free conductors), the wires are not smaller than 18 AWG (0.82 mm ²) and the free length of the wire inside an outlet box or wiring compartment is 150 mm or more.		N/A
Annex DVH (DVH.3.4)	Size of protective earthing conductors and terminals	(See sub-clause 5.6.5)	N/A
Annex DVH (DVH.4)	Permanently connected equipment is required to have a suitable wiring compartment and wire bending space.		N/A
Annex DVH (DVH.4.1)	Wire bending space		N/A
Annex DVH (DVH.4.2)	Volume of wiring compartment		N/A
Annex DVH (DVH.4.3)	Separation of circuits		N/A
Annex DVH (DVH.5)	Equipment markings and instructional safeguards		N/A
Annex DVH (DVH.5.1)	Identification of protective earthing terminal		N/A
Annex DVH (DVH.5.2)	Identification of terminal for earthed conductor (neutral)		N/A
Annex DVH (DVH.5.3)	Identification of terminals for aluminium conductors		N/A
Annex DVH (DVH.5.4)	Wire temperature ratings		N/A
Annex DVH (DVH 5.5)	Equipment connected to a centralized d.c. power system, and having one pole of the DC mains input terminal connected to the main protective earthing terminal in the equipment, is required to comply with special earthing, wiring, marking and installation instruction requirements.		N/A

IEC62368_1E ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
Annex DVI (6.7)	Equipment intended for connection to telecommunication network outside plant cable is required to be protected against overvoltage from power line crosses.		N/A
Annex DVJ (10.6.1)	Equipment connected to a telecommunication and cable distribution networks and supplied with an earphone intended to be held against, or in the ear is required to comply with special acoustic pressure requirements.		N/A

IEC62638_1E ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
ATTACHMENT TO TEST REPORT IEC 62368-1 SINGAPORE NATIONAL DIFFERENCES Audio/video, information and communication technology equipment - Part 1: Safety requirements			
Differences according to : Special National Conditions			
TRF template used: : IECEE OD-2020-F3, Ed. 1.1			
Attachment Form No. : SG_ND_IEC62368_1E			
Attachment Originator : Intertek Testing Services (Singapore) Pte Ltd			
Master Attachment : 2021-07-16			
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	National Differences		--
	Not Applicable		--
Chapter 7	Special national conditions (if any) Controlled goods under Consumer Protection (Safety Requirements) Registration Scheme (CPS) are required to be tested to additional requirements stipulated by Enterprise Singapore in Chapter 7 of the CPS information booklet. The CPS information booklet is updated on an ongoing basis. At the point of testing, refer to the latest copy of the CPS information booklet for the minimum edition of standard to apply for testing of products under the CPS scheme and any new requirements. Link to CPS information booklet: https://www.consumerproductsafety.gov.sg/files/cps-info-booklet.pdf		P
3	All appliances must be tested to 230 VAC, 50 Hz.		P
4	Appliance fitted with voltage selector shall be tested as follows: Connect appliance to 230 VAC mains with voltage selector switch to settings not suitable for operation at 230 VAC.		P
5	All appliances (with tropical test requirements in applicable Standards) shall comply with the tropical condition test as stated in the relevant IEC Standards.		N/A

IEC62638_1E ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
6	All Class I appliances (3-pin mains plug) must be fitted with 3-pin mains plugs complying with SS 145/SS 472 that are registered with the Authority.		P
7	a) All Class II appliances must be fitted with 2-pin mains plug complying with EN 50075. b) Class II appliances that are fitted with 3-pin mains plugs must use plugs that comply with SS 145 and registered with the Authority.		N/A
9	Detachable power cord set must be listed in the test report critical component list.		N/A
18	AC Adaptor incorporated with 13A socket-outlet to be tested to additional tests clauses 13, 17 and 18 of SS 246 (till 25/10/2021), or clauses 12.1 & 12.3 of SS 145 Part 3: 2020.		P
19	Supplier who is supplying AC adaptors with detachable interchangeable plug pins must include with its products, written instructions to inform customer on the type of detachable interchangeable plug pins that are approved and suitable to use in Singapore. These instructions are to be submitted to the Conformity Assessment Body for verification when applying for Certificate of Conformity.		N/A
20	For AC Adaptors supplied together with Personal Mobility Devices: 1. Registered Supplier to declare the model of the AC adaptor that is to be used with/ bundled together with the PMDs; 2. Registered Supplier to provide valid IEC 60950-1 or IEC 62368-1 test reports for certification and registration of the declared AC adaptor under the CPS scheme; and Registered Supplier to provide the UL 2272 test report as supporting document, showing that the listed AC adaptor in the UL 2272 test report is the model declared to be used with/ bundled together with the PMDs.		P
21	CD/ DVD ROMs (used in personal computers) to have test certificate showing that CD/DVD ROM has complied with IEC 60825- 1.		N/A

IEC62638_1E ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
22	Modem card incorporated in the personal computer must be tested at set level (sub-clauses 5.1 & 6 of IEC 60950) or at component level.		N/A
23	Powerline Ethernet Adaptor incorporated with 13A socket-outlet, to be tested to additional test clauses 13, 17 & 18 of SS 246 (till 25/10/2021), or clauses 13, 17 & 18 of SS 145 Part 3: 2020.		N/A
	Other additional requirements which may be included in Chapter 7 of the information booklet in ongoing basis at the time of testing.		N/A